

संदर्भ संख्या : - GRID-INDIA/NLDC/MR/

दिनांक : 23.02.2024

सेवा में,

वितरण सूची के अनुसार

विषय:- निष्पादन रिपोर्ट – जनवरी 2024

महोदय,

आई०ई०जी०सी०-2010 के प्रावधान के अनुसार, जनवरी 2024 माह की अखिल भारतीय प्रणाली की निष्पादन रिपोर्ट संलग्न है।

धन्यवाद

भवदीय



(सुरजीत बनर्जी)

मुख्य महाप्रबन्धक (प्रणाली प्रचालन)

(रा०भा०प्रे०के०)

संलग्नक : मासिक प्रणाली रिपोर्ट

वितरण सूची Distribution List

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- 2 मुख्य अभियंता (जी एम), के.वि.प्रा., सेवा भवन, आर. के. पुरम, नई दिल्ली-110066
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Chief Engineer, National Power Committee, NRPC Building, 3rd floor, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 9 कार्यपालक निदेशक, पू. क्षेत्र. भा. प्रे. के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14, Golf Club Road, Kolkata-700033
- 10 कार्यपालक निदेशक, प. क्षेत्र. भा. प्रे. के., एफ-3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई - 400093
Executive Director, WRLDC, F-3, MIDC Area, Andheri (East), Mumbai-400093
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Executive Director, NRPC, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
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Chief General Manager, NERLDC, Dongtiah, Lower Nongrah, Laplang, Shillong-793006



ग्रिड-इंडिया
GRID-INDIA

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

NATIONAL LOAD DESPATCH CENTRE –राष्ट्रीय भार प्रेषण केन्द्र

मासिक ऑपरेशन रिपोर्ट

Monthly
Operation Report



जनवरी 2024

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

NATIONAL LOAD DESPATCH CENTRE, NEW DELHI



माह जनवरी 2024 के लिए प्रचालन निष्पादन रिपोर्ट
OPERATIONAL PERFORMANCE REPORT FOR THE
MONTH OF JANUARY-2024

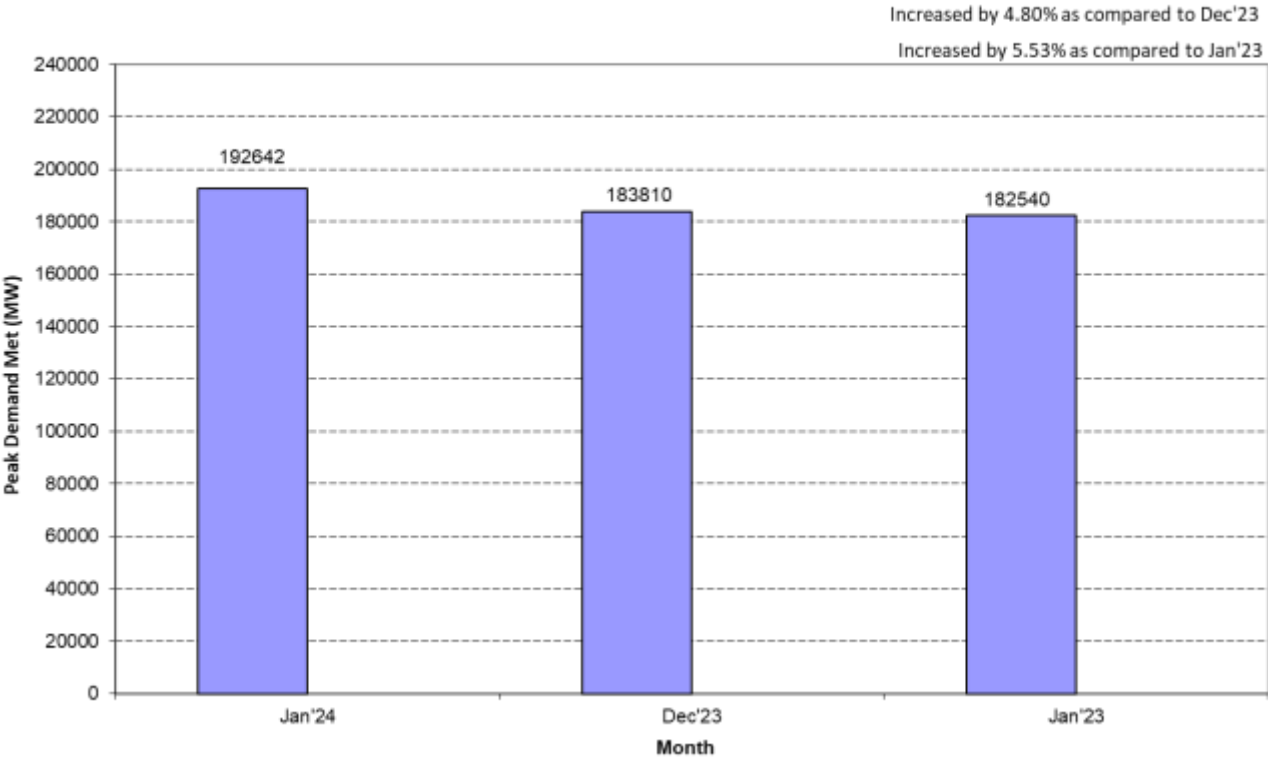
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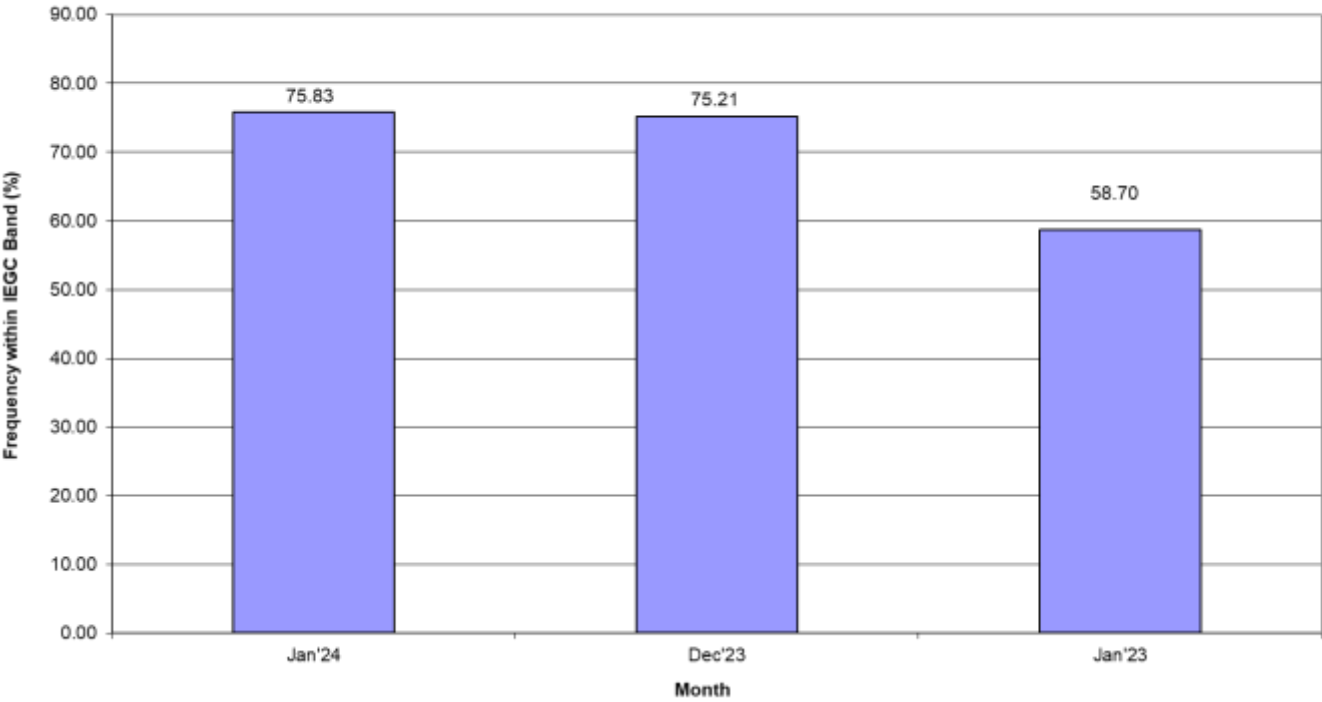
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1. SUMMARY OF REPORT FOR THE MONTH OF JANUARY-2024

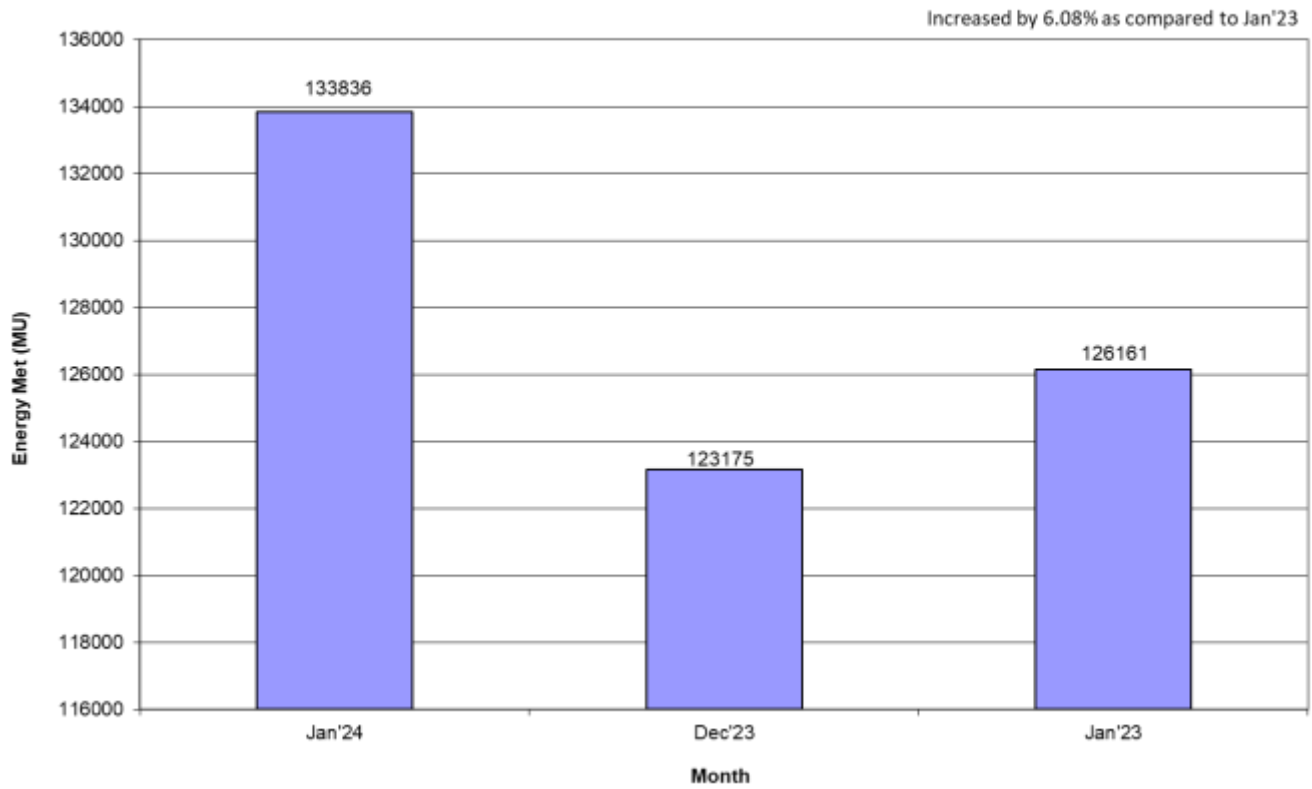
EVENING PEAK HOUR (at 1900hrs) DEMAND MET AT NATIONAL LEVEL (MW)



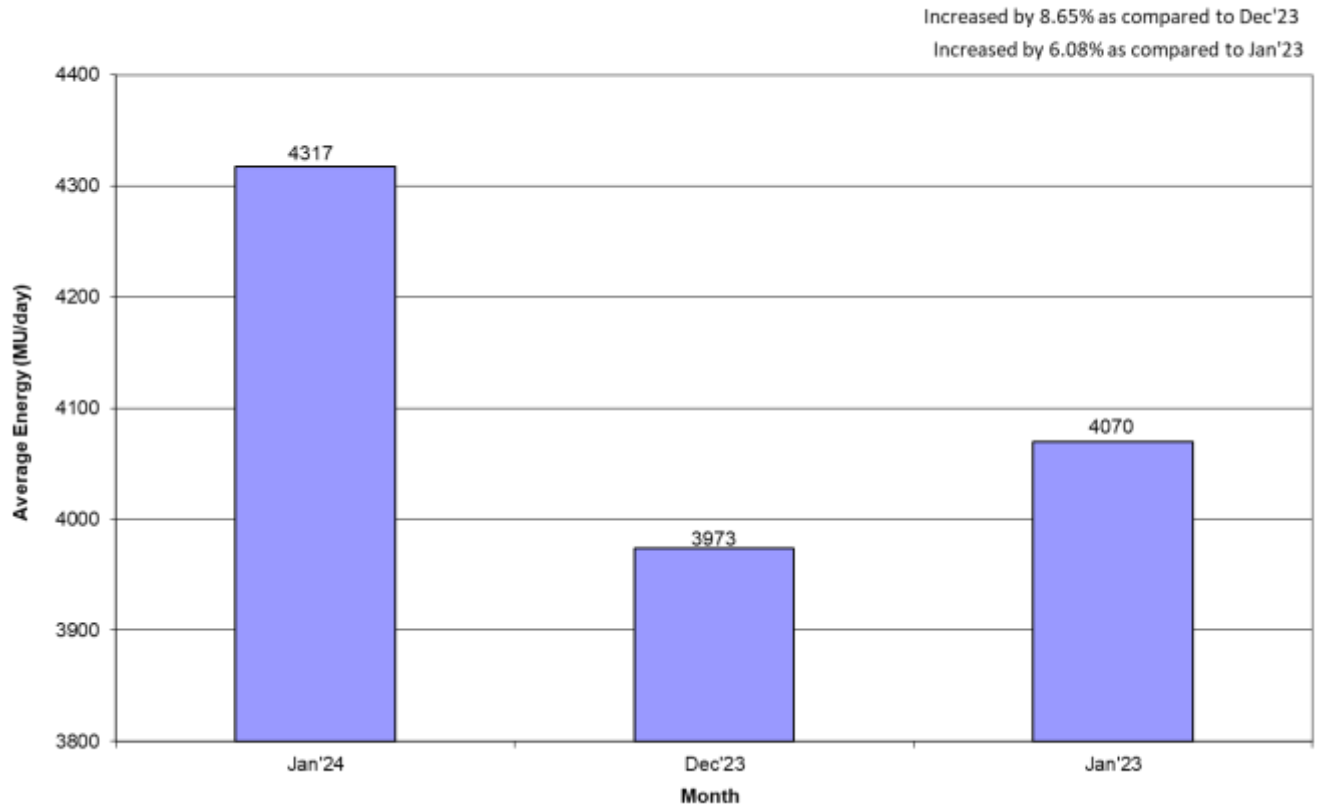
Frequency within IEGC Band



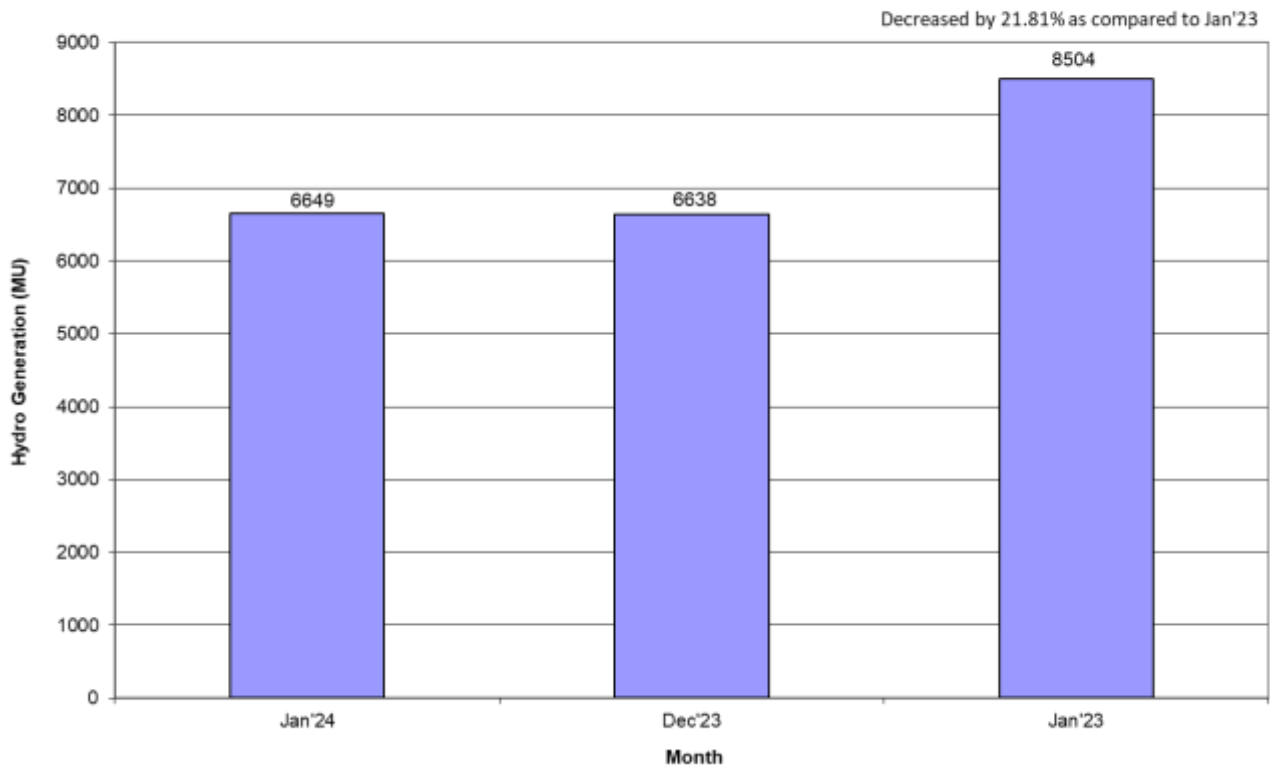
ENERGY MET AT NATIONAL LEVEL (MU)



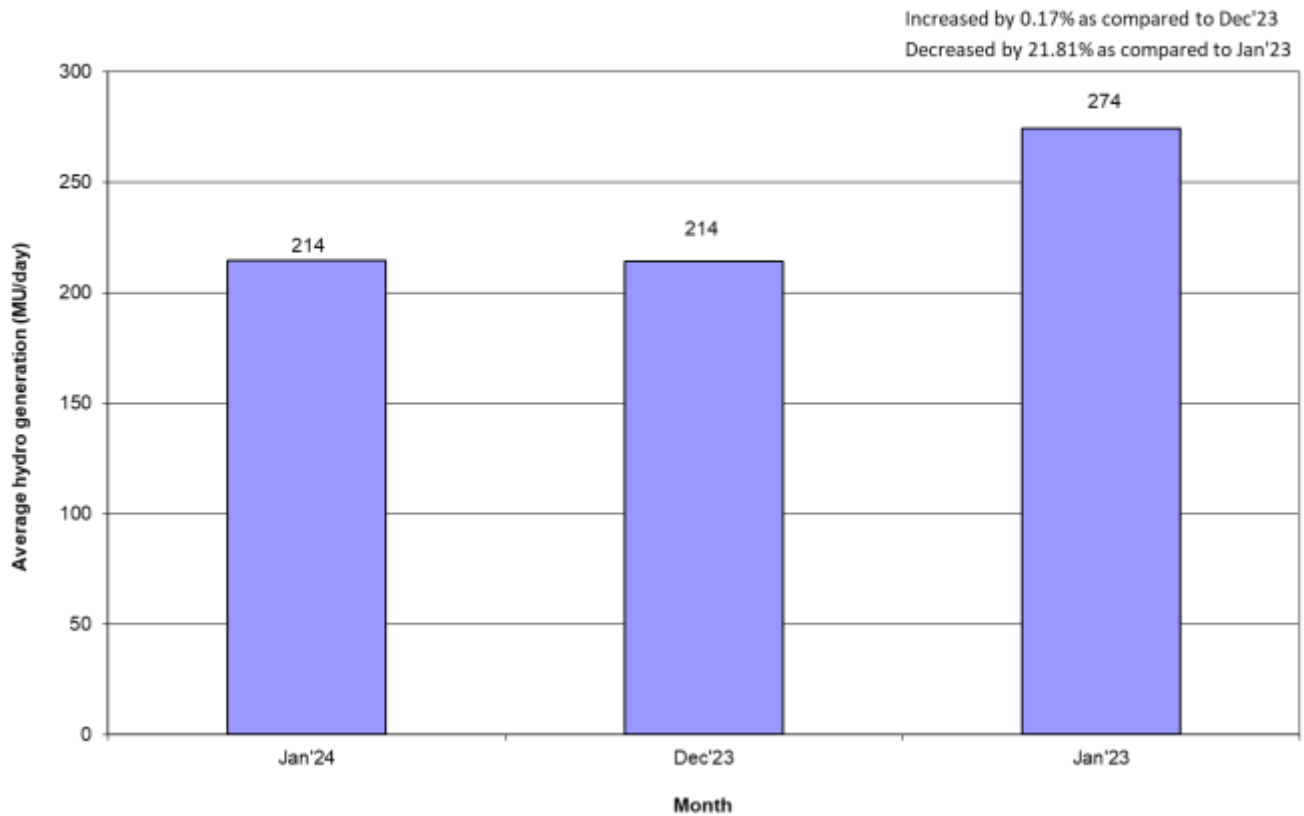
AVERAGE ENERGY MET AT NATIONAL LEVEL (MU/Day)



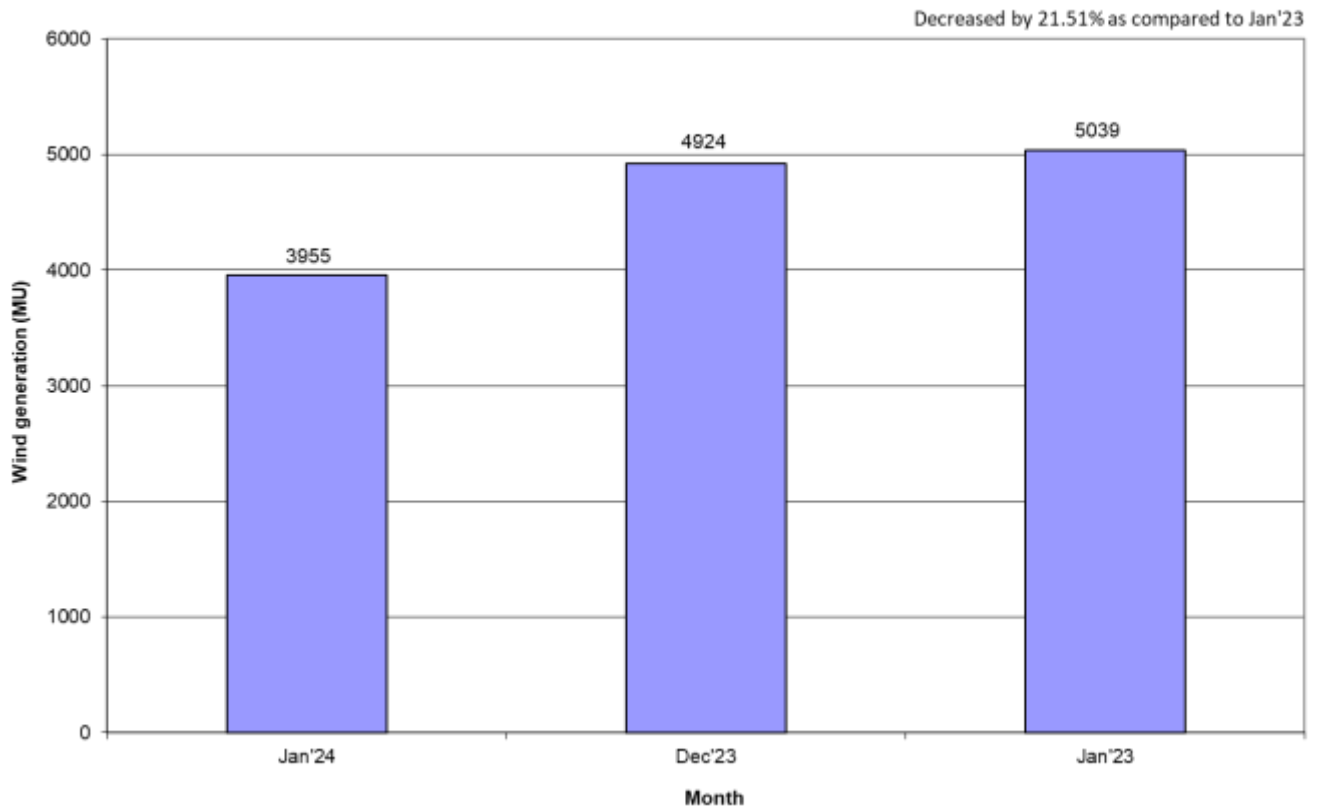
HYDRO GENERATION AT NATIONAL LEVEL (MU)



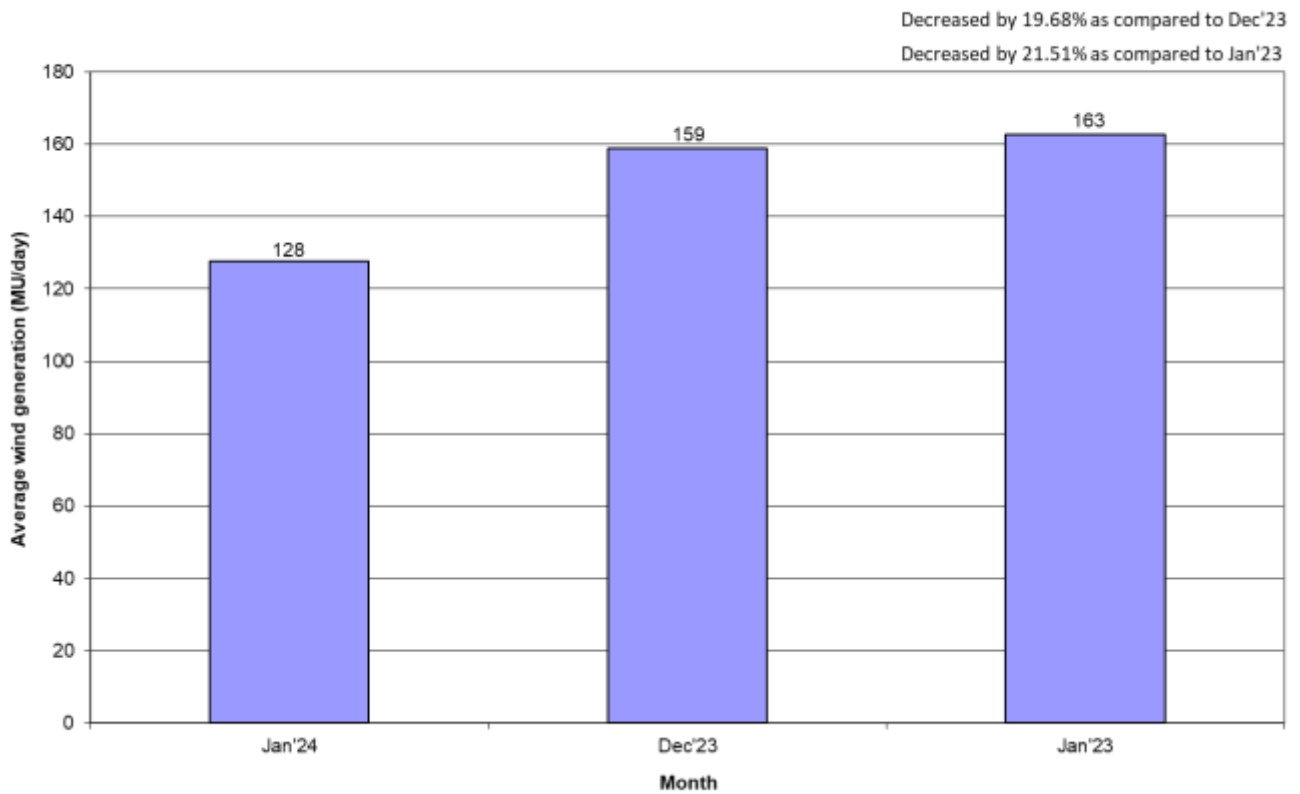
AVERAGE HYDRO GENERATION AT NATIONAL LEVEL (MU/Day)



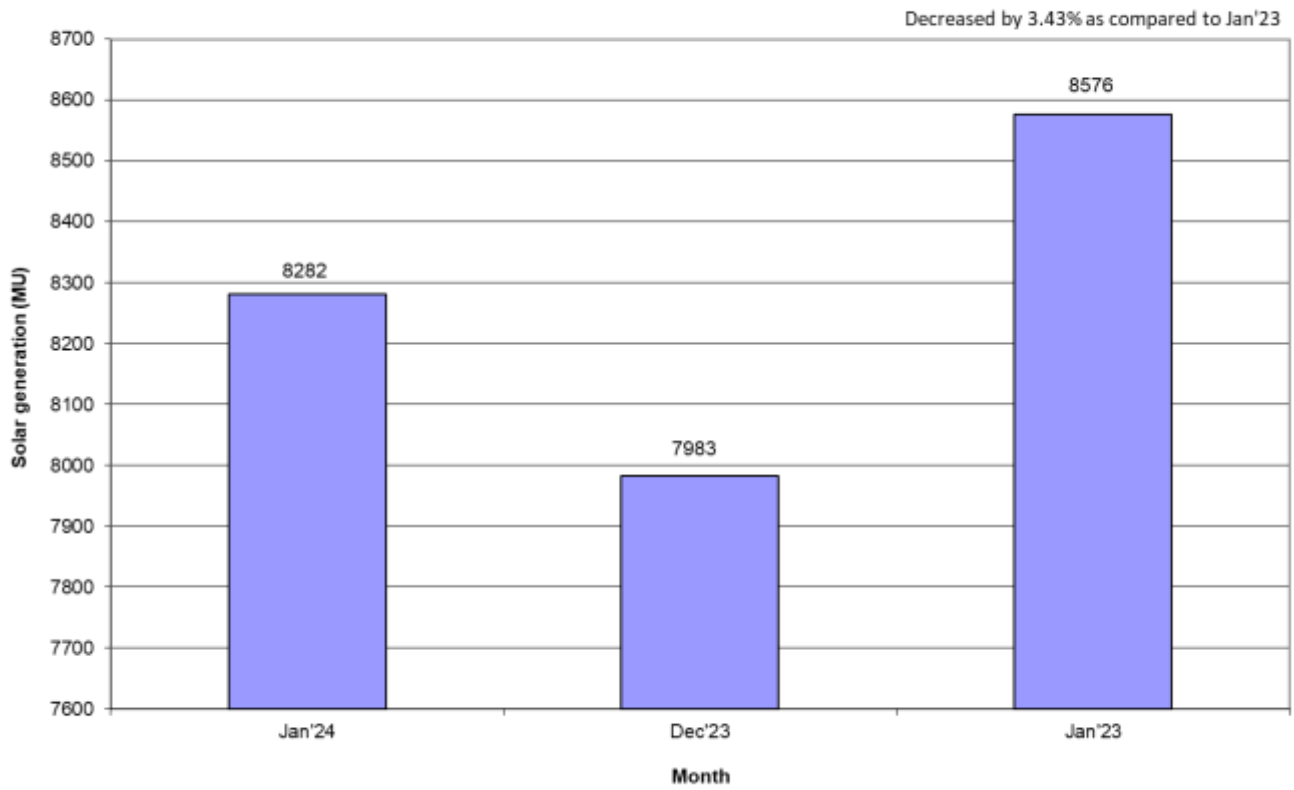
WIND GENERATION AT NATIONAL LEVEL (MU)



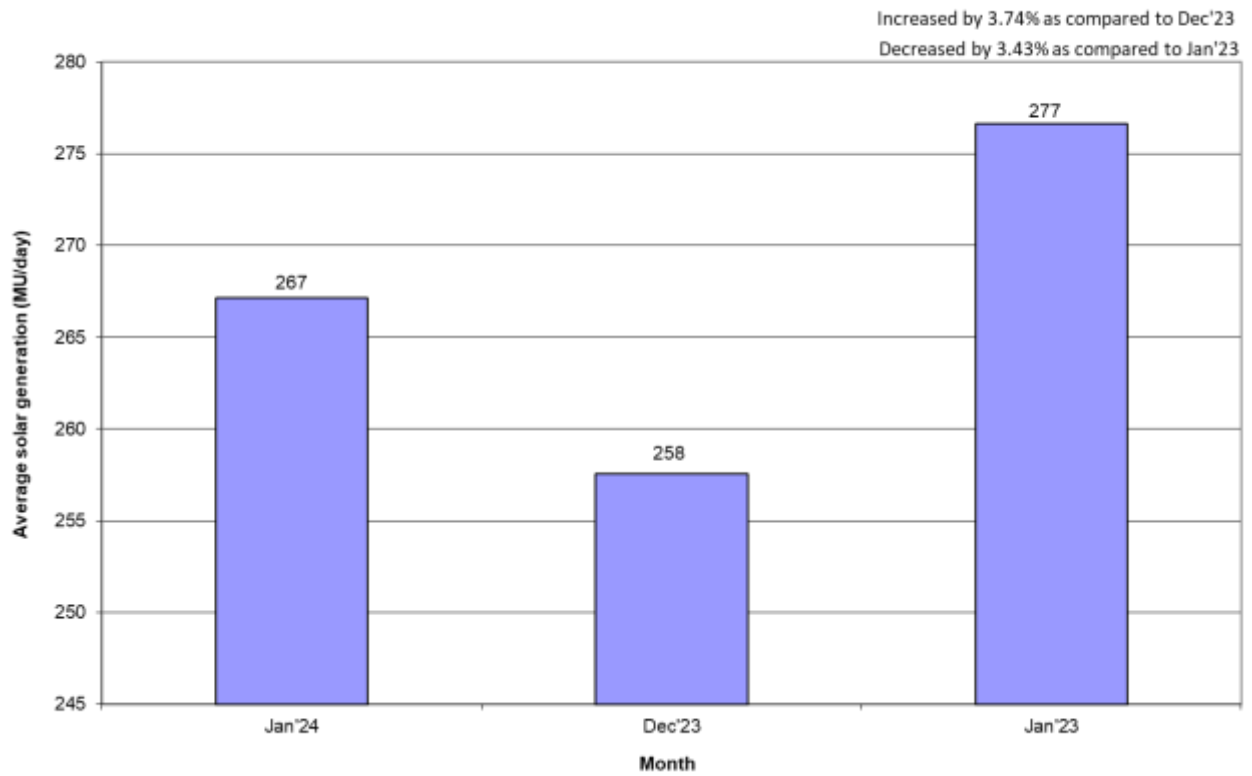
AVERAGE WIND GENERATION AT NATIONAL LEVEL (MU/Day)



SOLAR GENERATION AT NATIONAL LEVEL (MU)



AVERAGE SOLAR GENERATION AT NATIONAL LEVEL (MU/Day)



2. ALL INDIA REGIONWISE INSTALLED CAPACITY

As on 31.01.2024
(All figures are in MW)

S No	Region	THERMAL					NUCLEAR	HYDRO	RES @ MNRE	GRAND TOTAL
		COAL	LIGNITE	GAS	DIESEL	TOTAL				
1	NR	55631	1580	5995	0	63206	1620	20830	35509	121165
2	WR	74663	1400	10806	0	86869	2540	7563	43745	140716
3	SR	49094	3640	6492	434	59660	3320	11827	53342	128149
4	ER	27852	0	80	0	27932	0	4764	1906	34603
5	NER	949	0	1665	36	2650	0	1944	575	5169
6	ISLANDS	0	0	0	120	120	0	0	40	160
	ALL INDIA	208189	6620	25038	589	240437	7480	46928	135116	429961

Remarks: Data from CEA website

3. राष्ट्रीय स्तर पर संध्याकालीन शिखर विद्युत मांग पूर्ति
EVENING PEAK HOUR DEMAND MET AT NATIONAL LEVEL

माह: जनवरी 2024 MONTH:- JAN 2024

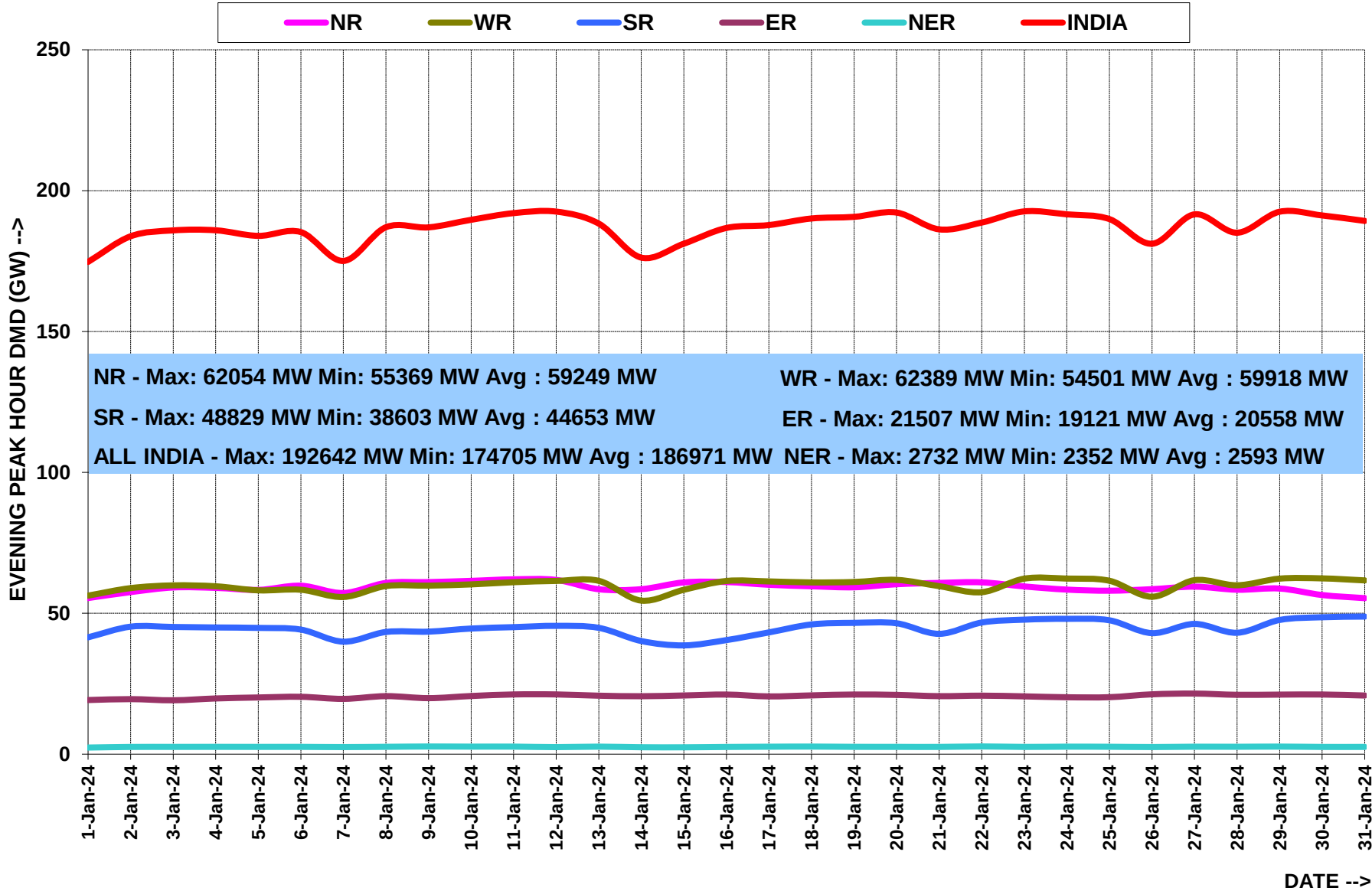
सभी आंकड़े मेगावाट मे All figures in MW

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
1-Jan-24	55410	56267	41466	19210	2352	174705
2-Jan-24	57519	58935	45256	19532	2564	183806
3-Jan-24	59160	59901	45134	19121	2578	185894
4-Jan-24	59020	59581	44964	19770	2597	185932
5-Jan-24	58299	58139	44810	20120	2580	183948
6-Jan-24	59783	58352	44211	20363	2587	185296
7-Jan-24	57203	55786	39922	19623	2518	175052
8-Jan-24	60775	59671	43373	20592	2614	187025
9-Jan-24	61071	59817	43483	19884	2713	186968
10-Jan-24	61503	60259	44590	20635	2666	189653
11-Jan-24	62054	61029	45079	21193	2672	192027
12-Jan-24	61844	61492	45538	21198	2500	192572
13-Jan-24	58535	61519	44834	20750	2670	188308
14-Jan-24	58556	54501	40154	20563	2457	176231
15-Jan-24	60960	58345	38603	20831	2438	181177
16-Jan-24	61093	61469	40482	21155	2557	186756
17-Jan-24	60118	61307	43216	20501	2646	187788
18-Jan-24	59591	60945	46043	20862	2687	190128
19-Jan-24	59232	61106	46594	21166	2604	190702
20-Jan-24	60312	61850	46448	21025	2586	192221
21-Jan-24	60779	59625	42703	20585	2573	186265
22-Jan-24	60980	57481	46726	20761	2732	188680
23-Jan-24	59523	62295	47738	20511	2575	192642
24-Jan-24	58401	62316	48027	20197	2661	191602
25-Jan-24	58012	61540	47512	20221	2625	189910
26-Jan-24	58553	55874	42957	21238	2534	181156
27-Jan-24	59452	61754	46235	21507	2647	191595
28-Jan-24	58357	59938	43090	21064	2623	185072
29-Jan-24	58776	62304	47628	21136	2685	192529
30-Jan-24	56479	62389	48594	21168	2568	191198
31-Jan-24	55369	61672	48829	20818	2571	189259
उच्चतम MAXIMUM	62054	62389	48829	21507	2732	192642
निम्नतम MINIMUM	55369	54501	38603	19121	2352	174705
औसत AVERAGE	59249	59918	44653	20558	2593	186971
अब तक का उच्चतम All Time Max.	76274	65518	52366	28579	3510	215592
दिनांक Date	04.09.23	13.10.23	18.04.23	18.04.23	12.08.22	28.08.23

नोट : सभी आंकड़े 1900 बजे की मांगपूर्ति पर आधारित है ।

Note: The above figures correspond to value at 1900 Hrs

EVENING PEAK HOUR DEMAND(at 1900hrs)MET DURING THE MONTH OF JANUARY 2024



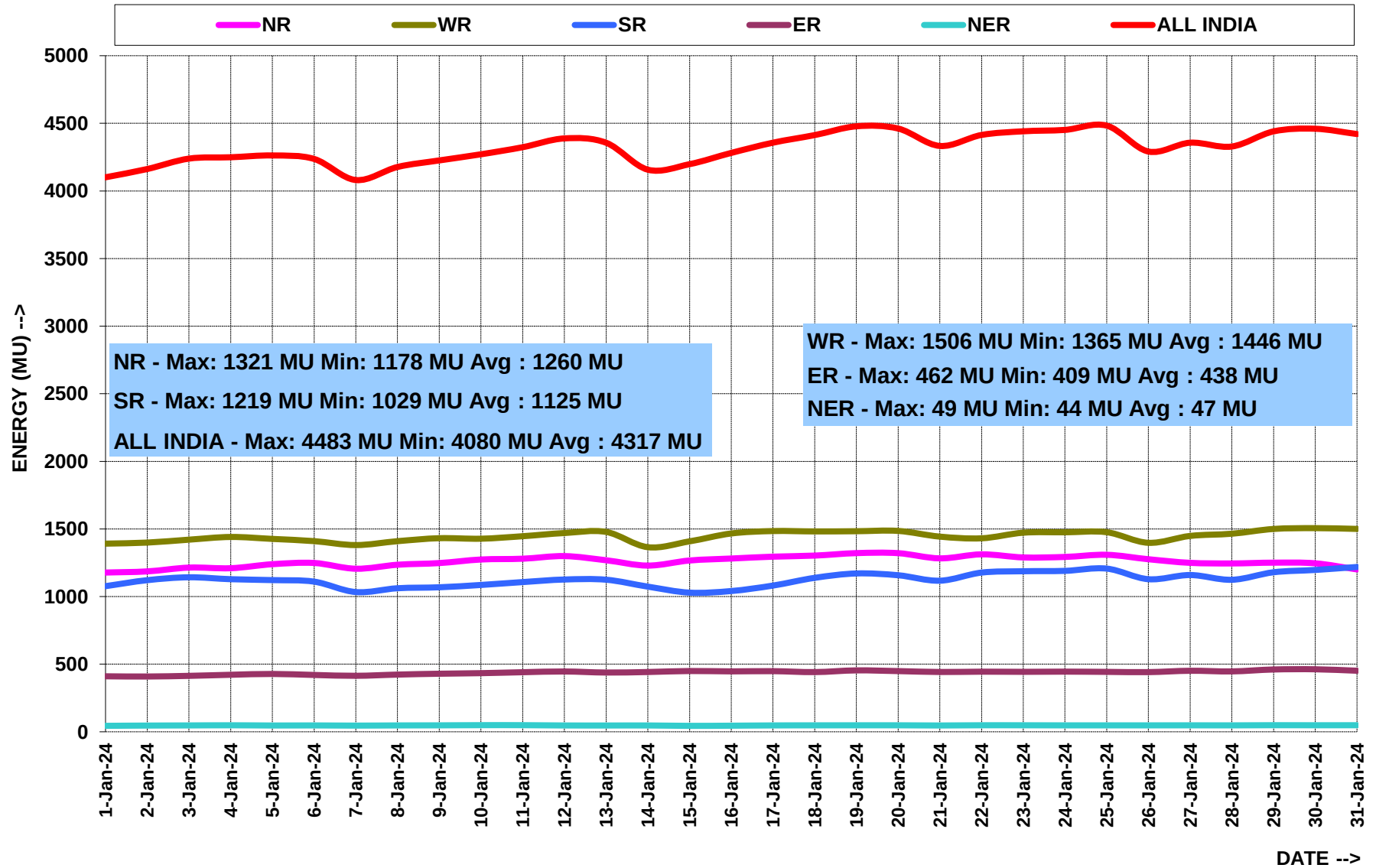
4. राष्ट्रीय स्तर पर विद्युत ऊर्जा आपूर्ति ENERGY MET AT NATIONAL LEVEL

माह: जनवरी 2024 MONTH:- JAN 2024

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jan-24	1178	1392	1077	410	44	4101
02-Jan-24	1186	1399	1121	409	46	4162
03-Jan-24	1214	1421	1143	414	47	4239
04-Jan-24	1210	1441	1128	422	48	4249
05-Jan-24	1240	1427	1122	428	47	4262
06-Jan-24	1249	1410	1111	420	47	4237
07-Jan-24	1206	1381	1033	414	46	4080
08-Jan-24	1236	1410	1062	423	47	4177
09-Jan-24	1248	1432	1069	429	48	4225
10-Jan-24	1274	1428	1086	434	49	4271
11-Jan-24	1280	1446	1107	441	49	4323
12-Jan-24	1299	1470	1126	446	47	4388
13-Jan-24	1269	1478	1125	438	46	4356
14-Jan-24	1229	1365	1074	442	47	4157
15-Jan-24	1267	1408	1029	450	44	4198
16-Jan-24	1281	1467	1041	447	45	4281
17-Jan-24	1295	1485	1082	448	47	4357
18-Jan-24	1303	1482	1139	442	48	4413
19-Jan-24	1321	1483	1171	454	48	4478
20-Jan-24	1321	1486	1158	449	48	4461
21-Jan-24	1283	1443	1117	442	47	4333
22-Jan-24	1312	1431	1178	445	48	4414
23-Jan-24	1290	1473	1187	443	48	4441
24-Jan-24	1293	1475	1190	445	48	4451
25-Jan-24	1309	1476	1207	443	48	4483
26-Jan-24	1276	1398	1128	441	48	4291
27-Jan-24	1249	1449	1159	451	48	4357
28-Jan-24	1245	1465	1124	447	48	4329
29-Jan-24	1251	1500	1180	461	49	4440
30-Jan-24	1246	1506	1197	462	48	4460
31-Jan-24	1201	1500	1219	452	49	4421
कुल TOTAL	39061	44826	34890	13593	1465	133836
उच्चतम MAXIMUM	1321	1506	1219	462	49	4483
निम्नतम MINIMUM	1178	1365	1029	409	44	4080
औसत AVERAGE	1260	1446	1125	438	47	4317
संचयी 2023-24 Cumulative 2023-24	406638	426757	344110	158477	16828	1352810
अब तक का उच्चतम All Time Max.	1796	1618	1329	639	72	5224
दिनांक Date	04.09.23	02.09.23	19.04.23	08.06.23	15.09.23	02.09.23

ENERGY MET DURING THE MONTH OF JANUARY' 2024



5. वर्ष 2023-24 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR YEAR 2023-24

राष्ट्रीय ग्रीड NATIONAL GRID								
फ्रिक्वेंसी रूपरेखा (Hz)			<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी
% समय	Apr-23	All India Grid	10.54	67.90	21.56	50.33	49.49	49.99
	May-23	All India Grid	9.83	68.48	21.69	50.43	49.48	49.99
	Jun-23	All India Grid	6.47	67.82	25.71	50.41	49.51	50.01
	Jul-23	All India Grid	4.60	74.96	20.44	50.23	49.76	50.01
	Aug-23	All India Grid	7.08	77.28	15.63	50.20	49.74	50.00
	Sep-23	All India Grid	5.26	77.90	16.84	50.21	49.77	50.00
	Oct-23	All India Grid	8.87	74.42	16.71	50.21	49.71	49.99
	Nov-23	All India Grid	6.83	74.36	18.81	50.26	49.73	50.00
	Dec-23	All India Grid	7.83	75.21	16.96	50.26	49.71	49.99
	Jan-24	All India Grid	6.80	75.83	17.37	50.25	49.73	50.00
	2023-24 (upto Jan)	All India Grid	7.41	73.44	19.15	50.43	49.48	50.00

5.1 जनवरी 2024 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR JANUARY 2024

फ्रिक्वेंसी रूपरेखा (Hz)		<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी	एफ़.वी.आई.
% समय	01-Jan-24	9.03	67.58	23.39	50.32	49.70	50.00	0.06
	02-Jan-24	8.95	78.51	12.55	50.20	49.70	49.99	0.05
	03-Jan-24	12.13	78.52	9.35	50.22	49.65	49.97	0.05
	04-Jan-24	7.95	78.63	13.41	50.24	49.76	49.99	0.04
	05-Jan-24	9.97	74.66	15.37	50.21	49.78	49.99	0.04
	06-Jan-24	9.04	69.26	21.70	50.29	49.69	50.00	0.06
	07-Jan-24	5.42	72.09	22.49	50.31	49.77	50.01	0.05
	08-Jan-24	10.78	72.09	17.13	50.22	49.66	49.99	0.06
	09-Jan-24	1.60	72.58	25.82	50.29	49.74	50.02	0.04
	10-Jan-24	2.67	75.73	21.60	50.20	49.77	50.01	0.03
	11-Jan-24	0.57	72.53	26.90	50.28	49.86	50.02	0.03
	12-Jan-24	5.54	74.05	20.41	50.24	49.67	50.00	0.04
	13-Jan-24	11.40	75.37	13.23	50.24	49.62	49.98	0.05
	14-Jan-24	8.29	74.56	17.15	50.25	49.66	49.99	0.05
	15-Jan-24	4.11	78.80	17.09	50.23	49.81	50.00	0.03
	16-Jan-24	8.78	74.07	17.14	50.22	49.76	49.99	0.04
	17-Jan-24	9.32	77.22	13.46	50.19	49.52	49.98	0.05
	18-Jan-24	1.71	81.05	17.23	50.24	49.83	50.01	0.03
	19-Jan-24	7.92	74.73	17.35	50.23	49.75	49.99	0.04
	20-Jan-24	3.78	80.13	16.09	50.26	49.79	50.00	0.03
	21-Jan-24	3.48	79.09	17.43	50.33	49.76	50.01	0.03
	22-Jan-24	7.42	72.06	20.52	50.27	49.58	50.00	0.05
	23-Jan-24	9.38	76.83	13.80	50.24	49.76	49.99	0.04
	24-Jan-24	7.84	79.61	12.56	50.21	49.74	49.99	0.04
	25-Jan-24	10.25	76.68	13.07	50.20	49.77	49.99	0.04
	26-Jan-24	5.15	74.80	20.05	50.25	49.81	50.01	0.04
	27-Jan-24	10.35	71.24	18.41	50.32	49.63	49.99	0.06
	28-Jan-24	5.95	80.24	13.81	50.27	49.67	50.00	0.04
	29-Jan-24	2.00	76.27	21.72	50.22	49.86	50.01	0.03
	30-Jan-24	6.77	83.70	9.53	50.22	49.70	49.99	0.04
	31-Jan-24	3.24	78.10	18.66	50.26	49.83	50.01	0.03

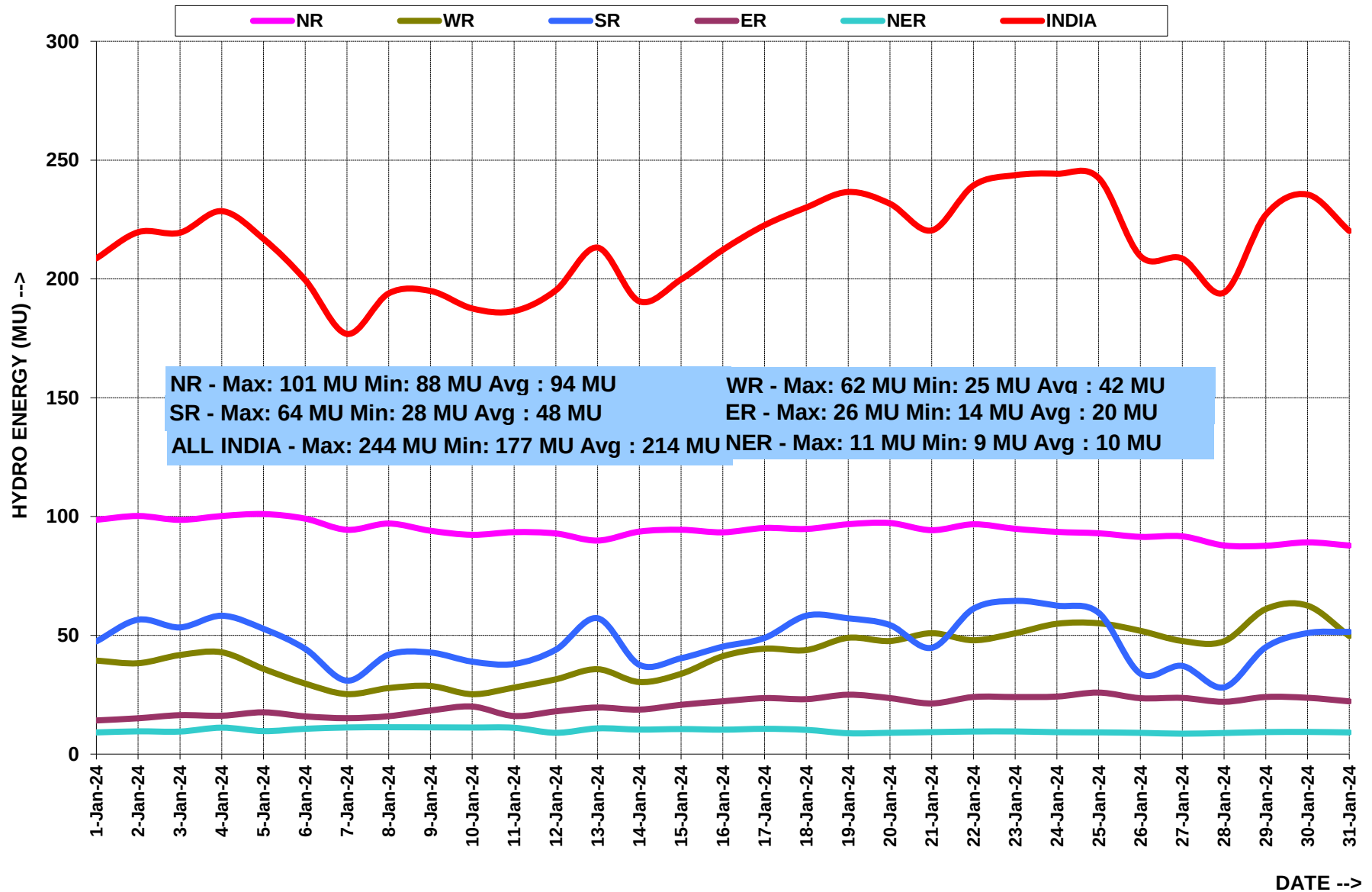
6.राष्ट्रीय स्तर पर जल विद्युत उत्पादन
HYDRO GENERATION AT NATIONAL LEVEL

माह: जनवरी 2024 MONTH:- JAN 2024

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jan-24	99	39	47	14	9	209
02-Jan-24	100	38	57	15	10	220
03-Jan-24	99	42	53	16	9	219
04-Jan-24	100	43	58	16	11	229
05-Jan-24	101	36	53	18	10	217
06-Jan-24	99	30	44	16	11	200
07-Jan-24	94	25	31	15	11	177
08-Jan-24	97	28	42	16	11	194
09-Jan-24	94	29	43	18	11	195
10-Jan-24	92	25	39	20	11	188
11-Jan-24	93	28	38	16	11	186
12-Jan-24	93	31	44	18	9	195
13-Jan-24	90	36	57	20	11	213
14-Jan-24	94	30	38	19	10	191
15-Jan-24	94	34	40	21	11	200
16-Jan-24	93	41	45	22	10	212
17-Jan-24	95	44	49	24	11	223
18-Jan-24	95	44	58	23	10	230
19-Jan-24	97	49	57	25	9	237
20-Jan-24	97	48	54	24	9	232
21-Jan-24	94	51	45	21	9	220
22-Jan-24	97	48	61	24	9	239
23-Jan-24	95	51	64	24	10	244
24-Jan-24	94	55	62	24	9	244
25-Jan-24	93	55	59	26	9	242
26-Jan-24	91	52	34	24	9	210
27-Jan-24	92	48	37	24	9	209
28-Jan-24	88	47	28	22	9	194
29-Jan-24	88	61	45	24	9	227
30-Jan-24	89	62	51	24	9	236
31-Jan-24	88	50	51	22	9	220
कुल TOTAL	2924	1299	1486	634	306	6649
उच्चतम MAXIMUM	101	62	64	26	11	244
निम्नतम MINIMUM	88	25	28	14	9	177
औसत AVERAGE	94	42	48	20	10	214
संचयी 2023-24 Cumulative 2023-24	69453	14912	18644	21444	6152	130606
अब तक का उच्चतम All Time Max.	443	167	208	157	41	877
दिनांक Date	01.08.23	18.12.14	31.08.18	14.09.22	29.08.23	30.08.22

HYDRO ENERGY DURING THE MONTH OF JANUARY' 2024



7. राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन
WIND GENERATION AT NATIONAL LEVEL

माह: जनवरी 2024 MONTH:- JAN 2024

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jan-24	21	68	55	-----	-----	143
02-Jan-24	16	42	59	-----	-----	116
03-Jan-24	20	52	57	-----	-----	129
04-Jan-24	18	76	48	-----	-----	142
05-Jan-24	14	65	45	-----	-----	124
06-Jan-24	17	111	57	-----	-----	186
07-Jan-24	18	142	74	-----	-----	233
08-Jan-24	16	126	72	-----	-----	214
09-Jan-24	19	174	82	-----	-----	275
10-Jan-24	16	120	84	-----	-----	220
11-Jan-24	6	80	102	-----	-----	188
12-Jan-24	9	49	114	-----	-----	172
13-Jan-24	5	30	46	-----	-----	82
14-Jan-24	8	39	40	-----	-----	88
15-Jan-24	4	37	30	-----	-----	71
16-Jan-24	4	28	27	-----	-----	59
17-Jan-24	15	31	33	-----	-----	79
18-Jan-24	15	64	29	-----	-----	108
19-Jan-24	11	65	35	-----	-----	111
20-Jan-24	10	50	49	-----	-----	109
21-Jan-24	9	46	37	-----	-----	91
22-Jan-24	6	43	44	-----	-----	93
23-Jan-24	10	46	42	-----	-----	98
24-Jan-24	6	40	28	-----	-----	74
25-Jan-24	14	39	34	-----	-----	88
26-Jan-24	12	65	59	-----	-----	136
27-Jan-24	8	75	71	-----	-----	153
28-Jan-24	6	37	58	-----	-----	101
29-Jan-24	3	33	63	-----	-----	99
30-Jan-24	9	33	42	-----	-----	84
31-Jan-24	7	54	30	-----	-----	91
कुल TOTAL	349	1960	1647	-----	-----	3955
उच्चतम MAXIMUM	21	174	114	-----	-----	275
निम्नतम MINIMUM	3	28	27	-----	-----	59
औसत AVERAGE	11	63	53	-----	-----	128
संचयी 2023-24 Cumulative 2023-24	7193	32048	31761	-----	-----	71003
अब तक का उच्चतम All Time Max.	86	273	298	-----	-----	611
दिनांक Date	07.08.23	07.08.23	21.07.23	-----	-----	02.08.23

*Source: As reported by SLDCs. Limited visibility of embedded wind generator data.

8. राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन
SOLAR GENERATION AT NATIONAL LEVEL

माह: जनवरी 2024 MONTH:- JAN 2024

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jan-24	84	52	113	2.1	1.2	251
02-Jan-24	90	48	100	2.4	1.1	241
03-Jan-24	97	50	94	2.5	1.1	244
04-Jan-24	85	50	78	5.3	1.1	219
05-Jan-24	96	49	87	4.7	1.2	237
06-Jan-24	95	52	95	5.1	1.1	248
07-Jan-24	76	52	93	5.3	1.2	226
08-Jan-24	69	52	61	2.2	1.1	185
09-Jan-24	80	44	60	4.9	0.9	190
10-Jan-24	103	46	82	2.4	1.0	234
11-Jan-24	126	58	113	2.2	0.8	300
12-Jan-24	123	64	115	5.5	0.7	308
13-Jan-24	104	66	111	4.7	0.8	286
14-Jan-24	106	68	111	5.3	1.1	291
15-Jan-24	116	65	113	5.1	1.1	300
16-Jan-24	110	68	112	3.1	0.8	294
17-Jan-24	90	60	115	4.5	0.6	270
18-Jan-24	95	60	104	1.7	0.7	261
19-Jan-24	112	62	106	2.1	1.0	284
20-Jan-24	110	61	98	4.6	1.2	276
21-Jan-24	95	65	112	4.6	0.9	278
22-Jan-24	106	65	116	4.2	1.0	293
23-Jan-24	93	63	120	4.7	1.1	281
24-Jan-24	100	67	116	1.6	0.9	286
25-Jan-24	107	69	117	2.0	0.9	296
26-Jan-24	97	67	124	2.6	0.9	291
27-Jan-24	93	65	119	2.6	0.9	280
28-Jan-24	84	66	122	5.2	1.1	277
29-Jan-24	104	63	119	6.0	1.1	293
30-Jan-24	66	61	129	5.3	1.1	263
31-Jan-24	107	56	127	5.1	0.8	296
कुल TOTAL	3017	1835	3280	120	30	8282
उच्चतम MAXIMUM	126	69	129	6	1.2	308
निम्नतम MINIMUM	66	44	60	2	0.6	185
औसत AVERAGE	97	59	106	4	1.0	267
संचयी 2023-24 Cumulative 2023-24	36204	16639	33389	998	295.8	87424
अब तक का उच्चतम All Time Max.	147	71	143	6.7	2.4	351
दिनांक Date	10.05.23	12.05.23	21.03.23	23.03.23	22.06.22	10.05.23

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

9. राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति
DAILY MAXIMUM DEMAND MET AT NATIONAL LEVEL

माह: जनवरी 2024 MONTH:- JAN 2024

सभी आंकड़े मेगावाट में All figures in MW

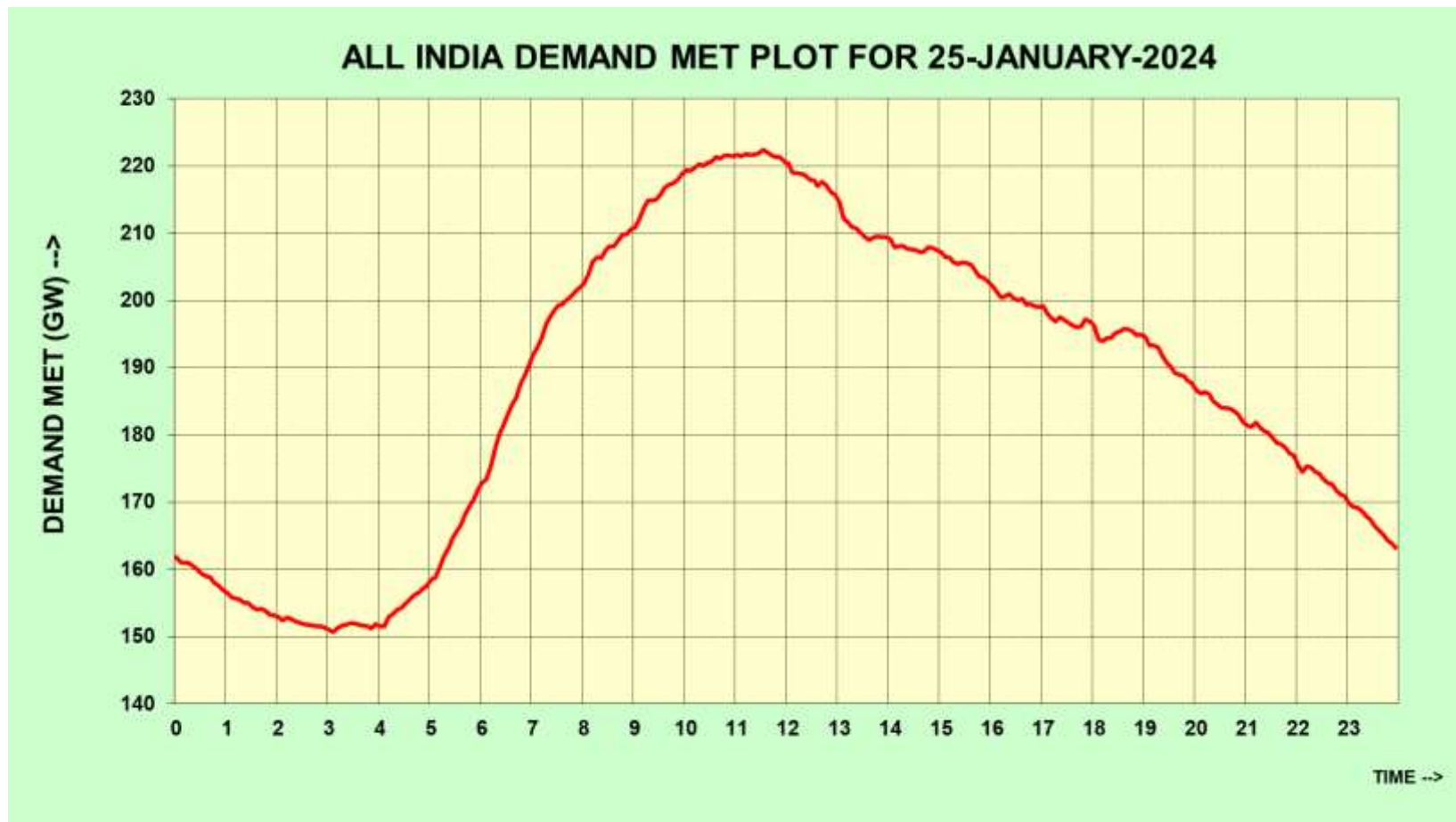
दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL	विभिन्नता फैक्टर Diversity Factor*
01-Jan-24	61335	70372	56770	20430	2434	207953	1.016
02-Jan-24	60422	72624	57505	20947	2647	209524	1.022
03-Jan-24	62059	72003	57623	20325	2739	210785	1.019
04-Jan-24	61670	72155	57476	20659	2707	208037	1.032
05-Jan-24	62979	72523	56332	21033	2685	210777	1.023
06-Jan-24	63598	71282	56505	20879	2732	211224	1.018
07-Jan-24	61095	69772	52344	20676	2583	203047	1.017
08-Jan-24	63734	70962	53548	21713	2711	204828	1.038
09-Jan-24	63796	71834	53984	21453	2742	207039	1.033
10-Jan-24	65371	71431	54608	21611	2762	212121	1.017
11-Jan-24	66712	71871	56168	22058	2755	215228	1.020
12-Jan-24	68801	73098	57344	22261	2763	219800	1.020
13-Jan-24	66230	73690	55790	21229	2765	217645	1.009
14-Jan-24	63667	67340	54783	21528	2599	207217	1.013
15-Jan-24	65117	72026	54170	21887	2509	212774	1.014
16-Jan-24	65633	73353	54277	22144	2598	215183	1.013
17-Jan-24	64517	74356	55640	21615	2668	214411	1.020
18-Jan-24	65894	74204	57110	21676	2666	214834	1.031
19-Jan-24	68515	73828	58816	22132	2664	219640	1.029
20-Jan-24	67118	74064	57180	21792	2600	218071	1.021
21-Jan-24	65473	72341	55755	21174	2676	213206	1.020
22-Jan-24	69524	71819	59522	21589	2755	221330	1.018
23-Jan-24	65588	73857	58387	21266	2651	217114	1.021
24-Jan-24	65446	73236	58998	21866	2718	217134	1.024
25-Jan-24	67208	73293	60860	21251	2627	222327	1.013
26-Jan-24	68258	67887	57363	22191	2595	215128	1.015
27-Jan-24	65204	73103	58951	23062	2666	219550	1.016
28-Jan-24	64982	73675	56039	22187	2652	215887	1.017
29-Jan-24	66432	74799	59165	22013	2802	222283	1.013
30-Jan-24	64525	74283	59346	21878	2603	219094	1.016
31-Jan-24	62879	73740	60804	21573	2609	220092	1.007
उच्चतम MAXIMUM	69524	74799	60860	23062	2802	222327	1.038
निम्नतम MINIMUM	60422	67340	52344	20325	2434	203047	1.007
औसत AVERAGE	64961	72414	56876	21552	2667	214299	1.020
अब तक का उच्चतम All Time Max.	81220	74799	64314	29641	3712	239978	
दिनांक Date	04.09.23	29.01.24	15.03.23	28.07.23	04.09.23	01.09.23	

* Diversity factor = (Sum of regional max demands) / All India max demand

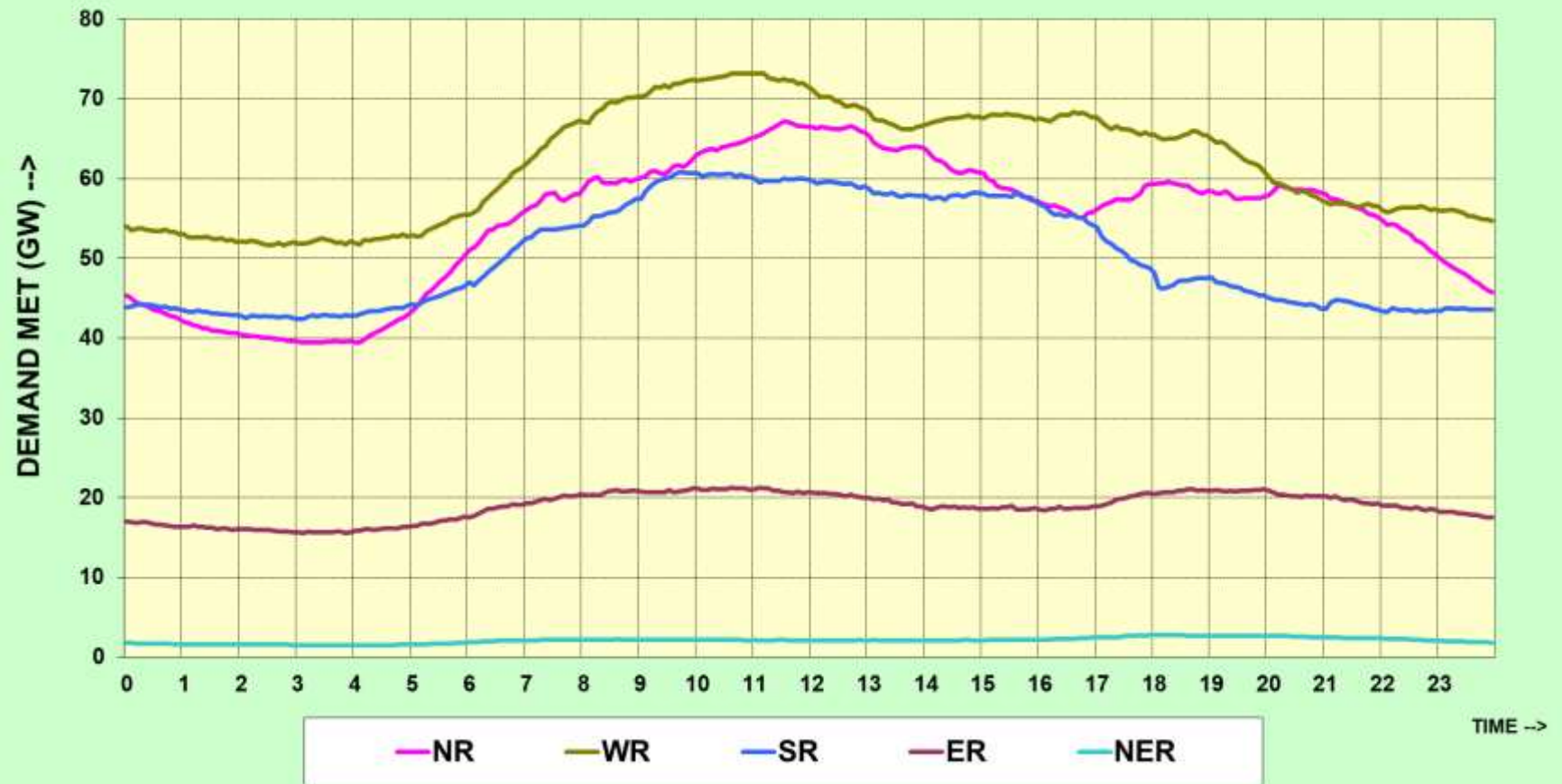
नोट : यह आंकड़े एनएलडीसी स्काडा प्रणाली में दर्ज दैनिक अधिकतम मांगपूर्ति दर्शाते हैं ।

Note: The above figures denote daily maximum demand met recorded in NLDC SCADA.

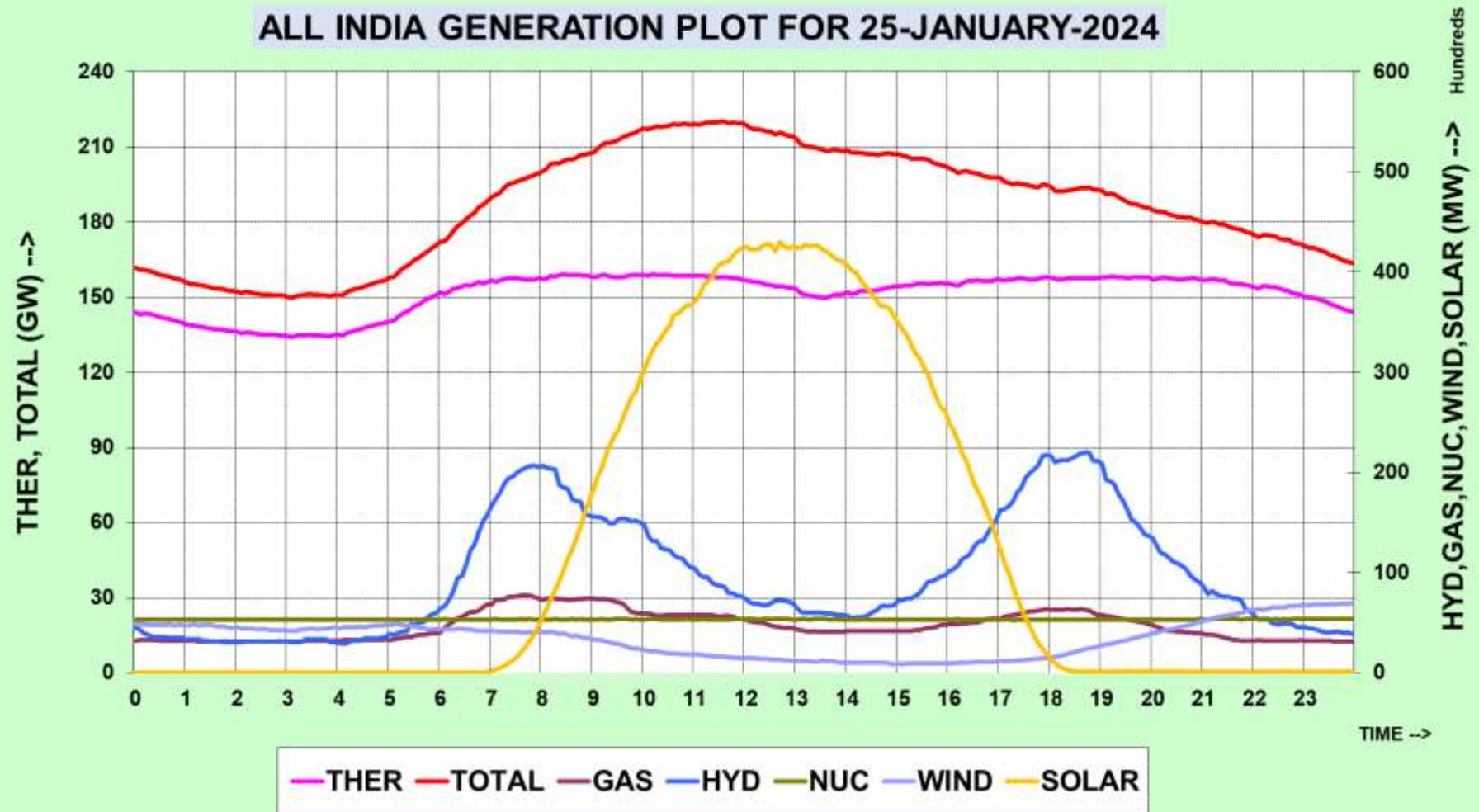
10. DEMAND AND GENERATION PLOTS FOR 25 JANUARY (MAXIMUM DEMAND MET)

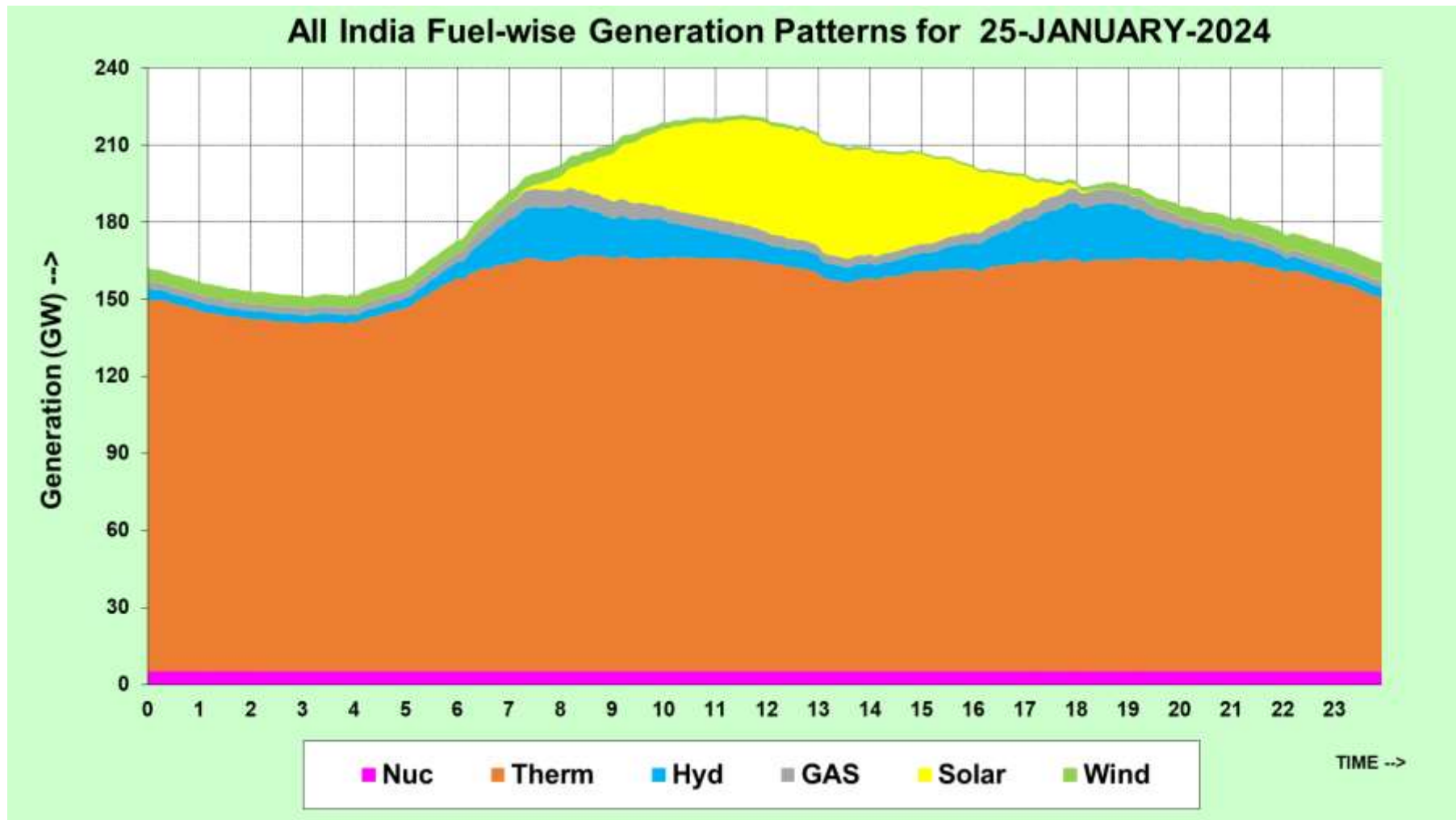


REGIONAL DEMAND MET PLOT FOR 25-JANUARY-2024



ALL INDIA GENERATION PLOT FOR 25-JANUARY-2024





25 जनवरी 2024 (अधिकतम माँग) की आल इंडिया माँग आपूर्ति
ALL INDIA DEMAND MET FOR 25 JANUARY 2024 (MAXIMUM DEMAND*)

Time (hrs)	All India Demand (MW)		Time (hrs)	All India Demand (MW)
00:05	161114		12:05	219001
00:20	160526		12:20	218653
00:35	158986		12:35	216987
00:50	157635		12:50	216089
01:05	155841		13:05	212174
01:20	155027		13:20	210703
01:35	154040		13:35	209024
01:50	153306		13:50	209389
02:05	152421		14:05	207934
02:20	152289		14:20	207691
02:35	151769		14:35	207189
02:50	151527		14:50	207795
03:05	150737		15:05	206403
03:20	151720		15:20	205449
03:35	151788		15:35	205259
03:50	151235		15:50	203360
04:05	151539		16:05	201328
04:20	154045		16:20	200973
04:35	155544		16:35	200209
04:50	157134		16:50	199198
05:05	158784		17:05	197946
05:20	163083		17:20	197512
05:35	166638		17:35	196180
05:50	170508		17:50	197187
06:05	173389		18:05	194152
06:20	180088		18:20	194422
06:35	184483		18:35	195767
06:50	188911		18:50	194784
07:05	192937		19:05	193270
07:20	197615		19:20	191877
07:35	199445		19:35	189242
07:50	201391		19:50	188063
08:05	203816		20:05	186123
08:20	206332		20:20	185131
08:35	207996		20:35	184022
08:50	209888		20:50	183017
09:05	211968		21:05	181210
09:20	214794		21:20	180553
09:35	216828		21:35	178800
09:50	217975		21:50	177197
10:05	219231		22:05	174503
10:20	219975		22:20	174495
10:35	221348		22:35	172885
10:50	221513		22:50	171138
11:05	221444		23:05	169343
11:20	221662		23:20	167932
11:35	222121		23:35	165829
11:50	221293		23:50	163870
Maximum Demand of 222327 MW met@11:30 hrs (from 1 min. interval SCADA DATA)				

* 15 minute interval SCADA DATA of instantaneous demand

11. ACTUAL POWER SUPPLY POSITION - JANUARY 2024

REGION	STATE	Requirement/Availability in MU/DAY			Requirement/Availability in MU				Peak Demand/Peak Met in MW			
		Requirement	Energy met	Surplus(+)/Deficit(-)	Requirement	Energy met	Surplus(+)/Deficit(-)	% Shortage	Requirement	Peak Demand Met	Surplus(+)/Deficit(-)	% Shortage
NR	Chandigarh	5	5	0.0	158	158	0	0.0	348	348	0	0.0
	Delhi	88	88	0.0	2732	2732	0	0.0	5816	5816	0	0.0
	Haryana	161	158	-2.5	4991	4913	-78	-1.6	9530	9530	0	0.0
	Himachal Pradesh	37	37	-0.2	1148	1142	-5	-0.5	2181	2181	0	0.0
	J&K(UT) and Ladakh(UT)	66	64	-1.1	2032	1997	-35	-1.7	3181	3133	-48	-1.5
	Punjab	159	159	0.0	4942	4942	0	0.0	9370	9370	0	0.0
	Rajasthan	328	325	-3.5	10171	10064	-107	-1.1	18128	18128	0	0.0
	Uttar Pradesh	375	374	-0.9	11619	11592	-27	-0.2	22703	22703	0	0.0
	Uttarakhand	46	46	-0.7	1439	1418	-21	-1.5	2635	2405	-230	-8.7
	Railway	3	3	0.0	104	104	0	0.0	208	208	0	0.0
	NFL	0	0	0.0	2	2	0	0.0	4	4	0	0.0
	TOTAL	1269	1260	-8.8	39337	39064	-273	-0.7	69559	69327	-232	-0.3
WR	Gujarat	399	399	0.0	12383	12383	0	0.0	21861	21861	0	0.0
	Madhya Pradesh	309	308	-1.4	9588	9544	-44	-0.5	18252	17817	-435	-2.4
	Chhattisgarh	103	103	-0.1	3203	3200	-3	-0.1	5338	5334	-4	-0.1
	Maharashtra	564	564	-0.1	17476	17474	-2	0.0	28651	28651	0	0.0
	Goa	13	13	0.0	401	401	0	0.0	721	720	-1	-0.1
	DNHDDPDCL	28	28	0.0	879	879	0	0.0	1294	1294	0	0.0
	AMNSIL	19	19	0.0	578	578	0	0.0	1083	1083	0	0.0
	BALCO	12	12	0.0	386	386	0	0.0	526	526	0	0.0
	TOTAL	1448	1447	-1.6	44895	44847	-48	-0.1	74674	74583	-91	-0.1
SR	Andhra Pradesh	209	208	-0.6	6482	6463	-19	-0.3	11052	11052	0	0.0
	Telangana	247	246	-0.7	7663	7641	-22	-0.3	12666	12666	0	0.0
	Karnataka	271	270	-0.7	8389	8367	-22	-0.3	15668	15668	0	0.0
	Kerala	82	82	-0.2	2550	2543	-7	-0.3	4354	4354	0	0.0
	Tamil Nadu	312	311	-0.9	9658	9630	-28	-0.3	15920	15920	0	0.0
	Puducherry	8	8	0.0	257	256	-1	-0.3	439	439	0	-0.1
	TOTAL	1129	1126	-3.2	34999	34901	-99	-0.3	60891	60891	0	0.0
ER	Bihar	99	97	-2.5	3077	2999	-78	-2.5	5625	5625	0	0.0
	Jharkhand	34	31	-2.6	1052	971	-81	-7.7	1805	1805	0	0.0
	DVC	71	71	0.0	2207	2207	0	0.0	3251	3251	0	0.0
	Odisha	94	94	0.0	2925	2925	0	0.0	5051	5051	0	0.0
	West Bengal	143	143	0.0	4428	4428	0	0.0	7281	7281	0	0.0
	Sikkim	2	2	0.0	61	61	0	0.0	137	137	0	0.0
	Railways_ER ISTS	0	0	0.0	4	4	0	0.0	20	20	0	0.0
	TOTAL	444	439	-5.1	13754	13595	-159	-1.2	23227	23044	-183	-0.8
NER	Arunachal Pradesh	3	3	0.0	89	89	0	0.0	181	181	0	0.0
	Assam	27	27	-0.4	837	824	-13	-1.6	1538	1538	0	0.0
	Manipur	3	3	-0.3	104	96	-8	-8.1	258	248	-10	-3.9
	Meghalaya	6	6	-0.1	201	196	-5	-2.3	405	405	0	0.0
	Mizoram	2	2	0.0	70	70	0	0.0	162	162	0	0.0
	Nagaland	2	2	0.0	71	71	0	0.0	162	162	0	0.0
	Tripura	4	4	0.0	122	122	0	0.0	235	235	0	0.0
	TOTAL	48	47	-0.8	1493	1467	-26	-1.7	2818	2808	-10	-0.4
ALL INDIA		4338	4318	-19.5	134478	133873	-605	-0.4				

NOTE: BASED ON THE DATA FURNISHED BY ENTITIES ON DAILY BASIS

12. ENERGY COMPARISON OF JANUARY 2024 vs JANUARY 2023

REGION	STATE	Energy Requirement (MU)					Energy Met (MU)				
		Jan-23	Jan-24	Difference	% Change	Average MU/day for Jan-24	Jan-23	Jan-24	Difference	% Change	Average MU/day for Jan-24
NR	Chandigarh	143	158	15	11	5	143	158	15	11	5
	Delhi	2481	2732	251	10	88	2481	2732	251	10	88
	Haryana	4420	4991	571	13	161	4386	4913	527	12	158
	Himachal Pradesh	1071	1148	77	7	37	1068	1142	74	7	37
	J&K(UT) and Ladakh(UT)	1926	2032	107	6	66	1918	1997	79	4	64
	Punjab	4626	4942	315	7	159	4612	4942	330	7	159
	Rajasthan	9534	10171	638	7	328	9199	10064	866	9	325
	Uttar Pradesh	10707	11619	912	9	375	10646	11592	946	9	374
	Uttarakhand	1360	1439	78	6	46	1325	1418	93	7	46
	Railway/NFL	92	106	14	15	3	92	106	14	15	3
	TOTAL	36360	39337	2977	8	1269	35869	39064	3195	9	1260
WR	Gujarat	11610	12383	773	7	399	11577	12383	806	7	399
	Madhya Pradesh	9569	9588	20	0	309	9569	9544	-24	0	308
	Chhattisgarh	3262	3203	-59	-2	103	3260	3200	-60	-2	103
	Maharashtra	16717	17476	760	5	564	16716	17474	758	5	564
	Goa	411	401	-10	-2	13	410	401	-10	-2	13
	DNHDDPDCL	847	879	32	4	28	847	879	32	4	28
	AMNSIL	503	578	76	15	19	503	578	76	15	19
	BALCO	382	386	4	1	12	382	386	4	1	12
	TOTAL	43299	44895	1596	4	1448	43265	44847	1582	4	1447
SR	Andhra Pradesh	6143	6482	339	6	209	6126	6463	337	6	208
	Telangana	7163	7663	500	7	247	7144	7641	498	7	246
	Karnataka	7383	8389	1006	14	271	7363	8367	1004	14	270
	Kerala	2313	2550	237	10	82	2306	2543	237	10	82
	Tamil Nadu	9230	9658	429	5	312	9203	9630	426	5	311
	Puducherry	250	257	7	3	8	249	256	7	3	8
	TOTAL	32482	34999	2518	8	1129	32392	34901	2508	8	1126
ER	Bihar	2962	3077	116	4	99	2936	2999	63	2	97
	Jharkhand	1019	1052	32	3	34	944	971	27	3	31
	DVC	2301	2207	-94	-4	71	2301	2207	-94	-4	71
	Odisha	2892	2925	33	1	94	2892	2925	33	1	94
	West Bengal	4088	4428	341	8	143	4086	4428	342	8	143
	Sikkim	59	61	2	3	2	59	61	2	3	2
	Railways_ER ISTS		4	-	-	0		4	-	-	0
	TOTAL	13321	13754	433	3	444	13219	13595	375	3	439
NER	Arunachal Pradesh	85	89	4	4	3	85	89	4	4	3
	Assam	818	837	19	2	27	814	824	9	1	27
	Manipur	107	104	-3	-3	3	106	96	-11	-10	3
	Meghalaya	221	201	-21	-9	6	221	196	-25	-11	6
	Mizoram	67	70	3	5	2	67	70	3	5	2
	Nagaland	66	71	5	8	2	66	71	5	8	2
	Tripura	118	122	4	3	4	118	122	4	3	4
	TOTAL	1482	1493	11	1	48	1478	1467	-11	-1	47
ALL INDIA		126944	134478	7534	6	4338	126223	133873	7650	6	4318

13. DEMAND COMPARISON OF JANUARY 2024 vs JANUARY 2023

REGION	STATE	Peak Requirement (MW)				Peak Met (MW)			
		Jan-23	Jan-24	Difference	% Change	Jan-23	Jan-24	Difference	% Change
NR	Chandigarh	323	348	25	7.7	323	348	25	7.7
	Delhi	5623	5816	193	3.4	5526	5816	290	5.2
	Haryana	8259	9530	1271	15.4	8259	9530	1271	15.4
	Himachal Pradesh	2071	2181	110	5.3	2071	2181	110	5.3
	J&K(UT) and Ladakh(UT)	3019	3181	162	5.4	3019	3133	114	3.8
	Punjab	9089	9370	281	3.1	9089	9370	281	3.1
	Rajasthan	17399	18128	729	4.2	17206	18128	922	5.4
	Uttar Pradesh	21342	22703	1361	6.4	21342	22703	1361	6.4
	Uttarakhand	2492	2635	143	5.7	2492	2405	-87	-3.5
	Railway/NFL	178	212	34	19.0	178	212	34	19.0
WR	Gujarat	19699	21861	2162	11	19658	21861	2203	11.2
	Madhya Pradesh	17345	18252	907	5	17238	17817	579	3.4
	Chhattisgarh	5142	5338	195	4	5130	5334	204	4.0
	Maharashtra	27860	28651	791	3	27860	28651	791	2.8
	Goa	677	721	44	6	676	720	44	6.5
	DNHDDPDCL	1240	1294	54	4.4	1240	1294	54	4.4
	AMNSIL	812	1083	272	33.5	812	1083	272	33.5
	BALCO	521	526	5	0.9	521	526	5	0.9
SR	Andhra Pradesh	11448	11052	-396	-3.5	11433	11052	-381	-3.3
	Telangana	13924	12666	-1258	-9.0	13906	12666	-1240	-8.9
	Karnataka	14972	15668	696	4.7	14972	15668	696	4.7
	Kerala	3993	4354	361	9.1	3993	4354	361	9.1
	Tamil Nadu	16063	15920	-143	-0.9	16063	15920	-143	-0.9
	Puducherry	406	439	33	8.2	406	439	33	8.1
ER	Bihar	5751	5625	-126	-2.2	5598	5625	27	0.5
	Jharkhand	1920	1805	-115	-6.0	1614	1805	191	11.8
	DVC	3442	3251	-191	-5.5	3442	3251	-191	-5.5
	Odisha	5053	5051	-2	0.0	5053	5051	-2	0.0
	West Bengal	7456	7281	-175	-2.3	7448	7281	-167	-2.2
	Sikkim	132	137	5	3.8	132	137	5	3.8
	Railways_ER ISTS*		20	20	-		20	20	-
NER	Arunachal Pradesh	166	181	15	9.0	166	181	15	9.0
	Assam	1643	1538	-105	-6.4	1643	1538	-105	-6.4
	Manipur	248	258	10	4.0	248	248	0	0.0
	Meghalaya	404	405	1	0.2	404	405	1	0.2
	Mizoram	159	162	3	1.9	159	162	3	1.9
	Nagaland	139	162	23	16.5	139	162	23	16.5
	Tripura	248	235	-13	-5.1	248	235	-13	-5.1

14. SCHEDULE AND DRAWAL OF CONSTITUENTS - JANUARY 2024

REGION	STATE	SCHEDULE (MU)	ACTUAL DRAWAL (MU)	Over drawal(+)/ Under Drawal(-) (MU)	% OD/UD	SCHEDULE (MU/DAY)	ACTUAL DRAWAL (MU/DAY)	Over drawal(+)/ Under Drawal(-) (MU/DAY)
NR	Chandigarh	149.02	158.01	8.99	6.03	4.81	5.10	0.29
	Delhi	2326.39	2321.65	-4.74	-0.20	75.04	74.89	-0.15
	Haryana	2993.45	2989.84	-3.61	-0.12	96.56	96.45	-0.12
	Himachal Pradesh	967.94	983.78	15.84	1.64	31.22	31.73	0.51
	J&K(UT) and Ladakh(UT)	1849.49	1858.90	9.41	0.51	59.66	59.96	0.30
	Punjab	1862.39	1864.86	2.47	0.13	60.08	60.16	0.08
	Rajasthan	3827.28	3788.09	-39.19	-1.02	123.46	122.20	-1.26
	Uttar Pradesh	3831.06	3811.72	-19.34	-0.50	123.58	122.96	-0.62
	Uttarakhand	1047.42	1062.24	14.82	1.41	33.79	34.27	0.48
	Railway	101.25	103.90	2.65	2.62	3.27	3.35	0.09
	NFL	2.35	2.32	-0.03	-1.19	0.08	0.07	0.00
	Total	18958.04	18945.31	-12.73	-0.07	611.55	611.14	-0.41
WR	Gujarat	4462.99	4431.82	-31.17	-0.70	143.97	142.96	-1.01
	Madhya Pradesh	5842.14	5758.35	-83.79	-1.43	188.46	185.75	-2.70
	Chhattisgarh	1413.88	1396.78	-17.10	-1.21	45.61	45.06	-0.55
	Maharashtra	5383.99	5361.56	-22.43	-0.42	173.68	172.95	-0.72
	Goa	399.60	385.17	-14.43	-3.61	12.89	12.42	-0.47
	DNHDDPDCL	881.54	878.99	-2.55	-0.29	28.44	28.35	-0.08
	AMNSIL	278.61	278.80	0.19	0.07	8.99	8.99	0.01
	BALCO*	387.32	386.09	-1.23	-0.32	12.49	12.45	-0.04
	Total	19050.07	18877.55	-172.52	-0.91	614.52	608.95	-5.57
SR	Andhra Pradesh	2387.62	2359.17	-28.45	-1.19	77.02	76.10	-0.92
	Telangana	3726.74	3723.39	-3.35	-0.09	120.22	120.11	-0.11
	Karnataka	3550.70	3506.11	-44.59	-1.26	114.54	113.10	-1.44
	Kerala	2041.99	2064.19	22.20	1.09	65.87	66.59	0.72
	Tamil Nadu	5368.91	5344.12	-24.79	-0.46	173.19	172.39	-0.80
	Puducherry	249.51	239.92	-9.59	-3.84	8.05	7.74	-0.31
	Goa (SR)	61.18	60.11	-1.07	-1.75	1.97	1.94	-0.03
	Total	17386.65	17297.01	-89.64	-0.52	560.86	557.97	-2.89
ER	Bihar	2702.14	2689.83	-12.32	-0.46	87.17	86.77	-0.40
	Jharkhand	717.93	739.25	21.32	2.97	23.16	23.85	0.69
	DVC	-1435.29	-1425.87	9.42	-0.66	-46.30	-46.00	0.30
	Odisha	568.68	512.17	-56.51	-9.94	18.34	16.52	-1.82
	West Bengal	746.73	689.22	-57.51	-7.70	24.09	22.23	-1.86
	Sikkim	62.33	60.51	-1.81	-2.91	2.01	1.95	-0.06
	Railways_ER ISTS*	3.69	3.68	0.00	-0.12	0.12	0.12	0.00
	Total	3366.21	3268.80	-97.41	-2.89	108.59	105.45	-3.14
NER	Arunachal Pradesh	86.13	85.75	-0.38	-0.44	2.78	2.77	-0.01
	Assam	663.16	678.17	15.01	2.26	21.39	21.88	0.48
	Manipur	91.84	95.58	3.74	4.07	2.96	3.08	0.12
	Meghalaya	157.08	151.36	-5.72	-3.64	5.07	4.88	-0.18
	Mizoram	52.64	47.17	-5.47	-10.39	1.70	1.52	-0.18
	Nagaland	69.09	68.04	-1.05	-1.52	2.23	2.19	-0.03
	Tripura	99.24	92.12	-7.12	-7.17	3.20	2.97	-0.23
	Total	1219.18	1218.19	-0.99	-0.08	39.33	39.30	-0.03
All India Total		59980.15	59606.86	-373.29	-0.62	1934.84	1922.80	-12.04

15. INTER REGIONAL EXCHANGES 2023-24 (All figures in MU)											
	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Dec'23	Jan'24	Fin. Year 2023-24
Name of Line											
Import of NR from WR (WR-NR)											
WR - NR 765 kV Gwalior - Agra 2xS/C	635.95	996.71	863.68	972.77	772.12	902.82	479.75	675.87	1109.33	1292.73	8701.73
WR-NR 765 kV Gwalior - Phagi 2xS/C	553.05	453.93	342.56	671.61	397.70	687.31	582.69	906.94	1039.97	857.36	6493.12
WR-NR 765 kV Jabalpur-Orai D/C	430.29	804.53	778.33	964.77	658.94	991.15	602.03	622.44	986.15	1084.38	7923.01
WR-NR 765 kV Satna-Orai	505.37	598.85	574.83	615.77	543.29	593.66	548.69	623.24	692.79	691.06	5987.55
WR-NR 765 kV Gwalior-Orai	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR - NR HVDC Champa - Kurukshetra	804.26	1126.68	2069.11	2235.18	1955.13	2230.56	739.22	674.04	450.87	1556.24	13841.29
WR - NR HVDC Vindhyachal	0.49	3.23	3.13	94.78	21.55	7.26	50.51	28.33	18.10	26.05	253.43
WR - NR 400 kV Zerda Kankroli	0.00	0.00	1.48	15.36	13.10	8.58	0.00	1.30	3.11	9.71	52.64
WR - NR 400 kV Zerda Bhinmal	0.00	0.00	0.00	12.54	13.56	23.81	0.34	3.28	11.42	20.30	85.25
WR - NR 400 kV RAPP C - Shujalpur D/C	2.28	31.04	16.05	164.49	80.51	147.53	18.12	25.82	78.97	145.10	709.91
WR - NR 220 kV Bhanpura - Ranpur	9.67	65.53	68.13	70.02	74.29	61.03	65.71	43.01	33.94	38.18	529.51
WR - NR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00	38.00	62.92	72.56	72.98	34.77	281.23
WR - NR 220 kV Auraiya - Malanpur/Mehgaon	0.00	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.71	0.00	0.86
WR - NR 400 kV Vindhyachal-Rihand D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.47	0.47
WR - NR HVDC Mundra - Mohindergarh	132.55	340.75	234.27	124.89	667.23	827.57	920.93	511.53	563.55	931.29	5254.56
WR-NR 765 kV Banaskata-Chittorgarh D/C	0.00	3.26	16.02	169.31	260.02	66.60	0.00	4.54	27.97	46.56	594.28
WR-NR 765 kV Varanasi-Vindhyachal	1030.05	1663.34	1854.30	1854.38	1551.38	1697.01	1174.50	1049.13	1321.93	1485.20	14681.22
Total WR - NR	4103.96	6088.00	6821.89	7965.87	7008.82	8282.89	5245.41	5242.03	6411.79	8219.40	65390.06
Export of NR to WR (NR-WR)											
NR - WR 765 kV Gwalior - Agra 2xS/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior - Phagi 2xS/C	0.00	5.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.41
NR - WR 765 kV Jabalpur-Orai D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Satna-Orai	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior-Orai	440.87	385.11	301.32	367.88	368.83	385.80	399.85	459.18	533.25	500.66	4142.75
NR - WR HVDC Champa - Kurukshetra	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR HVDC Vindhyachal	289.83	273.88	267.04	184.43	225.73	180.86	26.34	219.10	161.65	131.32	1960.18
NR - WR 400 kV Zerda Kankroli	185.75	94.43	108.34	24.78	31.17	41.36	96.48	75.31	44.39	13.91	715.92
NR - WR 400 kV Zerda Bhinmal	282.42	181.64	200.56	73.76	74.62	64.65	158.50	60.27	38.76	19.27	1154.45
NR - WR 400 kV RAPP C - Shujalpur D/C	85.74	45.64	45.64	0.00	3.36	0.00	46.93	23.25	0.00	0.00	250.56
NR - WR 220 kV Bhanpura - Ranpur	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.30	11.30
NR - WR 220 kV Auraiya - Malanpur/Mehgaon	56.56	28.58	50.82	47.16	97.78	68.03	111.08	74.65	30.32	32.76	597.74
NR-WR 400 kV Rihand-Vindhyachal D/C	644.30	672.49	642.32	664.33	653.25	658.85	669.33	655.41	657.92	560.83	6479.03
NR - WR HVDC Mundra - Mohindergarh	243.86	54.36	36.25	94.16	0.00	0.00	0.00	0.00	0.00	0.00	428.63
NR - WR 765 kV Banaskata-Chittorgarh D/C	976.83	428.19	421.52	94.53	51.63	180.84	597.81	524.56	292.98	164.35	3733.24
NR-WR 765 kV Varanasi-Vindhyachal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total NR - WR	3206.16	2169.73	2073.81	1551.03	1506.37	1580.39	2106.32	2091.73	1759.27	1434.40	19479.21

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Dec'23	Jan'24	Fin. Year 2023-24
Name of Line											
Import of NR from ER (ER-NR)											
ER - NR 765 kV Sasaram - Fatehpur	166.02	114.36	82.57	74.16	131.29	131.20	192.41	301.01	254.85	260.73	1708.60
ER - NR 765 kV Gaya - Varanasi 2*S/C	224.56	107.46	28.46	47.20	75.64	115.88	182.55	439.13	474.87	419.93	2115.68
ER - NR 765 kV Gaya - Balia	246.38	335.46	321.28	300.74	224.31	332.43	250.12	325.32	373.66	432.02	3141.72
ER - NR HVDC Alipurduar - Agra	0.00	0.00	375.05	784.99	444.13	750.81	497.43	168.00	0.00	1.34	3021.75
ER-NR 400 kV Patna - Balia D/C	219.16	246.32	242.35	242.23	300.66	262.01	203.70	186.43	90.17	338.23	2331.26
ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	127.03	223.63	277.35	376.44	415.21	330.20	255.04	312.10	270.56	284.40	2871.96
ER - NR 400 kV Biharshariff - Balia D/C	22.11	72.32	102.97	92.77	101.30	90.47	56.79	138.97	145.18	163.30	986.18
ER - NR 400 kV Motihari - Gorakhpur D/C	156.88	188.70	180.09	228.59	251.16	190.46	142.10	216.07	233.85	268.64	2056.54
ER - NR 400 kV Biharshariff - Varanasi D/C	67.56	47.21	34.66	24.65	44.14	47.88	63.00	164.47	171.28	183.46	848.31
ER - NR 400 kV Sasaram -Varanasi	74.99	38.68	46.09	55.64	28.71	57.56	16.47	4.09	2.63	7.91	332.77
ER - NR 400 kV Sasaram -Allahabad	51.91	33.39	21.29	28.77	17.25	52.61	27.20	32.87	36.28	31.03	332.60
ER - NR 220 kV Sahupuri-Karamnasa	78.64	82.22	82.32	43.45	51.19	29.26	36.16	53.46	34.10	44.57	535.37
ER - NR 132 kV Sahupuri - Karamnasa	1.92	0.00	0.96	1.12	0.96	2.02	3.28	5.11	1.62	1.44	18.43
ER - NR 132 kV 132 kV Rihand - Nagar Untari	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ER - NR 132 kV Rihand - Garhwa	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.64
ER-NR 400 kV Naubatpur - Balia D/C	211.13	243.59	239.56	236.53	300.12	264.57	198.04	328.08	343.58	353.99	2719.19
Total ER-NR	1648.75	1733.34	2035.00	2537.28	2386.07	2657.36	2124.29	2675.11	2432.81	2790.99	23021.00
Import of NR from NER (NER-NR)											
NER-NR HVDC Biswanath Chariali - Agra	0.00	0.00	153.62	339.28	465.76	447.38	470.05	190.40	0.00	0.00	2066.49
Total NER - NR	0.00	0.00	153.62	339.28	465.76	447.38	470.05	190.40	0.00	0.00	2066.49
Export of NR to ER (NR-ER)											
NR - ER 765 kV Sasaram - Fatehpur	0.00	2.65	7.74	3.45	0.00	4.62	0.00	0.00	0.00	0.00	18.46
NR - ER 765 kV Gaya - Varanasi 2*S/C	0.30	38.74	102.98	45.94	11.70	32.51	8.41	1.45	0.00	0.00	242.03
NR - ER 765 kV Gaya - Balia	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER HVDC Alipurduar - Agra	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Patna - Balia 2xD/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	3.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.47
NR - ER 400 kV Biharshariff - Balia D/C	17.15	4.18	13.06	0.00	0.00	3.12	4.14	0.75	0.00	0.00	42.40
ER - NR 400 kV Motihari - Gorakhpur D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Biharshariff - Varanasi D/C	2.90	7.31	20.53	10.39	0.22	12.39	4.69	0.00	0.00	0.00	58.43
NR - ER 400 kV Sasaram -Varanasi	0.00	0.00	0.00	0.00	1.67	0.00	0.85	5.47	4.05	3.29	15.33
NR - ER 400 kV Sasaram -Allahabad	2.25	0.00	0.00	0.82	2.28	0.00	0.23	0.00	0.00	0.24	5.82
NR - ER 220 kV Sahupuri-Karamnasa	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 132 kV Sahupuri - Karamnasa	0.01	0.96	0.48	0.96	0.00	0.58	0.48	0.00	1.44	0.96	5.87
NR - ER 132 kV 132 kV Rihand - Nagar Untari	1.74	2.20	1.08	3.37	2.39	3.46	1.80	2.75	0.00	1.23	20.02
ER - NR 132 kV Rihand - Garhwa	10.23	8.48	19.67	18.35	14.38	11.19	11.51	8.39	4.64	12.89	119.73
NR-ER 400 kV Balia- Naubatpur D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total NR - ER	38.05	64.52	165.54	83.28	32.64	67.87	32.11	18.81	10.13	18.61	531.56
Export of NR to NER (NER-NR)											
NR-NR HVDC Biswanath Chariali - Agra	261.93	291.71	104.47	0.00	0.00	0.00	0.00	257.02	0.00	486.81	1401.94
Total NR - NER	261.93	291.71	104.47	0.00	0.00	0.00	0.00	257.02	0.00	486.81	1401.94

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Dec'23	Jan'24	Fin. Year 2023-24
Name of Line											
Export of WR to ER (WR-ER)											
WR - ER 765 kV Ranchi - Dharamjaygarh 2x5/C	139.81	504.96	673.15	734.00	331.39	537.29	228.34	42.77	55.89	144.01	3391.61
WR - ER 765 kV Dharamjaygarh - Jharsuguda 2xD/C	652.31	469.82	490.99	593.61	768.89	177.22	238.94	52.91	224.71	96.99	3766.39
WR - ER 765 kV Durg - Jharsuguda D/C	0.00	0.64	1.75	26.78	7.42	1.32	0.00	0.00	0.00	0.00	37.91
WR - ER 400 kV Sipat - Ranchi D/C	15.97	93.84	118.29	133.75	52.57	87.51	30.29	8.20	13.05	16.97	570.44
WR - ER 400 kV Raigarh - Jharsuguda 2xD/C	0.44	6.23	8.79	6.26	28.69	11.41	0.82	0.78	0.99	0.01	64.42
WR - ER 220 kV Korba - Budhipadar D/C	36.08	80.04	73.19	22.96	28.75	29.18	26.25	36.70	47.92	14.51	395.58
WR - ER 220 kV Raigarh - Budhipadar	0.15	0.57	1.17	0.00	0.35	0.34	0.00	0.08	2.74	0.06	5.46
Total WR - ER	844.75	1156.11	1367.33	1517.34	1218.07	844.29	524.64	141.44	345.30	272.55	8231.82
Import of WR from ER (ER - WR)											
ER - WR 765 kV Ranchi - Dharamjaygarh 2x5/C	144.10	11.95	15.57	10.81	83.69	14.84	62.72	294.10	166.96	114.80	919.55
ER - WR 765 kV Dharamjaygarh - Jharsuguda 2XD/C	25.46	58.29	33.08	15.65	2.33	214.83	131.20	338.83	241.82	386.37	1447.86
ER - WR 765 kV Durg - Jharsuguda D/C	394.08	204.42	148.02	71.84	73.64	163.98	298.15	464.99	456.27	465.79	2741.18
ER - WR 400 kV Sipat - Ranchi D/C	55.56	5.71	9.07	7.22	29.65	8.91	40.59	105.31	60.55	61.48	384.04
ER - WR 400 kV Raigarh - Jharsuguda 2xD/C	248.78	148.94	125.69	125.00	127.87	110.94	267.35	221.93	208.02	310.82	1895.33
ER - WR 220 kV Korba - Budhipadar D/C	1.45	0.00	0.14	6.23	1.99	1.46	3.18	3.22	2.35	33.09	53.11
ER - WR 220 kV Raigarh - Budhipadar	55.98	38.46	41.41	66.29	74.76	49.37	81.54	41.32	24.30	48.09	521.51
Total ER - WR	925.41	467.76	372.98	303.04	393.92	564.32	884.73	1469.71	1160.27	1420.43	7962.57
Export of ER to NER (ER - NER)											
ER - NER 400 kV Binaguri - Bongaigaon D/C	6.99	16.15	105.81	142.19	71.19	215.10	134.98	60.68	0.64	4.12	757.85
ER - NER 400 kV Alipurduar - Bongaigaon D/C	4.62	34.31	135.94	83.03	84.82	161.59	95.15	85.48	0.00	1.71	686.65
ER - NER 220 kV Birpara - Salakati D/C	0.54	3.12	31.19	37.25	25.19	47.94	25.71	15.50	0.00	0.38	186.81
Total ER - NER	12.16	53.58	272.93	262.48	181.20	424.63	255.83	161.66	0.64	6.20	1631.31
Import of ER from NER (NER - ER)											
NER - ER 400 kV Binaguri - Bongaigaon D/C	52.83	40.19	37.75	2.10	10.62	0.07	5.53	48.07	75.32	53.10	325.59
NER - ER 400 kV Alipurduar - Bongaigaon D/C	182.37	135.22	72.81	20.59	29.76	3.08	23.52	172.08	294.27	231.86	1165.56
NER - ER 220 kV Birpara - Salakati D/C	34.59	31.32	10.78	0.62	2.83	0.14	2.65	38.72	64.62	39.51	225.77
Total NER - ER	269.78	206.73	121.33	23.31	43.22	3.29	31.70	258.87	434.22	324.47	1716.92
Export of ER to SR (ER - SR)											
ER - SR HVDC Gazuwaka	375.93	395.23	208.61	178.44	321.36	201.99	378.26	352.24	239.96	240.59	2892.60
ER - SR HVDC Talchar - Kolar	1233.20	1204.28	984.75	977.61	1284.58	1076.42	1294.48	1214.02	1257.33	611.06	11137.72
ER - SR 765 kV Angul- Srikakulam D/C	1683.42	1561.33	1412.42	1184.37	1380.27	1160.32	1534.32	1466.97	1513.04	1468.02	14364.50
Total ER - SR	3292.55	3160.85	2605.77	2340.42	2986.21	2438.74	3207.06	3033.23	3010.33	2319.67	28394.83
Import of ER from SR (SR - ER)											
SR - ER HVDC Gazuwaka	0.00	0.00	55.34	114.11	32.48	0.00	0.00	6.03	0.00	0.00	207.97
SR - ER HVDC Talchar - Kolar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
SR - ER 765 kV Angul- Srikakulam D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total SR - ER	0.00	0.00	55.34	114.11	32.48	0.00	0.00	6.03	0.00	0.00	207.97

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Dec'23	Jan'24	Fin. Year 2023-24
Name of Line											
Export of WR to SR (WR-SR)											
WR - SR HVDC Bhadrawati	544.41	411.52	171.68	139.64	464.81	222.80	593.89	435.49	593.74	438.40	4016.39
WR - SR 765 kV Sholapur - Raichur 2x5/C	417.69	222.76	146.07	83.93	451.70	125.10	226.89	116.70	185.60	358.47	2334.91
WR - SR 765 kV Wardha - Nizamabad D/C	1323.51	873.44	1018.25	852.05	1313.80	864.24	1099.04	930.66	873.02	1038.49	10186.50
WR - SR 400KV Kolhapur-Kudgi D/C	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00	0.00	0.03	0.09
WR-SR HVDC Raigarh-Pugalur	3004.35	1791.86	1182.91	885.62	2135.69	1450.16	2583.88	1858.37	1697.06	1365.48	17955.37
WR-SR 220kV Xeldem - Ambewadi S/C	77.16	70.18	68.17	61.74	68.61	67.55	69.49	67.29	61.47	59.32	670.97
WR-SR 220kV Ponda - Ambewadi S/C	0.29	1.54	0.28	0.16	5.23	0.04	0.08	0.97	0.04	1.38	10.00
WR-SR 765 kV Warora - Warangal D/C	-	-	-	-	-	-	-	-	-	1060.47	1060.47
Total WR - SR	5367.41	3371.30	2587.37	2023.14	4439.89	2729.89	4573.27	3409.48	3410.93	4322.03	36234.69
Import of WR from SR (SR - WR)											
SR - WR HVDC Bhadrawati	4.11	25.09	178.09	308.60	10.36	114.86	0.00	0.00	0.00	0.00	641.13
SR - WR 765 kV Raichur - Sholapur 2x5/C	29.13	205.27	344.82	478.51	54.56	420.76	215.80	334.15	175.90	151.44	2410.33
SR - WR 765 kV Wardha - Nizamabad D/C	0.00	3.90	0.32	18.33	0.00	19.87	0.03	0.89	0.00	0.53	43.88
SR - WR 400KV Kolhapur-Kudgi D/C	623.59	719.30	784.98	796.02	623.43	850.54	765.68	701.36	642.95	615.99	7123.85
SR-WR HVDC Pugalur-Raigarh	0.00	3.73	14.87	16.65	0.00	0.09	0.00	0.00	0.00	0.00	35.33
WR-SR 220kV Ambewadi - Xeldem S/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR-SR 220kV Ambewadi-Ponda S/C	0.00	0.00	0.01	0.00	0.01	0.01	0.02	0.01	0.02	0.02	0.10
SR-WR 765 kV Warangal - Warora D/C										0.26	0.26
Total SR - WR	656.84	957.30	1323.09	1618.10	688.36	1406.13	981.53	1036.41	818.88	768.24	10254.88
TOTAL ALL INDIA	20628	19721	20060	20679	21383	21447	20437	19992	19795	22384	206525
*In case of mutiple ckt / DC pole year corres. to the commissioning of final element											

Date	15.1 Import-Export of NR with WR during January 2024																	
	Import of NR from WR (WR-NR)																	
	WR - NR 765 kV Gwalior - Agra 2x5/C	WR-NR 765 kV Gwalior-Phagi 2x5/C	WR-NR 765 kV Jabalpur- Orai D/C	WR-NR 765 kV Satna- Orai	WR-NR 765 kV Gwalior-Orai	WR - NR HVDC Champa- Kurukshetra	WR - NR HVDC Vindhyachal	WR - NR 400 kV Zerda- Kankroli	WR - NR 400 kV Zerda- Bhinmal	WR-NR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	WR - NR 220 kV Auraiya - Malanpur / Mehgaon	WR-NR 400 kV Rihand Vindhyachal D/C	WR - NR HVDC Mundra - M'garh	WR -NR 765 kV Banaskata - Chittorgarh D/C	WR-NR 765 kV Varanasi- Vindhyachal	Total WR - NR
1-Jan-24	44.64	43.20	40.37	24.94	0.00		0.00	0.92	1.64	6.90	1.19	2.36	0.00	0.00	25.66	4.45	47.08	243.35
2-Jan-24	41.02	40.10	39.10	24.12	0.00		0.00	0.00	0.00	5.11	1.18	2.28	0.00	0.00	28.13	0.00	45.81	226.85
3-Jan-24	45.54	39.30	40.94	26.03	0.00		0.00	0.00	0.00	5.44	1.25	2.67	0.00	0.00	33.08	0.00	48.56	242.81
4-Jan-24	45.51	40.39	42.54	24.80	0.00		4.75	0.00	0.40	6.16	1.18	2.43	0.00	0.00	34.94	0.00	48.65	251.75
5-Jan-24	42.83	39.36	41.06	24.03	0.00	0.45	4.88	0.00	0.00	7.69	1.48	2.45	0.00	0.00	47.01	0.00	52.01	263.25
6-Jan-24	40.56	28.16	35.24	22.36	0.00	50.75	4.76	0.17	0.00	7.27	1.17	2.38	0.00	0.00	51.96	0.00	44.38	289.16
7-Jan-24	46.08	20.49	35.91	22.54	0.00	76.99	1.20	2.14	3.58	8.06	1.10	2.36	0.00	0.00	32.55	10.42	57.33	320.75
8-Jan-24	44.10	18.82	34.43	22.12	0.00	68.23	0.00	0.46	2.54	4.48	1.12	2.28	0.00	0.00	38.50	4.39	53.55	295.02
9-Jan-24	46.19	16.19	33.84	21.53	0.00	58.84	0.00	2.31	3.83	2.56	1.02	2.38	0.00	0.00	31.23	17.01	50.79	287.72
10-Jan-24	44.81	17.12	33.98	21.46	0.00	59.96	0.00	1.20	1.26	1.84	1.08	2.38	0.00	0.00	27.00	9.09	54.21	275.39
11-Jan-24	48.74	20.22	36.02	22.91	0.00	59.50	0.00	0.00	0.00	3.19	1.10	2.40	0.00	0.00	30.45	0.00	55.75	280.28
12-Jan-24	49.00	22.66	38.16	23.80	0.00	64.89	0.00	0.00	0.00	4.00	1.18	2.61	0.00	0.00	26.17	0.00	54.02	286.49
13-Jan-24	40.30	22.28	25.83	23.84	0.00	69.99	0.00	0.00	0.00	3.62	1.17	2.56	0.00	0.00	34.73	0.00	46.08	270.40
14-Jan-24	36.49	19.35	28.27	21.21	0.00	69.50	0.00	0.00	0.00	2.65	1.14	2.16	0.00	0.00	30.64	0.00	49.89	261.30
15-Jan-24	46.69	20.55	31.62	23.43	0.00	55.23	1.71	0.00	0.00	4.94	1.12	0.00	0.00	0.00	24.90	0.00	60.20	270.39
16-Jan-24	46.04	20.32	35.97	22.10	0.00	61.64	3.35	0.00	0.00	4.40	1.13	0.00	0.00	0.00	42.81	0.00	57.40	295.16
17-Jan-24	47.85	22.60	37.18	22.07	0.00	43.94	2.40	0.00	0.00	4.92	1.18	0.00	0.00	0.00	38.67	0.00	58.25	279.06
18-Jan-24	38.76	17.34	30.92	21.18	0.00	70.58	2.41	0.00	0.09	3.09	1.19	0.00	0.00	0.00	26.79	0.00	48.21	260.56
19-Jan-24	37.04	16.62	29.47	21.31	0.00	72.43	0.00	0.00	0.00	2.81	1.26	0.00	0.00	0.00	18.33	0.00	43.37	242.64
20-Jan-24	41.55	18.62	27.32	22.03	0.00	73.92	0.00	0.00	0.00	3.77	1.21	0.00	0.00	0.00	17.33	0.00	48.68	254.43
21-Jan-24	45.27	21.50	35.27	22.81	0.00	58.57	0.00	0.00	0.00	6.07	1.37	0.00	0.00	0.00	21.40	0.00	44.51	256.77
22-Jan-24	44.91	19.80	35.36	21.70	0.00	57.91	0.00	0.23	0.87	5.38	0.65	0.00	0.00	0.00	20.06	0.00	45.80	252.67
23-Jan-24	37.36	35.74	34.66	21.54	0.00	66.14	0.59	0.00	0.00	4.92	1.25	0.00	0.00	0.00	32.33	0.00	42.85	277.38
24-Jan-24	37.57	35.50	34.68	21.25	0.00	47.07	0.00	0.00	0.00	4.23	1.30	0.00	0.00	0.00	29.51	0.00	39.87	250.98
25-Jan-24	40.11	33.87	36.33	20.43	0.00	46.70	0.00	0.00	0.00	4.06	1.40	0.00	0.00	0.00	25.82	0.00	37.81	246.53
26-Jan-24	41.09	31.82	38.59	20.74	0.00	45.07	0.00	0.93	2.10	5.19	1.36	0.00	0.00	0.00	23.76	1.20	42.09	253.94
27-Jan-24	35.54	34.86	36.80	21.84	0.00	45.80	0.00	0.98	1.07	4.95	1.53	0.46	0.00	0.47	29.05	0.00	42.31	255.66
28-Jan-24	38.45	39.99	37.36	21.62	0.00	49.96	0.00	0.37	1.66	6.46	1.45	0.84	0.00	0.00	23.16	0.00	42.22	263.54
29-Jan-24	32.50	34.60	32.20	20.56	0.00	69.45	0.00	0.00	0.00	3.93	1.43	0.09	0.00	0.00	22.42	0.00	40.80	257.98
30-Jan-24	35.82	35.86	34.58	20.60	0.00	66.20	0.00	0.00	1.26	5.21	1.44	0.00	0.00	0.00	31.45	0.00	43.96	276.38
31-Jan-24	30.37	30.13	30.38	20.16	0.00	46.53	0.00	0.00	0.00	1.80	1.55	-0.32	0.00	0.00	31.45	0.00	38.76	230.81
Total	1292.73	857.36	1084.38	691.06	0.00	1556.24	26.05	9.71	20.30	145.10	38.18	34.77	0.00	0.47	931.29	46.56	1485.20	8219.40
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																		

Date	15.1 Import-Export of NR with WR during January 2024																	
	Export of NR to WR (NR- WR)																	
	NR - WR 765 kV Gwalior - Agra 2x5/C	NR-WR 765 kV Gwalior-Phagi 2x5/C	NR-WR 765 kV Jabalpur- Orai D/C	NR-WR 765 kV Satna- Orai	NR-WR 765 kV Gwalior-Orai	NR - WR HVDC Champa- Kurukshetra	NR - WR HVDC Vindhyachal	NR - WR 400 kV Zerda- Kankroli	NR - WR 400 kV Zerda- Bhinmal	NR-WR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	NR - WR 220 kV Auraiya - Malanpur / Mehgaon	NR-WR 400 kV Rihand- Vindhyachal D/C	NR - WR HVDC Mundra - M'garh	NR - WR 765 kV Banaskata - Chittorgarh D/C	NR-WR 765 kV Varanasi- Vindhyachal	Total NR - WR
1-Jan-24	0.00	0.00	0.00	0.00	19.03		4.84	0.00	0.00	0.00	0.00	0.00	1.12	22.28	0.00	0.00	0.00	47.27
2-Jan-24	0.00	0.00	0.00	0.00	18.78		4.84	1.05	1.47	0.00	0.00	0.00	1.22	22.37	0.00	8.85	0.00	58.58
3-Jan-24	0.00	0.00	0.00	0.00	17.86		3.69	1.20	1.73	0.00	0.00	0.00	0.94	22.65	0.00	9.63	0.00	57.70
4-Jan-24	0.00	0.00	0.00	0.00	19.15		0.00	0.17	0.00	0.00	0.00	0.00	1.20	22.38	0.00	2.91	0.00	45.81
5-Jan-24	0.00	0.00	0.00	0.00	19.18	0.00	0.00	0.54	1.29	0.00	0.00	0.00	1.10	22.41	0.00	7.75	0.00	52.27
6-Jan-24	0.00	0.00	0.00	0.00	16.13	0.00	0.00	0.00	0.24	0.00	0.00	0.00	1.05	22.31	0.00	1.68	0.00	41.41
7-Jan-24	0.00	0.00	0.00	0.00	14.21	0.00	1.90	0.00	0.00	0.00	0.00	0.00	0.75	21.63	0.00	0.00	0.00	38.49
8-Jan-24	0.00	0.00	0.00	0.00	14.10	0.00	2.43	0.00	0.00	0.00	0.00	0.00	0.99	21.36	0.00	0.00	0.00	38.88
9-Jan-24	0.00	0.00	0.00	0.00	12.11	0.00	2.43	0.00	0.00	0.00	0.00	0.00	0.05	22.12	0.00	0.00	0.00	36.71
10-Jan-24	0.00	0.00	0.00	0.00	12.85	0.00	2.91	0.00	0.00	0.00	0.00	0.00	0.42	21.89	0.00	0.00	0.00	38.07
11-Jan-24	0.00	0.00	0.00	0.00	14.35	0.00	2.43	0.12	0.55	0.00	0.00	0.00	0.20	22.17	0.00	1.18	0.00	41.00
12-Jan-24	0.00	0.00	0.00	0.00	14.54	0.00	2.44	0.66	0.93	0.00	0.00	0.00	0.39	22.23	0.00	6.40	0.00	47.59
13-Jan-24	0.00	0.00	0.00	0.00	13.33	0.00	2.43	1.56	1.48	0.00	0.00	0.00	0.66	22.41	0.00	13.16	0.00	55.03
14-Jan-24	0.00	0.00	0.00	0.00	15.35	0.00	2.42	1.41	1.98	0.00	0.00	0.00	1.13	22.32	0.00	12.49	0.00	57.10
15-Jan-24	0.00	0.00	0.00	0.00	12.76	0.00	1.99	0.81	1.10	0.00	0.00	1.42	0.63	22.38	0.00	8.70	0.00	49.79
16-Jan-24	0.00	0.00	0.00	0.00	15.36	0.00	0.00	1.05	1.24	0.00	0.00	1.96	0.56	22.60	0.00	9.99	0.00	52.76
17-Jan-24	0.00	0.00	0.00	0.00	15.21	0.00	0.00	0.50	0.72	0.00	0.00	1.25	0.73	22.42	0.00	6.76	0.00	47.59
18-Jan-24	0.00	0.00	0.00	0.00	14.29	0.00	0.00	0.10	0.00	0.00	0.00	0.76	1.09	22.29	0.00	2.01	0.00	40.54
19-Jan-24	0.00	0.00	0.00	0.00	13.90	0.00	4.38	0.39	0.78	0.00	0.00	0.97	1.31	22.42	0.00	4.41	0.00	48.56
20-Jan-24	0.00	0.00	0.00	0.00	13.20	0.00	8.57	0.49	0.95	0.00	0.00	1.03	1.08	22.22	0.00	6.92	0.00	54.46
21-Jan-24	0.00	0.00	0.00	0.00	16.40	0.00	12.12	0.27	0.22	0.00	0.00	0.70	1.00	22.13	0.00	6.16	0.00	59.00
22-Jan-24	0.00	0.00	0.00	0.00	14.91	0.00	6.60	0.00	0.00	0.00	0.00	0.62	1.23	22.26	0.00	1.31	0.00	46.93
23-Jan-24	0.00	0.00	0.00	0.00	18.01	0.00	0.79	0.52	0.77	0.00	0.00	0.98	1.54	21.83	0.00	10.90	0.00	55.34
24-Jan-24	0.00	0.00	0.00	0.00	18.62	0.00	2.43	0.59	0.60	0.00	0.00	1.10	1.45	10.09	0.00	2.92	0.00	37.80
25-Jan-24	0.00	0.00	0.00	0.00	17.86	0.00	5.85	0.42	0.76	0.00	0.00	0.45	1.79	-0.49	0.00	7.79	0.00	34.43
26-Jan-24	0.00	0.00	0.00	0.00	17.06	0.00	10.93	0.00	0.00	0.00	0.00	0.06	1.05	0.45	0.00	0.00	0.00	29.55
27-Jan-24	0.00	0.00	0.00	0.00	17.54	0.00	12.15	0.00	0.00	0.00	0.00	0.00	1.22	0.00	0.00	0.08	0.00	30.99
28-Jan-24	0.00	0.00	0.00	0.00	19.26	0.00	12.12	0.00	0.00	0.00	0.00	0.00	1.23	6.42	0.00	5.36	0.00	44.39
29-Jan-24	0.00	0.00	0.00	0.00	18.40	0.00	8.42	0.49	0.60	0.00	0.00	0.00	1.91	11.05	0.00	9.98	0.00	50.85
30-Jan-24	0.00	0.00	0.00	0.00	18.82	0.00	6.14	0.40	0.00	0.00	0.00	0.00	1.72	11.18	0.00	6.94	0.00	45.20
31-Jan-24	0.00	0.00	0.00	0.00	18.09	0.00	6.07	1.17	1.86	0.00	0.00	0.00	2.00	11.05	0.00	10.07	0.00	50.31
Total	0.00	0.00	0.00	0.00	500.66	0.00	131.32	13.91	19.27	0.00	0.00	11.30	32.76	560.83	0.00	164.35	0.00	1434.40

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

Date	15.2 Import-Export of NR with ER & NER during January 2024																		
	Import of NR from ER (ER-NR)																	Import of NR from NER (NER-NR)	
	ER - NR 765 kV Sasaram - Fatehpur	ER - NR 765 kV Gaya - Varanasi 2*5/C	ER - NR 765 kV Gaya Balia	ER - NR HVDC Alipurduar - Agra	ER-NR 400 kV Patna - Balia D/C	ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Balia D/C	ER - NR 400 kV Motihari - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Varanasi D/C	ER-NR 400 kV Sasaram - Varanasi	ER-NR 400 kV Sasaram - Allahabad	ER - NR 220 kV Sahupuri - Karamnasa	ER - NR 132 kV Sahupuri - Karamnasa	ER - NR 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	ER-NR 400 kV Naubatpur - Balia D/C	Total ER -NR	NER-NR HVDC Biswanath Chariaili - Agra	Total NER-NR
1-Jan-24	9.70	17.80	12.24		9.66	10.16	2.94	9.18	7.68	0.00	1.62	1.56	0.00		0.00	9.98	92.52	0.00	0.00
2-Jan-24	9.62	21.31	7.33		11.34	10.86	3.83	9.27	8.20	0.00	1.59	1.39			0.00	11.95	96.69	0.00	0.00
3-Jan-24	10.22	17.94	13.35		8.89	11.37	5.24	8.23	8.33	0.00	1.59	1.66			0.00	9.26	96.08	0.00	0.00
4-Jan-24	8.15	17.25	13.40		9.41	9.30	4.03	7.64	7.43	0.00	1.59	0.89			0.00	9.83	88.92	0.00	0.00
5-Jan-24	7.38	15.81	16.21		11.96	9.34	6.64	7.84	7.54	0.00	1.58	1.07			0.00	12.31	97.68	0.00	0.00
6-Jan-24	7.75	15.44	13.76		10.95	9.23	5.68	7.54	6.82	0.00	1.23	0.78	0.00		0.00	11.62	90.80	0.00	0.00
7-Jan-24	6.28	10.35	13.75		9.05	8.20	4.81	6.64	5.21	0.00	1.29	1.17		0.00	0.00	9.36	76.11	0.00	0.00
8-Jan-24	7.05	10.09	12.89		9.89	7.05	4.40	6.95	3.39	0.00	1.29	0.79			0.00	10.30	74.09	0.00	0.00
9-Jan-24	6.58	10.02	12.23		10.02	6.97	4.36	6.85	3.32	0.02	1.15	1.39	0.48		0.00	10.44	73.83	0.00	0.00
10-Jan-24	6.43	7.65	11.77		9.66	6.92	4.67	7.50	3.10	0.06	1.09	1.39			0.00	10.01	70.25	0.00	0.00
11-Jan-24	6.71	9.87	12.88		10.04	7.31	4.56	7.41	5.88	0.06	1.07	1.47		0.00		10.49	77.75	0.00	0.00
12-Jan-24	7.61	10.33	13.37		10.57	8.60	4.98	8.56	5.57	0.00	1.21	0.95		0.00	0.00	11.07	82.82	0.00	0.00
13-Jan-24	7.50	10.85	13.38		11.50	7.95	4.78	7.84	5.84	0.15	0.97	1.11			0.00	11.69	83.56	0.00	0.00
14-Jan-24	6.73	11.91	13.80		12.21	8.91	5.30	9.40	3.80	0.46	0.71	1.48			0.00	12.88	87.59	0.00	0.00
15-Jan-24	6.85	12.26	15.30		12.01	8.51	5.70	8.74	3.15	0.13	0.99	1.64			0.00	12.47	87.75	0.00	0.00
16-Jan-24	6.88	11.47	14.88		11.57	8.12	4.72	8.91	2.60	0.28	0.74	1.50			0.00	12.01	83.68	0.00	0.00
17-Jan-24	7.00	10.42	14.55		11.68	8.35	4.88	10.04	3.80	0.00	0.59	1.79	0.48		0.00	12.23	85.81	0.00	0.00
18-Jan-24	9.17	8.32	15.48		11.85	9.46	5.36	9.93	4.76	2.55	0.82	1.81			0.00	12.96	92.47	0.00	0.00
19-Jan-24	9.32	14.36	15.19	1.34	12.27	9.79	5.59	9.91	7.16	0.22	0.94	2.02			0.00	12.92	101.03	0.00	0.00
20-Jan-24	9.86	14.93	14.70		12.15	10.42	6.33	10.44	7.29	0.00	0.85	1.71	0.00		0.00	12.82	101.50	0.00	0.00
21-Jan-24	9.80	15.63	15.67		12.97	10.81	6.49	10.77	7.86	0.04	1.02		0.48		0.00	13.75	105.29	0.00	0.00
22-Jan-24	9.72	16.01	16.34		11.81	11.84	7.03	10.18	7.78	0.16	0.95	2.10			0.00	12.52	106.44	0.00	0.00
23-Jan-24	9.97	15.00	15.15		10.69	10.08	6.03	9.21	6.76	0.30	0.78	1.57	0.00		0.00	10.95	96.49	0.00	0.00
24-Jan-24	10.01	15.17	14.97		12.55	10.46	5.99	10.10	7.01	0.57	0.50	1.85	0.00	0.00		12.55	101.73	0.00	0.00
25-Jan-24	10.57	17.04	15.43		10.61	10.88	7.07	9.12	7.67	0.43	0.73	2.05		0.00	0.00	11.37	102.97	0.00	0.00
26-Jan-24	9.50	16.76	15.82		12.30	11.25	6.74	10.38	7.77	0.26	0.89	2.14			0.00	13.21	107.02	0.00	0.00
27-Jan-24	8.53	14.35	14.32		11.61	11.80	5.17	9.00	6.38	0.30	0.89	1.62			0.00	12.29	96.26	0.00	0.00
28-Jan-24	10.16	17.42	15.30		12.64	10.21	5.72	9.95	6.54	0.09	1.06	1.62			0.00	13.41	104.12	0.00	0.00
29-Jan-24	8.72	12.08	13.28		9.92	8.68	4.98	8.49	5.18	0.53	0.56	1.28	0.00		0.00	10.34	84.04	0.00	0.00
30-Jan-24	7.82	9.80	12.80		8.20	5.80	2.72	6.38	2.62	0.89	0.00	1.15			0.00	7.98	66.16	0.00	0.00
31-Jan-24	9.14	12.29	12.48		8.25	5.77	6.56	6.24	7.02	0.41	0.74	1.62			0.00	9.02	79.54	0.00	0.00
Total	260.73	419.93	432.02	1.34	338.23	284.40	163.30	268.64	183.46	7.91	31.03	44.57	1.44	0.00	0.00	353.99	2790.99	0.00	0.00

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

Date	15.2 Import-Export of NR with ER & NER during January 2024																		
	Export of NR to ER (NR- ER)																	Export of NR to NER (NR-NER)	
	NR - ER 765 kV Sasaram - Fatehpur	NR - ER 765 kV Gaya - Varanasi 2*5/C	NR - ER 765 kV Gaya Ballia	NR - ER HVDC Alipurduar - Agra	NR-ER 400 kV Patna - Ballia D/C	NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Ballia D/C	NR - ER 400 kV Motihari - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Varanasi D/C	NR-ER 400 kV Sasaram - Varanasi	NR-ER 400 kV Sasaram - Allahabad	NR - ER 220 kV Sahupuri- Karamnasa	NR - ER 132 kV Sahupuri - Karamnasa	NR - ER 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	NR-ER 400 kV Ballia- Naubatpur D/C	Total NR - ER	NR-NER HVDC Biswanath Chariali - Agra	Total NR-NER
1-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.46	0.00	0.00	0.48		0.42	0.00	1.36	13.21	13.21
2-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.46	0.00	0.00			0.41	0.00	0.87	12.21	12.21
3-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.49	0.00	0.00			0.43	0.00	0.92	16.46	16.46
4-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.54	0.00	0.00			0.44	0.00	0.98	15.77	15.77
5-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.55	0.00	0.00			0.52	0.00	1.07	10.27	10.27
6-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.00	0.00		0.35	0.00	0.43	14.47	14.47
7-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.13	0.00	0.00		0.09	0.26	0.00	0.48	15.14	15.14
8-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00			0.27	0.00	0.45	15.95	15.95
9-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.41	0.00	0.41	16.39	16.39
10-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.40	0.00	0.40	16.97	16.97
11-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.54		0.00	0.54	15.66	15.66
12-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00		0.05	0.44	0.00	0.53	16.23	16.23
13-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.53	0.00	0.53	11.85	11.85
14-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.46	0.00	0.46	17.08	17.08
15-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.43	0.00	0.43	15.82	15.82
16-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.37	0.00	0.37	15.68	15.68
17-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.26	0.00	0.00	0.00		0.45	0.00	0.71	17.12	17.12
18-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.42	0.00	0.42	17.03	17.03
19-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.30	0.00	0.30	10.90	10.90
20-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.10	0.00	0.00	0.00		0.31	0.00	0.41	16.85	16.85
21-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00		0.42	0.00	0.42	15.70	15.70
22-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.51	0.00	0.51	15.70	15.70
23-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.56	0.00	0.56	16.89	16.89
24-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.55		0.00	0.55	16.81	16.81
25-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.57	0.00	0.57	17.11	17.11
26-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.54	0.00	0.54	16.00	16.00
27-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.54	0.00	0.54	18.04	18.04
28-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.56	0.00	0.56	17.40	17.40
29-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.48		0.58	0.00	1.06	17.28	17.28
30-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.24	0.00			0.54	0.00	0.78	17.04	17.04
31-Jan-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.45	0.00	0.45	17.78	17.78
Total	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.29	0.24	0.00	0.96	1.23	12.89	0.00	18.61	486.81	486.81
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																			

Date	15.3 Import and Export of ER with WR & NER during January 2024																							
	Export of WR to ER (WR-ER)								Import of WR from ER (ER - WR)								Export of ER to NER (ER-NER)				Import of ER from NER (NER-ER)			
	WR - ER 765 kV Dharamjaygarh - Ranchi 2x5/C	WR - ER 765 kV Dharamjaygarh - Jharsuguda Q/C	WR - ER 765 kV Durg - Jharsuguda D/C	WR - ER 400 kV Sipat - Ranchi D/C	WR - ER 400 kV Raigarh - Jharsuguda- 2xb/C	WR - ER 220 kV Korba - Budhipadar D/C	WR - ER 220 kV Raigarh - Budhipadar	Total WR - ER	ER - WR 765 kV Dharamjaygarh - Ranchi 2x5/C	ER-WR 765 kV Dharamjaygarh - Jharsuguda D/C	ER-WR 765 kV Durg - Jharsuguda D/C	ER - WR 400 kV Sipat - Ranchi D/C	ER - WR 400 kV Raigarh - Jharsuguda- 2xD/C	ER - WR 220 kV Korba - Budhipadara D/C	ER - WR 220 kV Raigarh - Budhipadar	Total ER - WR	ER - NER 400 kV Binaguri - Bongaigaon D/C	ER - NER 400 kV Alipurduar - Bongaigaon D/C	ER - NER 220 kV Birpara - Salakati D/C	Total ER - NER	NER - ER 400 kV Binaguri - Bongaigaon D/C	NER - ER 400 kV Alipurduar - Bongaigaon 2xD/C	NER - ER 220 kV Birpara - Salakati D/C	Total NER - ER
1-Jan-24	3.5	11.8	0.0	0.6	0.0	0.8	0.0	16.82	2.5	5.8	19.8	1.5	11.2	0.4	0.6	41.80	0.13	0.03	0.00	0.16	2.14	8.63	1.69	12.46
2-Jan-24	2.3	9.9	0.0	0.5	0.0	0.7	0.0	13.33	5.1	6.9	22.0	2.3	10.6	0.5	1.1	48.32	0.12	0.01	0.00	0.13	1.83	7.70	1.55	11.08
3-Jan-24	2.5	9.0	0.0	0.2	0.0	0.4	0.0	12.14	5.2	5.2	22.6	2.6	10.9	0.8	1.1	48.45	0.00	0.00	0.00	0.00	2.64	10.29	2.00	14.93
4-Jan-24	4.4	10.9	0.0	0.4	0.0	0.5	0.0	16.27	2.7	5.6	19.4	1.5	10.4	0.6	1.0	41.27	0.23	0.17	0.01	0.42	1.81	7.60	1.49	10.90
5-Jan-24	4.6	9.4	0.0	0.5	0.0	0.5	0.0	14.94	2.5	9.6	18.7	1.4	11.7	0.8	1.0	45.66	0.20	0.03	0.00	0.24	1.35	6.44	1.31	9.10
6-Jan-24	3.1	6.5	0.0	0.6	0.0	0.5	0.0	10.74	5.0	3.5	17.2	2.1	9.4	0.8	1.6	39.49	0.10	0.00	0.00	0.10	2.25	9.01	1.73	12.99
7-Jan-24	4.8	3.1	0.0	1.1	0.0	1.2	0.0	10.08	2.2	6.1	13.9	0.7	5.3	0.2	1.4	29.92	0.04	0.00	0.00	0.04	2.65	10.21	1.95	14.81
8-Jan-24	5.8	5.8	0.0	1.2	0.0	1.2	0.0	13.92	1.4	3.6	14.7	0.7	5.8	0.1	1.4	27.67	0.07	0.03	0.00	0.10	2.31	9.38	1.78	13.48
9-Jan-24	7.0	8.8	0.0	1.4	0.0	1.4	0.0	18.64	1.7	4.2	15.7	0.7	6.6	0.1	0.9	29.88	0.03	0.00	0.00	0.03	2.28	9.10	1.51	12.88
10-Jan-24	8.2	5.8	0.0	1.2	0.0	1.4	0.0	16.68	1.8	4.3	11.2	1.6	7.7	0.2	0.9	27.75	0.08	0.00	0.01	0.08	2.25	9.02	1.51	12.78
11-Jan-24	8.9	3.4	0.0	1.5	0.0	1.0	0.0	14.84	0.8	4.2	11.6	0.7	8.4	0.4	1.3	27.37	0.18	0.19	0.05	0.42	2.12	8.52	1.42	12.06
12-Jan-24	6.0	0.2	0.0	1.0	0.0	0.7	0.0	7.89	2.1	10.9	12.2	0.9	8.7	0.4	1.2	36.54	0.14	0.11	0.04	0.29	2.59	9.48	1.60	13.67
13-Jan-24	2.6	0.1	0.0	0.3	0.0	0.9	0.0	3.86	5.1	14.3	13.9	2.0	8.1	0.3	1.4	45.10	0.16	0.04	0.01	0.21	1.71	7.19	0.94	9.84
14-Jan-24	2.7	0.5	0.0	0.2	0.0	1.0	0.0	4.45	5.8	13.9	14.3	2.3	8.1	0.2	1.6	46.30	0.00	0.00	0.00	0.00	3.21	11.98	1.13	16.31
15-Jan-24	8.0	1.0	0.0	0.8	0.0	0.8	0.0	10.61	0.3	11.1	12.2	0.8	8.0	0.2	1.7	34.31	0.12	0.06	0.00	0.18	2.32	9.33	0.68	12.32
16-Jan-24	8.7	1.9	0.0	0.7	0.0	0.4	0.0	11.72	0.2	11.8	10.5	1.0	10.0	0.3	1.5	35.23	0.07	0.03	0.00	0.10	1.70	7.59	1.66	10.95
17-Jan-24	9.7	1.7	0.0	0.6	0.0	0.4	0.0	12.49	0.0	11.8	12.3	0.9	13.1	0.6	1.4	40.16	0.05	0.00	0.00	0.05	1.52	7.28	1.45	10.25
18-Jan-24	0.4	0.8	0.0	0.0	0.0	0.4	0.0	1.66	6.7	17.7	14.7	3.5	13.5	0.7	1.7	58.56	0.07	0.00	0.00	0.07	1.43	6.88	1.34	9.65
19-Jan-24	1.2	0.4	0.0	0.1	0.0	0.0	0.0	1.68	11.5	18.9	17.1	4.5	11.9	1.6	2.0	67.39	0.98	0.90	0.13	2.01	0.65	3.72	0.73	5.10
20-Jan-24	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.70	9.5	22.0	15.4	3.8	11.2	2.2	1.7	65.85	0.10	0.00	0.00	0.10	1.17	6.23	1.16	8.56
21-Jan-24	1.8	0.0	0.0	0.2	0.0	0.1	0.0	2.17	8.8	23.9	14.9	3.3	11.2	1.3	1.4	64.80	0.06	0.00	0.00	0.06	1.22	6.56	1.31	9.10
22-Jan-24	4.0	1.1	0.0	0.4	0.0	0.0	0.0	5.45	4.8	12.1	14.5	2.8	11.2	1.5	1.5	48.35	0.10	0.00	0.00	0.10	1.07	6.03	1.06	8.16
23-Jan-24	2.6	0.2	0.0	0.2	0.0	0.0	0.0	2.97	6.4	22.5	14.8	3.1	11.4	1.8	1.5	61.53	0.08	0.00	0.00	0.08	1.33	6.69	1.24	9.26
24-Jan-24	4.0	0.6	0.0	0.3	0.0	0.0	0.0	4.98	4.6	15.5	15.1	2.7	12.0	2.1	1.7	53.63	0.04	0.00	0.00	0.04	1.29	6.60	0.85	8.73
25-Jan-24	6.9	1.0	0.0	0.7	0.0	0.0	0.0	8.61	3.0	14.6	14.6	1.8	10.7	2.6	1.5	48.95	0.10	0.00	0.02	0.12	1.39	6.50	0.78	8.67
26-Jan-24	6.0	1.2	0.0	0.8	0.0	0.0	0.0	7.97	2.0	11.1	12.4	1.3	8.8	2.7	1.9	40.15	0.09	0.00	0.02	0.12	1.21	5.68	0.78	7.67
27-Jan-24	6.4	0.3	0.0	0.7	0.0	0.0	0.0	7.41	0.8	14.3	13.4	1.0	9.1	1.9	2.2	42.70	0.11	0.05	0.02	0.18	1.28	6.15	1.14	8.57
28-Jan-24	4.4	0.6	0.0	0.3	0.0	0.0	0.0	5.18	3.7	18.9	13.7	2.2	10.7	1.8	2.3	53.32	0.06	0.00	0.00	0.06	1.44	6.71	1.28	9.43
29-Jan-24	4.1	0.1	0.0	0.2	0.0	0.0	0.0	4.34	3.1	20.6	14.1	2.5	10.5	1.8	2.5	55.18	0.18	0.02	0.01	0.21	1.17	5.19	0.94	7.30
30-Jan-24	6.5	0.4	0.0	0.3	0.0	0.0	0.0	7.20	2.0	20.2	14.3	2.1	10.7	1.9	2.6	53.73	0.18	0.00	0.01	0.19	0.93	5.32	0.81	7.07
31-Jan-24	2.4	0.4	0.0	0.0	0.0	0.0	0.0	2.80	3.6	21.4	14.7	3.2	13.5	2.1	2.4	61.05	0.27	0.03	0.04	0.34	0.85	4.86	0.67	6.38
Total	144.01	96.99	0.00	16.97	0.01	14.51	0.06	272.55	114.80	386.37	465.79	61.48	310.82	33.09	48.09	1420.43	4.12	1.71	0.38	6.20	53.10	231.86	39.51	324.47

Date	15.4 Import and Export of SR with ER & WR during January 2024																									
	Export of ER to SR (ER - SR)				Import of ER from SR (SR - ER)				Export of WR to SR (WR-SR)								Import of WR from SR (SR - WR)									
	ER - SR HVDC Gazuwaka	ER - SR HVDC Talchar -Kolar	ER-SR 765 kV Angul- Srikakulam D/C	Total ER - SR	SR - ER HVDC Gazuwaka	SR - ER HVDC Talchar -	SR-ER 765 kV Angul- Srikakulam D/C	Total SR - ER	WR - SR HVDC Bhadrawati	WR - SR 765 kV Sholapur- Raichur 2x5/C	WR-SR 765 kV Wardha - Nizamabad D/C	WR-SR 400kV Kolhapur-Kudgi D/C	WR-SR HVDC Raigarh- Puglur	WR-SR 220kV Keldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	WR-SR 765 kV Warora - Warangal D/C	Total WR - SR	SR - WR HVDC Bhadrawati	SR - WR 765 kV Raichur - Sholapur 2x5/C	SR-WR 765 kV Wardha - Nizamabad D/C	SR-WR 400kV Kolhapur- Kudgi D/C	SR-WR HVDC Raigarh- Puglur	WR-SR 220kV Keldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	SR-WR 765 kV Warangal - Warora D/C	Total SR - WR
1-Jan-24	0.00	32.30	52.29	84.59	0.00	0.00	0.00	0.00	18.35	8.55	29.62	0.00	36.23	2.08	0.00	32.89	127.71	0.00	3.02	0.00	16.95	0.00	0.00	0.00	0.00	19.97
2-Jan-24	0.00	32.82	53.27	86.09	0.00	0.00	0.00	0.00	21.32	9.91	33.69	0.00	60.10	2.19	0.00	35.07	162.28	0.00	3.11	0.00	17.96	0.00	0.00	0.00	0.00	21.07
3-Jan-24	0.00	31.30	55.58	86.88	0.00	0.00	0.00	0.00	16.77	11.13	37.89	0.00	46.61	2.24	0.00	38.19	152.84	0.00	2.41	0.00	19.09	0.00	0.00	0.00	0.00	21.50
4-Jan-24	0.00	42.60	52.72	95.31	0.00	0.00	0.00	0.00	19.90	10.24	36.64	0.00	45.41	2.23	0.00	36.30	150.72	0.00	3.70	0.00	20.72	0.00	0.00	0.00	0.00	24.42
5-Jan-24	0.00	36.15	50.27	86.42	0.00	0.00	0.00	0.00	14.84	6.73	33.80	0.00	62.46	2.30	0.00	32.15	152.29	0.00	7.29	0.00	22.22	0.00	0.00	0.00	0.00	29.51
6-Jan-24	0.00	38.01	44.48	82.49	0.00	0.00	0.00	0.00	12.80	9.56	30.14	0.00	51.53	2.24	0.00	35.10	141.37	0.00	5.88	0.00	20.38	0.00	0.00	0.00	0.00	26.27
7-Jan-24	0.00	37.38	36.98	74.36	0.00	0.00	0.00	0.00	16.11	8.67	19.57	0.00	46.25	2.13	0.00	29.09	121.83	0.00	11.21	0.02	22.56	0.00	0.00	0.00	0.03	33.82
8-Jan-24	0.05	37.90	43.52	81.48	0.00	0.00	0.00	0.00	12.14	11.47	35.28	0.00	37.51	2.25	0.00	36.16	134.81	0.00	2.00	0.00	18.84	0.00	0.00	0.00	0.00	20.84
9-Jan-24	5.42	34.77	44.02	84.22	0.00	0.00	0.00	0.00	22.17	10.00	40.69	0.00	45.36	2.26	0.00	36.66	157.14	0.00	3.19	0.00	18.00	0.00	0.00	0.00	0.00	21.20
10-Jan-24	7.30	31.32	45.82	84.44	0.00	0.00	0.00	0.00	17.26	9.92	43.00	0.00	28.16	2.23	0.00	40.06	140.63	0.00	2.22	0.00	19.90	0.00	0.00	0.00	0.00	22.13
11-Jan-24	7.29	31.31	44.75	83.35	0.00	0.00	0.00	0.00	12.74	5.43	37.62	0.00	23.81	2.18	0.00	35.67	117.43	0.00	5.22	0.00	22.80	0.00	0.00	0.00	0.00	28.03
12-Jan-24	7.30	37.84	42.46	87.60	0.00	0.00	0.00	0.00	12.37	2.73	29.74	0.00	32.06	1.86	0.00	28.92	107.68	0.00	10.56	0.00	26.32	0.00	0.00	0.00	0.00	36.88
13-Jan-24	7.30	31.30	46.55	85.15	0.00	0.00	0.00	0.00	10.55	8.95	35.74	0.00	26.47	1.63	0.00	36.23	119.57	0.00	4.56	0.00	22.29	0.00	0.00	0.00	0.00	26.85
14-Jan-24	7.30	31.31	46.47	85.09	0.00	0.00	0.00	0.00	7.35	12.62	37.42	0.00	14.61	2.13	0.68	37.60	112.41	0.00	2.26	0.00	16.76	0.00	0.00	0.00	0.00	19.02
15-Jan-24	7.30	37.42	38.84	83.56	0.00	0.00	0.00	0.00	7.34	8.58	27.01	0.00	14.61	2.09	0.00	27.30	86.93	0.00	11.64	0.24	23.36	0.00	0.00	0.00	0.12	35.35
16-Jan-24	7.30	41.59	35.63	84.51	0.00	0.00	0.00	0.00	7.33	1.98	21.53	0.00	14.61	2.19	0.66	22.28	70.58	0.00	11.69	0.28	22.41	0.00	0.00	0.00	0.12	34.50
17-Jan-24	7.29	33.82	38.84	79.95	0.00	0.00	0.00	0.00	7.32	4.96	22.90	0.00	30.67	2.08	0.00	24.33	92.26	0.00	9.78	0.00	22.52	0.00	0.00	0.00	0.00	32.31
18-Jan-24	9.96	11.92	49.36	71.25	0.00	0.00	0.00	0.00	11.62	13.55	37.43	0.00	38.87	2.08	0.00	37.11	140.66	0.00	1.12	0.00	17.34	0.00	0.00	0.00	0.00	18.47
19-Jan-24	14.96	0.00	52.22	67.18	0.00	0.00	0.00	0.00	14.54	19.20	39.05	0.00	49.94	2.07	0.00	40.96	165.76	0.00	1.04	0.00	17.24	0.00	0.00	0.00	0.00	18.28
20-Jan-24	15.05	0.00	47.98	63.03	0.00	0.00	0.00	0.00	14.23	14.82	32.57	0.00	63.26	0.55	0.00	34.37	159.80	0.00	3.98	0.00	19.44	0.00	0.00	0.00	0.00	23.43
21-Jan-24	15.16	0.00	45.21	60.36	0.00	0.00	0.00	0.00	12.89	13.07	31.15	0.00	60.35	0.00	0.00	30.06	147.52	0.00	4.76	0.00	19.15	0.00	0.00	0.00	0.00	23.92
22-Jan-24	14.39	0.00	51.99	66.38	0.00	0.00	0.00	0.00	7.31	23.71	43.30	0.00	44.09	0.17	0.00	41.51	160.11	0.00	0.53	0.00	14.97	0.00	0.00	0.00	0.00	15.50
23-Jan-24	14.30	0.00	47.95	62.25	0.00	0.00	0.00	0.00	7.31	16.60	38.38	0.00	75.59	2.17	0.00	36.01	176.06	0.00	5.22	0.00	19.64	0.00	0.00	0.00	0.00	24.86
24-Jan-24	13.23	0.00	53.30	66.53	0.00	0.00	0.00	0.00	19.63	20.22	40.06	0.03	52.46	2.16	0.00	41.09	175.64	0.00	0.60	0.00	15.84	0.00	0.00	0.00	0.00	16.44
25-Jan-24	11.41	0.00	53.21	64.62	0.00	0.00	0.00	0.00	7.30	22.28	38.83	0.00	46.28	2.07	0.00	41.12	157.88	0.00	0.84	0.00	16.14	0.00	0.00	0.00	0.00	16.98
26-Jan-24	11.39	0.00	48.03	59.42	0.00	0.00	0.00	0.00	7.29	15.61	32.94	0.00	46.23	1.99	0.00	33.96	138.03	0.00	3.89	0.00	17.19	0.00	0.00	0.00	0.00	21.08
27-Jan-24	11.38	0.00	50.70	62.08	0.00	0.00	0.00	0.00	7.30	13.32	35.56	0.00	40.69	2.10	0.00	36.25	135.22	0.00	4.02	0.00	20.73	0.00	0.00	0.00	0.00	24.75
28-Jan-24	11.40	0.00	46.81	58.21	0.00	0.00	0.00	0.00	21.30	8.74	26.95	0.00	49.06	1.87	0.00	27.77	135.70	0.00	10.45	0.00	24.21	0.00	0.00	0.00	0.00	34.65
29-Jan-24	11.40	0.00	48.34	59.74	0.00	0.00	0.00	0.00	24.15	8.81	28.29	0.00	54.11	2.09	0.00	29.97	147.43	0.00	5.98	0.00	21.96	0.00	0.00	0.00	0.00	27.95
30-Jan-24	11.40	0.00	48.39	59.79	0.00	0.00	0.00	0.00	24.13	11.61	29.68	0.00	49.19	1.85	0.00	30.81	147.26	0.00	5.24	0.00	21.09	0.00	0.00	0.00	0.00	26.33
31-Jan-24	11.31	0.00	52.04	63.35	0.00	0.00	0.00	0.00	22.72	15.52	32.01	0.00	78.90	1.82	0.00	35.49	186.46	0.00	3.99	0.00	17.94	0.00	0.00	0.00	0.00	21.94
TOTAL	240.59	611.06	1468.02	2319.67	0.00	0.01	0.00	0.01	438.40	358.47	1038.49	0.03	1365.48	59.32	1.38	1060.47	4322.03	0.00	151.44	0.53	615.99	0.00	0.00	0.02	0.26	768.24

16. भूटान , नेपाल, बांग्लादेश एवं म्यान्मार के साथ अंतरराष्ट्रीय विद्युत विनिमय

अप्रैल 2023 से मार्च 2024 April 2023 to March 2024

अंतरराष्ट्रीय विद्युत विनिमय [भारत 'से निर्यात'/'को आयात']

Transnational Exchange ('Export from'/'Import to' India)

माह MONTH	भूटान BHUTAN		नेपाल NEPAL		बांग्लादेश BANGLADESH		म्यान्मार MYANMAR	
	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)
अप्रैल 2023 APR'23	19.40	41.40	357.99	0.00	687.30	0.00	0.73	0.00
मई 2023 MAY'23	2.27	117.99	271.80	0.34	775.10	0.00	0.61	0.00
जून 2023 JUN'23	0.00	626.15	80.00	88.86	743.92	0.00	0.77	0.00
जुलाई 2023 JUL'23	0.00	1291.66	0.00	223.72	780.77	0.00	1.02	0.00
अगस्त 2023 AUG'23	0.00	1244.52	0.00	281.61	767.79	0.00	0.94	0.00
सितंबर 2023 SEP'23	0.00	924.25	0.00	323.59	754.88	0.00	0.77	0.00
अक्टूबर 2023 OCT'23	0.00	588.64	0.00	460.97	755.92	0.00	0.84	0.00
नवंबर 2023 NOV'23	51.41	36.77	0.00	271.72	628.24	0.00	0.72	0.00
दिसम्बर 2023 DEC'23	212.51	0.00	88.76	5.09	593.51	0.00	0.70	0.00
जनवरी 2024 JAN'24	287.94	0.00	271.79	0.00	601.77	0.00	0.65	0.00
कुल Total	573.53	4871.38	1070.33	1655.89	7089.20	0.00	7.75	0.00

* Based on daily operational data

Date	16.1 Import from neighbouring countries during January 2024													
	(All figures in MU)													
	Import from Bhutan								Import from Bangladesh		Import from Nepal			Import from Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurwar D/C*	400 kV Jigmeling-Alipurwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
10-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
11-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
12-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00
13-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00
14-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
17-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
18-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
19-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00
21-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00
22-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00
23-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00
24-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00
25-Jan-24	0.19	0.00	0.62	0.00	0.00	0.00	0.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00
26-Jan-24	0.48	0.00	0.03	0.00	0.00	0.00	0.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00
27-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00
28-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00
29-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00
30-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00
31-Jan-24	0.00	0.00	0.00	0.00	0.00	0.00	0.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total	0.67	0.00	0.65	0.00	0.00	0.00	2.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Based on SEM/Energy meter data for links where available (*Mangdechu generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)														

Date	16.2 Export to neighbouring countries during January 2024													
	(All figures in MU)													
	Export to Bhutan								Export to Bangladesh		Export to Nepal			Export to Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurwar D/C*	400 kV Jigmeling-Alipurwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jan-24	0.46	1.47	2.83	1.53	1.39	2.05	0.09	0.68	15.06	2.29	1.07	4.01	0.00	0.02
2-Jan-24	0.36	1.41	2.84	1.41	1.29	1.88	0.00	0.64	17.22	2.24	1.19	4.76	0.00	0.02
3-Jan-24	0.25	1.27	2.70	1.28	2.45	0.72	0.13	0.49	18.97	2.16	1.17	5.00	0.00	0.02
4-Jan-24	0.13	1.19	2.67	1.27	3.53	0.00	0.06	0.48	18.36	2.18	1.17	5.15	0.00	0.03
5-Jan-24	0.06	1.16	2.72	1.30	3.55	0.00	0.08	0.43	14.69	2.08	1.20	5.13	0.00	0.02
6-Jan-24	0.28	1.31	2.84	1.45	3.28	0.00	0.07	0.66	16.68	2.08	1.13	5.11	0.00	0.02
7-Jan-24	0.54	1.52	3.00	1.58	3.27	0.00	0.09	0.73	14.84	1.71	1.34	5.93	0.00	0.02
8-Jan-24	0.30	1.34	2.96	1.36	2.25	1.48	0.10	0.47	17.25	2.07	1.18	6.13	0.00	0.02
9-Jan-24	0.28	1.34	2.78	1.30	1.36	2.02	0.10	0.68	18.47	2.29	1.33	6.25	0.00	0.02
10-Jan-24	0.38	1.45	2.73	1.42	1.52	2.26	0.10	0.43	18.40	2.31	1.41	6.67	0.00	0.01
11-Jan-24	0.55	1.62	2.65	1.48	1.42	2.11	0.10	0.39	18.51	2.28	1.38	7.09	0.00	0.01
12-Jan-24	0.50	1.56	2.88	1.38	1.02	1.51	0.00	0.59	14.63	2.04	1.32	6.56	0.00	0.01
13-Jan-24	0.30	1.44	2.76	1.38	1.24	1.85	0.00	0.59	16.60	2.19	1.34	6.44	0.00	0.01
14-Jan-24	0.73	1.78	3.02	1.55	1.40	2.07	0.00	0.79	19.57	2.28	1.38	7.53	0.00	0.01
15-Jan-24	0.55	1.64	2.84	1.33	1.34	2.00	0.03	0.74	18.95	2.33	1.27	6.90	0.00	0.00
16-Jan-24	0.66	1.75	2.76	1.46	1.35	1.99	0.07	0.62	18.82	1.78	1.20	7.21	0.00	0.01
17-Jan-24	0.56	1.65	2.60	1.47	1.46	2.16	0.04	0.62	16.56	1.94	1.35	7.69	0.00	0.01
18-Jan-24	0.70	1.77	2.72	1.38	1.44	2.12	0.04	0.64	16.48	2.27	1.44	7.88	1.06	0.01
19-Jan-24	0.59	1.76	2.52	1.33	1.46	2.18	0.00	0.54	13.36	2.11	1.54	7.85	1.06	0.01
20-Jan-24	0.77	1.60	2.65	1.39	1.50	2.23	0.00	0.63	14.84	2.26	1.40	7.62	1.06	0.10
21-Jan-24	1.07	2.07	3.08	1.49	1.58	2.34	0.00	0.66	16.07	2.23	1.52	8.69	1.06	0.02
22-Jan-24	0.44	1.46	2.28	1.44	1.46	2.47	0.00	0.62	15.80	2.43	1.51	8.27	1.06	0.02
23-Jan-24	0.56	1.54	2.40	1.37	1.53	2.28	0.00	0.39	16.86	2.39	1.50	8.05	1.06	0.02
24-Jan-24	0.11	0.80	0.28	1.37	1.75	2.60	0.00	0.73	18.05	2.40	1.41	8.21	1.06	0.02
25-Jan-24	0.00	0.40	0.00	1.22	1.71	2.47	0.00	0.74	18.12	2.47	1.50	8.27	1.06	0.02
26-Jan-24	0.00	0.25	0.00	1.32	1.14	2.13	0.00	0.64	14.99	2.24	1.62	9.53	1.06	0.01
27-Jan-24	0.32	1.22	2.14	1.27	1.18	1.75	0.00	0.55	17.74	2.35	1.46	8.08	1.06	0.02
28-Jan-24	0.52	1.39	2.22	1.29	1.28	1.71	0.00	0.42	18.67	2.46	1.56	9.16	1.06	0.02
29-Jan-24	0.44	1.33	2.04	1.18	1.32	1.96	0.00	0.49	18.34	2.47	1.48	8.56	1.06	0.02
30-Jan-24	0.31	1.24	2.05	1.15	1.39	2.05	0.00	0.56	18.16	2.52	1.50	8.44	1.06	0.02
31-Jan-24	0.42	1.34	2.04	1.18	1.34	1.99	0.00	0.52	18.12	2.58	1.44	8.63	1.06	0.02
Total	13.14	43.09	73.99	42.31	53.18	54.38	1.10	18.16	529.21	69.42	42.31	220.81	14.86	0.65
Based on SEM/Energy meter data for links where available (*Mangdechu generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)														

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जनवरी 2024 EXCHANGES THROUGH POWER EXCHANGES - JANUARY 2024				
क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
1	AD HYDRO POWER LIMITED	उत्तरी क्षेत्र NR	12.75	0.00
2	ADANI SOLAR ENERGY JAISALMER TWO PRIVATE LIMITED(Project-2)		14.02	0.00
3	ADANI SOLAR ENERGY JODHPUR TWO LIMITED		9.09	0.00
4	ALTRA XERGI POWER PRIVATE LIMITED		43.04	0.00
5	Amp Energy Green Six Private Limited		14.84	0.00
6	Amplus Ages Private Limited		7.99	0.00
7	Budhil HEP (Greenko Budhil Hydro Power Pvt. Ltd.)		0.54	0.00
8	Chandigarh (UT)		2.94	23.00
9	Delhi		186.88	185.28
10	Grian Energy Private Limited		9.23	0.00
11	Haryana		41.04	361.03
12	Himachal Pradesh		67.64	76.34
13	IGSTPS JHAJJAR		4.06	0.00
14	Jammu Kashmir		30.42	495.30
15	KARCHAM WANGTOO HYDRO ELECTRIC PLANT.		17.01	0.00
16	NEPAL ELECTRICITY AUTHORITY TANAKPUR		0.00	30.78
17	NTPC Dadri Stage I		15.41	0.00
18	NTPC Dadri Stage II		2.18	0.00
19	NTPC Rihand stage I		9.43	0.00
20	NTPC Rihand stage II		5.03	0.00
21	NTPC Rihand stage III		10.32	0.00
22	NTPC Singrauli		9.15	0.00
23	NTPC Tanda Stage II		3.46	0.00
24	NTPC Unchahar Stage I		1.07	0.00
25	NTPC Unchahar Stage II		0.97	0.00
26	NTPC Unchahar Stage III		0.42	0.00
27	NTPC Unchahar Stage IV		0.94	0.00
28	North Central Railway Prayagraj		0.00	25.05
29	Onevolt Energy Private Limited		9.49	0.00
30	Punjab		17.78	1091.33
31	RENEW SURYA PRATAP PRIVATE LIMITED		9.69	0.00
32	RENEW SURYA VIHAAN PRIVATE LIMITED		3.52	0.00
33	Rajasthan		78.58	463.34
34	ReNew Surya Ravi Private Limited		34.75	0.00
35	SHREE CEMENT LIMITED TPS		30.56	0.00
36	Sainj HEP		5.49	0.00
37	Singoli Bhatwari HEP		5.05	0.00
38	Sorang HEP (Himachal Sorang Power Pvt. Ltd.))		6.92	0.00
39	Uttar Pradesh		409.27	637.36
40	Uttarakhand		5.85	236.80

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जनवरी 2024 EXCHANGES THROUGH POWER EXCHANGES - JANUARY 2024				
क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
41	ACB (INDIA) LIMITED	पश्चिमी क्षेत्र WR	84.18	0.23
42	ADANI RENEWABLE ENERGY HOLDING FOUR LIMITED		15.60	0.00
43	ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana		50.82	0.00
44	Adani Power Limited - Raigarh TPP		256.43	0.00
45	Adani Power Limited-Raipur TPP		395.27	2.23
46	ArcelorMittal Nippon Steel India Limited		0.00	151.32
47	BHARAT ALUMINIUM COMPANY LTD		44.20	0.00
48	BHARAT ALUMINIUM COMPANY LTD (Bulk Consumer)		0.00	9.93
49	CONTINUUM POWER TRADING (TN) PRIVATE LIMITED		10.19	0.00
50	Chhattisgarh		593.60	12.28
51	D B Power Limited		168.79	0.00
52	DHARIWAL STU OTH		14.79	0.00
53	Daman and Diu - Dadra and Nagar Haveli		4.33	79.35
54	Dhariwal ISTS		12.64	0.00
55	GMR Warora Energy Limited		62.27	0.23
56	Goa WR		3.69	11.93
57	Gujarat		335.50	474.30
58	INOX GREEN ENERGY SERVICES LIMITED		0.85	0.00
59	Jaypee Nigrie Super Thermal Power Plant		202.38	0.00
60	Jhabua Power Limited		24.25	0.21
61	Jindal Power Limited, Stage-1		195.13	2.23
62	Jindal Power Limited, Stage-2		436.09	1.03
63	Jindal Steel & Power Ltd , DCP		95.38	0.00
64	MAHAN ENERGEN LIMITED.		395.28	7.16
65	MB POWER (MADHYA PRADESH) LIMITED		66.35	0.98
66	Madhya Pradesh		474.77	96.40
67	Maharashtra		208.44	420.36
68	Maruti Clean Coal and Power Limited		19.51	0.00
69	NREL-Dayapar Wind		3.66	0.00
70	NTPC Gadawara		9.55	0.00
71	NTPC Jhanor Gandhar GPS		0.00	0.78
72	NTPC Kawas GPS		0.00	0.64
73	NTPC Korba Stage I &II		1.06	0.00
74	NTPC Korba Stage III		0.42	0.00
75	NTPC Lara Stage I		0.28	0.00
76	NTPC Mouda Stage I		3.94	0.00
77	NTPC Mouda Stage II		6.36	0.00
78	NTPC SAIL POWER COMPANY LIMITED		0.10	0.00
79	NTPC Sipat Stage I		9.07	0.00
80	NTPC Sipat Stage II		4.77	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जनवरी 2024
EXCHANGES THROUGH POWER EXCHANGES - JANUARY 2024

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
81	NTPC Solapur		6.77	0.00
82	NTPC Vindhyachal Stage I		4.62	0.00
83	NTPC Vindhyachal Stage II		2.91	0.00
84	NTPC Vindhyachal Stage III		2.77	0.00
85	NTPC Vindhyachal Stage IV		1.94	0.00
86	NTPC Vindhyachal Stage V		1.35	0.00
87	NTPC khargone		7.12	0.00
88	R.K.M POWERGEN PRIVATE LIMITED		27.66	0.00
89	Ratnagiri Gas & Power Private Limited		1.56	0.71
90	SKS Power Generation Chhattisgarh Limited		40.05	0.00
91	Sasan Power Limited		3.04	0.00
92	TRN ENERGY PRIVATE LIMITED		15.29	0.00
93	The Tata Power Co Ltd (MTPS)		26.21	0.00
94	WIND FIVE RENERGY LIMITED		7.62	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जनवरी 2024
EXCHANGES THROUGH POWER EXCHANGES - JANUARY 2024

क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
95	Andhra Pradesh	दक्षिणी क्षेत्र SR	112.46	446.17
96	Ayana Renewable Power Six Private Limited		20.57	0.00
97	COASTAL ENERGEN PRIVATE LIMITED		122.21	0.00
98	Goa SR		0.04	3.50
99	IL&FS TAMIL NADU POWER COMPANY LIMITED		36.70	0.00
100	Karnataka		1.38	57.79
101	Kerala		24.92	205.90
102	NLC INDIA LIMITED NEVELI NEW THERMAL POWER STATION		5.00	0.00
103	NLC INDIA LIMITED THERMAL POWER STATION I EXPANSION		2.53	0.00
104	NLC INDIA LIMITED THERMAL POWER STATION II EXPANSION		0.00	0.00
105	NLC INDIA LIMITED THERMAL POWER STATION II STAGE I		0.00	0.00
106	NLC INDIA LIMITED THERMAL POWER STATION II STAGE II		0.00	0.00
107	NTPC KUDGI		12.00	0.00
108	NTPC Ramagundam Stage I &II		7.37	0.00
109	NTPC Ramagundam Stage III		1.89	0.00
110	NTPC Simhadri Stage I		0.04	0.00
111	NTPC Simhadri Stage II		3.97	0.00
112	NTPC Talcher Super Thermal Power Station Stage II		1.48	0.00
113	NTPC Telangana		2.37	0.00
114	OSTRO KANNADA POWER PRIVATE LIMITED		17.63	0.00
115	Pondicherry UT		95.85	0.01
116	RENEW SURYA ROSHNI PRIVATE LIMITED Koppal PS		18.93	0.00
117	Ramagundam Floating solar		7.16	0.00
118	SEIL ENERGY INDIA LIMITED		54.46	0.00
119	SEIL Energy India Limited Project II		93.33	0.00
120	SIMHAPURI ENERGY LIMITED		57.39	3.33
121	Simhadri FSP 15 MW		2.04	0.00
122	Simhadri Floating solar (10 MW)		1.40	0.00
123	Tamil Nadu		214.24	256.94
124	Telangana		133.49	1345.46

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जनवरी 2024 EXCHANGES THROUGH POWER EXCHANGES - JANUARY 2024				
क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
125	ADHUNIK POWER & NATURAL RESOURCES LIMITED	पूर्वी क्षेत्र ER	39.20	0.00
126	Bihar		217.45	61.81
127	Damodar Valley Corporation		32.23	102.43
128	Druk Green Power Corporation Limited (DGPCL-Bhutan)		0.00	336.79
129	GMR KAMALANGA ENERGY LTD-CTU		0.02	0.00
130	IND BARATH ENERGY UTKAL LIMITED		14.21	1.01
131	JORETHANG LOOP HEP, DANS ENERGY PRIVATE LIMITED		0.15	0.00
132	Jharkhand		49.28	28.26
133	Jindal India Thermal Power Limited		334.07	0.00
134	KANTI BIJLEE UTPADAN NIGAM LIMITED		0.89	0.00
135	MAITHON POWER LIMITED		0.40	0.00
136	NABINAGAR POWER GENERATING COMPANY LIMITED		15.86	0.00
137	NEA Nepal Solu Khola (Dudhkoshi) Hydropower Project		0.15	0.00
138	NEPAL ELECTRICITY AUTHORITY MUZAFFARPUR		0.00	197.71
139	NTPC BARH Stage I		3.56	0.00
140	NTPC BARH Stage II		27.49	0.00
141	NTPC Darlipali		3.56	0.00
142	NTPC Farakka stage I		15.03	0.00
143	NTPC Farakka stage III		3.90	0.00
144	NTPC Kahalgaon stage I		6.47	0.00
145	NTPC Kahalgaon stage II		11.57	0.00
146	NTPC North Karanpura STPS		0.02	0.00
147	NTPC North Karanpura STPS infirm (UNIT # 2)		14.66	0.00
148	NTPC Talcher Stage I		0.46	0.00
149	Odisha		471.68	53.48
150	Sikkim		7.85	0.06
151	Tashiding HEP, Shiga Energy Private Limited		0.20	0.00
152	West Bengal		249.14	127.17

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जनवरी 2024
EXCHANGES THROUGH POWER EXCHANGES - JANUARY 2024

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
153	AGARTALA GAS BASED POWER STATION	पूर्वोत्तर क्षेत्र NER	0.03	0.00
154	ASSAM GAS BASED POWER STATION		2.79	0.00
155	Arunachal Pradesh		4.06	11.28
156	Assam		59.47	14.58
157	Bongaigaon Thermal Power Station NTPC		3.35	0.00
158	Doyang HPS, NEEPCO		0.15	0.00
159	KAMENG HYDRO POWER STATION		8.36	0.00
160	Kopili Power Station NEEPCO		0.36	0.00
161	Manipur		1.63	0.16
162	Meghalaya		5.66	30.61
163	Nagaland		1.74	1.59
164	Palatana Plant		1.85	0.00
165	Pare Hydro Power Station		0.04	0.00
166	Ranganadi Hydro Power Station		0.04	0.00
167	Tripura		28.48	0.10
Total			8184	8184

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM
MONTH : JANUARY 2024

RE Source & Unit wise break up (01.01.2024-31.01.2024)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	52	3	68	7
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	8	4	27	2
5	Small Hydro	0	0	0	0
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	60	7	95	9

RECs Issued (01.01.2024-31.01.2024)

Sr.No.	Non Solar	Solar	Total
1	2933453	66554	3000007

Redemption of REC (01.01.2024-31.01.2024)

Sr.No.	Non Solar	Solar	Total
1	2603104	46200	2649304

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM
MONTH : JANUARY 2024

RE Source & Unit wise break up (Apr'23-Jan'24)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	418	24	522	28
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	324	55	268	10
5	Small Hydro	101	8	42	7
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	5	1	5	1
9	Bio-fuel cogeneration	8	2	23	5
	Total	856	90	860	51

RECs Issued (Apr'23-Jan'24)

Sr.No.	Non Solar	Solar	Total
1	21802376	915238	22717614

Redemption of REC (Apr'23-Jan'24)

Sr.No.	Non Solar	Solar	Total
1	8373328	713800	9087128

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM

MONTH : JANUARY 2024

RE Source & Unit wise break up Since Inception to Jan'24

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	3131	529	2975	530
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	1593	547	1453	475
5	Small Hydro	745	41	698	42
6	Others	4	2	3	1
7	Geothermal	0	0	0	0
8	Biomass	402	37	378	35
9	Bio-fuel cogeneration	823	91	383	55
	Total	6698	1247	5890	1138

RECs Issued since Inception to Jan'24

Sr.No.	Non Solar	Solar	Total
1	109682516	13325976	123008492

Redemption of REC since Inception to Jan'24

Sr.No.	Non Solar	Solar	Total
1	75901513	12342188	88243701

REC Closing balance as on 31.01.2024

Sr.No.	Non Solar	Solar	Total
1	33781003	983788	34764791

20.Details of Grid Events during the Month of January 2024 in Northern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HEMM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t. Antecedent Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GI-2	Rajasthan	02-Jan-24 07:28	02-Jan-24 10:00	02:32	690	160	1.611	0.325	42839	49201	i) 400/220kV Akal(RS) has one and half breaker scheme at 400kV level and double main transfer bus scheme at 220kV level. ii) As reported, at 07:28hrs, 220 kV Akal(RS) Bhur(RS) Ckt-1 tripped on B-N phase to earth fault due to heavy fog. iii) At the same time, 400/220 kV 500 MVA ICT-1, 2 & 4 and 315MVA ICT-3 at Akal(RS) also tripped. (Exact reason yet to be shared, but it is suspected that there is delay in CB opening due to which ICTs also got tripped. Also O/C protection settings of ICTs need to be shared.) iv) As per DR of 400/220 kV 500 MVA ICT 2 at Akal(RS), O/C I/F protection operated and fault current was I _B =5.157kA. v) As per DR of 400/220 kV 315 MVA ICT 3 at Akal(RS), O/C I/F protection operated and fault current was I _B =2.288kA. vi) As per SCADA SOE, 220kV Akal(RS)-Barmer(RS) Ckt also tripped during the same time. (Exact reason yet to be shared) vii) As per PMU at Jodhpur(RS), two consecutive B-N phase to earth faults are observed with delayed fault clearance time of 320ms and 1400ms respectively. viii) As per SCADA, change in demand of approx. 160MW is observed in Rajasthan control area. ix) As per SCADA, change in Rajasthan wind generation of approx. 690MW is observed.	1) 400/220 kV 500 MVA ICT 2 at Akal(RS) 2) 400/220 kV 500 MVA ICT 2 at Akal(RS) 3) 400/220 kV 315 MVA ICT 3 at Akal(RS) 4) 400/220 kV 500 MVA ICT 4 at Akal(RS)
2	GD-1	Rajasthan	05-Jan-24 05:16	05-Jan-24 06:43	01:27	1817	410	5.726	0.965	31732	42496	i) During the antecedent condition, 220kV Kota(PG)-KTPS(RVUN) (RS) Ckt-1 & 2, 220kV KTPS-HeeraPura Ckt, 220kV KTPS-Beawar Ckt, 220 kV Debari(RS)-RAPS_A(NP) (RS) Ckt and 220 kV Anta(NP)-Sakapura(RS) (RS) Ckt were out of service and 110MW Unit-1, 210MW Unit-4 & 5 and 195MW Unit-6 & 7 at 220kV KTPS (generating total "803MW", 200MW Unit-2 at RAPS-A (generating "200MW", 220MW Unit-3 at RAPS-B (generating "185MW", 220MW Unit-5 & 6 at RAPS-C (generating total "452MW", 43MW Unit-1, 3 & 4 at RPS HE generating total "51MW", 13MW Unit-1, 2 & 3 at 15 HE (generating total "91MW") were in service. ii) Before the event, at 05:15 hrs, 220kV KTPS-Bundi Ckt, 220kV KTPS-Vaika Ckt, 220kV RAPS-A-RAPS-B Ckt and 220kV RAPS-C-Anta20 Ckt were already carrying 288MW, 214MW, 273MW and 289MW respectively and limited connectivity with the grid was available (220kV Vaika, Bundi and Anta). iii) At 05:15:56 hrs, 220kV Bundi-Gulabpura (RS) ckt was manually opened on SLOC instruction to avoid line tripping on overloading. iv) After this, as per DR, 220kV KTPS-Vaika Ckt current reached "950A" (~120MW) and tripped on over-current protection operation from KPS Unit. At the same time, as per DR, 220kV RAPS-C(NP)-Anta(NP) ckt current reached "1800A" (~600MW) and tripped due to over-loading. v) Due to tripping of above lines, KTPS-RAPS-A-RAPS-B got isolated from the grid with only radial load available at 220kV Debari, 220 kV Rangpur and 220kV Chittorgarh. This led to over frequency; frequency reached to 51.6Hz. vi) Subsequent to this, RAPS-B unit-4 tripped on over frequency and came to house load. RAPS-B unit-4 operated on house load for approx. 1.5 hours but couldn't able to come back to grid mode due to issue in governing system (as intimated by RAPS). vii) RAPS-A unit-2 (no provision to come to house load) and KTPS Units also tripped on over frequency protection. viii) RAPS-C unit-5&6 tripped due to loss of auxiliary supply (auxiliary supply chnageover from 220kV to 400kV blocked due to frequency mismatch). ix) As per PMU, no fault is observed in the system. x) As per SCADA, load loss of approx. 410 MW is observed in Rajasthan control area and total change in generation of approx. 1817MW is observed.	1) 220kV KTPS-Beawar ckt 2) 220 kV Anta(NP)-Sakapura(RS) (RS) Ckt 3) 220 kV Debari(RS)-RAPS_A(NP) (RS) Ckt 4) 220kV KTPS-Vaika ckt 5) 220kV RAPS_C(NP)-Anta(NP) ckt 6) 200 MW RAPS-A Unit-2 7) 220 MW RAPS-B Unit-4 8) 220 MW RAPS-C Unit-5 9) 220 MW RAPS-C Unit-6 10) 110 MW Kota TPS Unit-1 11) 110 MW Kota TPS Unit-4 12) 110 MW Kota TPS Unit-5 13) 195 MW Kota TPS Unit-6 14) 185 MW Kota TPS Unit-7
3	GI-1	Rajasthan	05-Jan-24 12:19	05-Jan-24 14:07	01:48	385	0	0.786	0.000	48977	62168	i) Total MW generation of CSP Bhadla and SBEP(L) is evacuated through five 220/33kV 125MVA ICTs at Saurya Urja(U) which are again connected to 220kV Saurya Urja(U)-Bhadla(PG) Ckt-1 & 2. During antecedent condition, total MW generation of Saurya Urja(U) (CSP Bhadla+SBEP(L)) was "500MW and ICTs were carrying "100MW each. ii) As reported, at 12:19hrs, 220/33 kV 125 MVA ICT-1 at Saurya Urja Solar(U) tripped due to earth fault in LV side of ICT (exact location and nature of fault yet to be shared). iii) As per PMU at Saurya Urja(U), Y-B phase to phase fault with fault clearance time of 80ms is observed in the system. iv) As per PMU, no generation loss occurred at Saurya Urja(U). But reduction in solar generation of approx. 385MW in total is observed (150MW at AHEJL, 100MW at RSEPL and 135MW at RSUPJ).	1) 220/33 kV 125 MVA ICT-1 at Saurya Urja Solar(U)
4	GI-2	Delhi	09-Jan-24 11:03	09-Jan-24 12:36	01:33	0	360	0.000	0.589	45066	61169	i) 400/220kV Dwarka(PG/DTL) has one and half breaker scheme at 400kV level (owned by PG upto ICT) and double main bus scheme at 220kV level (owned by DTL). ii) As reported, at 11:03 Hrs, during the maintenance work of 220kV bus section/liaison 1 at Dwarka(DTL), bus-bar protection operated at 220kV Dwarka(DTL). iii) Due to bus bar protection operation, 400/220 kV 500 MVA ICT-1 & 2 at Dwarka (PG) tripped from 220kV side only. "400kV side remained connected. 220kV Dwarka-Pappankal(DTL) Ckt-1 & 2 and 220kV Dwarka-Naraina (DTL) Ckt-1 & 2 also tripped. Both the buses at 220kV level became dead. iv) As per PMU, no fault is observed in the system. v) As per SCADA, change in demand of approx. 360MW is observed in Delhi control area. vi) As reported, after tripping power was restored from Ridge valley and Bannaji within 10 minutes.	1) 400/220 kV 500 MVA ICT-1 at Dwarka (PG) 2) 400/220 kV 500 MVA ICT-2 at Dwarka (PG) 3) 220kV Dwarka-Pappankal(DTL) Ckt-1 4) 220kV Dwarka-Pappankal(DTL) Ckt-2 5) 220kV Dwarka-Naraina (DTL) Ckt-1 6) 220kV Dwarka-Naraina (DTL) Ckt-2
5	GI-2	Haryana	09-Jan-24 14:01	09-Jan-24 14:32	00:31	0	0	0.000	0.000	47825	58747	i) During antecedent condition, 800kV HVDC Champa-Kushabheta Bipoles was carrying total 2500MW (625MW each pole). ii) As reported at 14:00:20hrs, "commutation failure detected" and "Pole-4 Instability Detected by SSAD" protection latched in Pole 4 which initiated CAT A2 sequence for blocking of Pole 4 and isolated Pole 4 from parallel Pole 2. iii) Further after "800ms of isolation of CAT A2 sequence by Pole 4 on instability protection, opening sequence to HVHS at both ends didn't initiate which led to failure of protective isolation of faulty Pole 4 and generated CAT B alarm leading to tripping of parallel Pole 2 also. iv) Further at 14:01:17 hrs, "Instability detected" protection latched in Pole 1 also which initiated CAT A2 sequence for protective isolation from Pole 3. v) Further at 14:01:18hrs, like Pole 4, CAT A2 sequence in Pole 1 also failed to initiate HVHS opening leading to protective sequence failure which generated CAT B alarm that resulted in tripping of parallel Pole 3. vi) Due to tripping of all four (04) poles, power order reduced from 2500MW to 0MW. vii) As per PMU, fluctuation in power order was observed.	1) 800 kV HVDC Kurukshetra(PG) Pole-01 2) 800 kV HVDC Kurukshetra(PG) Pole-02 3) 800 kV HVDC Kurukshetra(PG) Pole-03 4) 800 kV HVDC Kurukshetra(PG) Pole-04
6	GI-2	Rajasthan	10-Jan-24 12:19	10-01-2024 13:51	01:32	1360	0	2.542	0.000	53506	64858	i) As reported, at 12:19hrs, 400kV Bhadla(RS)-Bikaner(RS) Ckt-2 tripped on Y-B phase to phase fault with fault distance of 9.176kA and fault current of 8.29kA in Y and B phase respectively from Bhadla(RS). As per information received from SLOC Rajasthan, fault was traced by Patrolling from team of M/s Ramelex Pvt. Ltd. and the broken conductor was repaired after taking shutdown of 400kV Bhadla(RS)-Bikaner(RS) Ckt-1 & 2. (Exact location of conductor snapping need to be shared). ii) As per DR at Bikaner(RS) end of 400kV Bhadla(RS)-Bikaner(RS) Ckt-2, fault was sensed in zone-1; fault current was 4.799kA and 5.723kA in Y and B phase respectively from Bikaner(RS) end and fault clearing time was "55ms". iii) As per SCADA, change in NR total solar generation of approx. 1360MW is observed out of which approx. 900MW is recovered within 2 minutes. iv) As per PMU at Bhadla(RS), Y-B phase to phase fault is observed with fault clearance time of 120 ms. v) As per SCADA SOE, 220/33kV 100MVA ICT-1 & 3 at RSKPL(PG) also tripped during the same time. (Exact reason yet to be shared)	1) 400kV Bhadla(RS)-Bikaner(RS) Ckt-2 2) 220/33kV 100MVA ICT-1 at RSKPL(PG) 3) 220/33kV 100MVA ICT-3 at RSKPL(PG)
7	GD-1	Himachal Pradesh	10-Jan-24 20:29	10-Jan-24 23:31	03:02	0	0	0.000	0.000	40641	58974	i) 400kV Koldam(NT) has one and half breaker scheme. ii) As reported, at 20:29 Hrs, during hand tripping of 400kV Ludhiana(PG)-Koldam(NT) (PG) Ckt-1 from Koldam(NT) end, one pole of the CB 1432 failed to open which resulted in bus bar protection operation. iii) Fault was severe and got extended to both the buses and all the elements connected to Bus-1 & 2 tripped and 400kV Koldam(NT) S/s became dead. iv) As per PMU, R-Y phase to phase fault with fault clearing time of 80ms is observed in the system. v) As per SCADA, no change in demand is observed in HP control area.	1) 400kV Ludhiana(PG)-Koldam(NT) (PG) Ckt-1 2) 400 kV Hallargarh(PG)-Koldam(NT) (PG) Ckt 3) 400 kV Koldam(NT)-Parbati Pooling Banah(PG) (PKTCL) Ckt 4) 400kV Bus 1 at Koldam(NT) 5) 400kV Bus 2 at Koldam(NT)
8	GI-2	Uttar Pradesh	11-Jan-24 19:12	11-Jan-24 19:49	00:37	0	150	0.000	0.238	43682	63083	i) As reported, at 19:12hrs, B-phase isolator jumper broke at 400/220kV Muzaffarnagar(U) and bus-bar protection operated at 220kV Bus-1 at Muzaffarnagar(U). ii) Due to bus bar protection operation, all the elements connected to 220kV Bus-1 at Muzaffarnagar(U) tripped and Bus-1 became dead. iii) As reported, 132kV Muzaffarnagar-Khatauli Ckt and 132kV Muzaffarnagar-Buland Ckt also tripped during the same time (exact reason yet to be shared). iv) As per PMU at Muzaffarnagar(U), B-N phase to earth fault with fault clearance time of 80ms is observed. v) As per SCADA, change in demand of approx. 150 MW is observed in UP control area.	1) 400/220 kV 115 MVA ICT-3 at Muzaffarnagar(U) 2) 400/220 kV 500 MVA ICT-4 at Muzaffarnagar(U) 3) 220/132kV 160MVA ICT-4 at Muzaffarnagar(U) 4) 220 kV Muzaffarnagar(U)-Buland Ckt 5) 220 kV Muzaffarnagar(U)-Jansath Ckt 6) 220 kV Muzaffarnagar(U)-Khatauli Ckt
9	GI-2	Rajasthan	15-Jan-24 13:59	19-Jan-24 21:45		2020	0	4.027	0.000	50162	61237	i) As reported, at 13:59hrs on 15th January, 2024, 400kV Bhadla(RS)-Bikaner(RS) Ckt-1 tripped on R-B phase to phase fault with fault distance of 17.88km and fault current of 13.38kA and 12.73kA in R and B phase respectively from Bikaner(RS) and fault distance of 147.5km and fault current of 2.351kA and 3.572kA in R and B phase respectively from Bhadla(RS); fault was observed in zone-1 at both the ends. ii) As per information received from SLOC Rajasthan, line patrolling was done by representatives of M/s Ramelex Pvt. Ltd. and it was observed that conductor was broken between tower location no. 455-457. iii) As per SCADA, change in NR total solar generation of approx. 2020MW is observed at 13:59hrs. iv) As per PMU at Bhadla(RS), R-B phase to phase fault is observed with fault clearance time of 80 ms at 13:59hrs.	1) 400 kV Bikaner-Bhadla (RS) Ckt-1
10	GI-2	Rajasthan	15-Jan-24 14:06	19-Jan-24 21:50		1760	0	3.539	0.000	49728	60568	i) As reported, at 14:06hrs, 400kV Bhadla(RS)-Bikaner(RS) Ckt-2 tripped on R-Y phase to phase fault with fault distance of 147.1km and fault current of 3.868kA and 4.393kA in R and Y phase respectively from Bikaner(RS) and fault distance of 34.9km and fault current of 11.07kA and 10.64kA in R and Y phase respectively from Bhadla(RS); fault was observed in zone-1 at both the ends. ii) As per information received from SLOC Rajasthan, line patrolling was done by representatives of M/s Ramelex Pvt. Ltd. and it was observed that conductor was broken between tower location no. 120-121. iii) As per DR at Bikaner(RS) end of 400kV Bhadla(RS)-Bikaner(RS) Ckt-2, fault was sensed in zone-1; fault current was 4.55kA and 5.14kA in R and Y phase respectively from Bikaner(RS) end and fault clearing time was "55ms". iv) At the same time, 220/33kV 100MVA ICT-1 at Rising Sun(RSDCL) tripped due to relay mal-operation (exact reason yet to be shared). v) As per SCADA, change in NR total solar generation of approx. 1760MW is observed at 14:06hrs. vi) As per PMU at Bhadla(PG), R-Y phase to phase fault is observed with fault clearance time of 80 ms at 14:06hrs.	1) 400 kV Bikaner-Bhadla (RS) Ckt-2 2) 220/33kV 100MVA ICT-1 at Rising Sun(RSDCL)
11	GI-1	Punjab	23-Jan-24 11:38	23-Jan-24 13:45		0	440	0.000	0.683	48653	64418	i) As reported, at 11:38hrs, LBB protection of Bus Coupler-1 CB operated at 220kV Jamalpur(BBMB). On inspection at switch yard, kite thread was found near 220kV Bus Coupler-1 (A-7) and Bus Coupler-2 (A-17). ii) Due to LBB protection operation, all the elements connected to 220kV Bus-1 at Jamalpur(BBMB) tripped and 220kV Bus-1 at Jamalpur(BBMB) became dead (Bus-wise arrangement of elements yet to be shared). iii) As per SCADA SOE, main CBs of all the elements connected to 400kV Bus-1 at Amritsar(PG) opened at the same time and 400kV Bus-1 at Amritsar(PG) became dead. However, lines remained connected through the CBs. Hence it is suspected that bus bar protection operated at 400kV Bus-1 at Amritsar(PG) (Exact reason of the same yet to be shared). iv) As per PMU at Bhakra(BB), no fault is observed in the system. v) As per SCADA, load loss of approx. 440 MW is observed in Punjab control area.	1) 220kV Bus-1 at Jamalpur(BB) 2) 220 kV Jalandhar-Jamalpur (BB) Ckt-1 3) 220 kV Bhakra_R-Jamalpur (BB) Ckt-1 4) 220 kV Gangwal-Jamalpur (BB) Ckt-1 5) 220 kV Gangwal-Jamalpur (BB) Ckt-2 6) 220 kV Jamalpur(BB)-Dandhari Kalan(PG) (PSTCL) Ckt-2 7) 220/66kV 100 MVA ICT-1 at Jamalpur(BBMB) 8) 220/66kV 160 MVA ICT-3 at Jamalpur(BBMB) 9) 220/132kV 100 MVA ICT-1 at Jamalpur(BBMB)

20.Details of Grid Events during the Month of January 2024 in Northern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (H:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
12	GD-1	Rajasthan	24-Jan-24 10:44	24-Jan-24 12:17		260	0	0.564	0.000	46099	61089	i) During antecedent condition, total generation of 220kV AHEJ3L was evacuating through 220 KV Fatehgarh_II(PG)-AHEJ3L PSS HB_FGRAH_PG (AHEJ3L) (AHEJ3L) Ckt which was carrying approx. 260 MW. ii) As reported, at 10:44 hrs, 220 KV Fatehgarh_II(PG)-AHEJ3L PSS HB_FGRAH_PG (AHEJ3L) (AHEJ3L) Ckt tripped on Y-N phase to earth fault, fault occurred due to snapping of jumper. iii) As per PMU plots of voltage at Fatehgarh2 end of Fatehgarh-Fatehgarh II ckt-1, Y-N phase to earth fault is observed which cleared within 100msec. Voltage dropped to "0.96pu during the fault. iv) Due to tripping of 220 KV Fatehgarh_II(PG)-AHEJ3L PSS HB_FGRAH_PG (AHEJ3L) (AHEJ3L) Ckt, RE generation of AHEJ3L affected. As per SCADA, total reduction in NR RE generation of approx. 260MW is observed. v) As per PMU at 220kV AHEJ3L, MW generation loss of approx. 260MW is observed at AHEJ3L.	1) 220 KV Fatehgarh_II(PG)-AHEJ3L PSS HB_FGRAH_PG (AHEJ3L) (AHEJ3L) Ckt-1
13	GI-2	Rajasthan	24-Jan-24 12:16	27-Jan-24 19:49		1225	0	2.300	0.000	53261	64337	i) As reported, at 12:16hrs, 400kV Bhadla(RS) Bikaner(RS) Ckt-1 tripped on Y-B phase to phase fault with fault distance of 70.45 km and fault current of 6.947 kA and 6.167 kA in Y and B phase respectively from Bhadla(RS) end and fault distance of 103.6 km and fault current of 4.568 kA and 5.264 kA in Y and B phase respectively from Bikaner(RS) end. ii) As per information received from SLDC Rajasthan, line patrolling was done by representatives of M/s Ramelex Pvt. Ltd. and it was observed that conductor was broken between tower location no. 242-243. iii) As per SCADA, change in NR total solar generation of approx. 1225MW is observed out of which approx. 1090MW is recovered within 3 minutes. iv) As per PMU at Bhadla(PG), Y-B phase to phase fault is observed with fault clearance time of 120 ms.	1) 400kV Bhadla(RS) Bikaner(RS) Ckt-1
14	GI-2	Rajasthan	28-Jan-24 16:32	28-Jan-24 18:44		0	160	0.000	0.311	39146	51415	i) As reported, at 16:32hrs, 220kV side of 400/220 KV 315 MVA ICT 1 at Ajmer(RS) tripped on operation of 86.2 relay due to inter trip cable fault. ii) As per scheme "SPS for Transformers at 400kV Ajmer (RVNP) substation", if any of the two 400/220kV 315 MVA ICT trips on fault/protection then tripping command will be extended from B6(Master trip) of that ICT to following feeders: 220kV Ajmer-Bewar (RS) Ckt, 220kV Ajmer-Kishangarh (RS) Ckt, 220kV Ajmer-Bherunda (RS) Ckt-1 & 2. Hence, the above said lines tripped on SPS operation due to tripping of ICT-1. iii) Further as reported, as remedial action taken, the faulty cable was already removed. iv) As per PMU at Bhadla(PG), no fault is observed in the system. v) As per SCADA, load loss of approx. 160MW is observed in Rajasthan control area.	1) 400/220 KV 315 MVA ICT 1 at Ajmer(RS) 2) 220kV Ajmer-Bewar (RS) Ckt 3) 220kV Ajmer-Kishangarh (RS) Ckt 4) 220kV Ajmer-Bherunda (RS) Ckt-1 5) 220kV Ajmer-Bherunda (RS) Ckt-2
15	GI-2	Rajasthan	28-Jan-24 14:58	28-Jan-24 16:24		0	540	0.000	1.014	41369	53271	i) As reported, at 14:58hrs, 220kV Isolator (488B) B-phase jumper of 220kV Bus Coupler-1 broke and the fault reflected on the 220kV bus bar at Ratangarh(RS). ii) Due to this fault, 400/220 KV 315 MVA ICT 1 at Ratangarh(RS), 220 KV Ratangarh(RS) Silar(PG) (PG) Ckt-1, 220 KV Ratangarh(RS) Sri Dungargarh (RS) Ckt, 220kV Ratangarh-Ratangarh220 (RS) Ckt-1 & 2 tripped (Bus-wise arrangement of elements yet to be shared). iii) As per SCADA SDE, 220kV Ratangarh220-Jhunjhunu (RS) Ckt also tripped during the same time. (Exact reason yet to be shared). iv) As per PMU at Silar(PG), Y-N phase to earth fault is observed with delayed fault clearance time of 280 ms (Phase sequence issue is observed). v) As per SCADA, load loss of approx. 540MW is observed in Rajasthan control area. vi) Further as reported, broken 220kV Isolator (488B) B-phase jumper of 220kV Bus Coupler-1 was already replaced.	1) 400/220 KV 315 MVA ICT 1 at Ratangarh(RS) 2) 220 KV Ratangarh(RS) Silar(PG) (PG) Ckt-1 3) 220 KV Ratangarh(RS) Sri Dungargarh (RS) Ckt 4) 220kV Ratangarh-Ratangarh220 (RS) Ckt-1 5) 220kV Ratangarh-Ratangarh220 (RS) Ckt-2
16	GD-1	Rajasthan	30-Jan-24 07:13	30-Jan-24 07:49		77	0	0.167	0.000	46099	61089	i) Generation of 220kV AHEJ4L PSS IV RE stations evacuates through 220 KV Adani RenewPark_SL_FGRAH_FBT1 (AREPR1)-AHEJ4L PSS 4 HB_FGRAH_FBT1 (AHEJ4L) (AREPR1) Ckt. During antecedent condition, AHEJ4L PSS IV RE station was generating approx. 77MW. ii) As reported, at 07:13hrs, 220 KV Adani RenewPark_SL_FGRAH_FBT1 (AREPR1)-AHEJ4L PSS 4 HB_FGRAH_FBT1 (AHEJ4L) (AREPR1) Ckt tripped on B-N phase to earth fault, fault cleared by distance protection in zone-1 (10km ~45%). As per DR, successful A/R operation is observed at AFSPS end and no A/R operated at AHEJ4L end. iii) As per PMU at Fatehgarh2(PG) end, R-N phase to earth fault is observed. As per DR at AFSPS end, R-N fault is observed. Phase sequence mapping issue either at Fatehgarh2 end or at AFSPS end is suspected. iv) As per PMU data, due to tripping of 220kV AHEJ4L PSS IV line, RE (wind) generation (17MW) of the RE station lost due to loss of evacuation path. v) As per PMU plots of voltage at Fatehgarh2 end of Fatehgarh-Fatehgarh II ckt-1, phase to earth fault cleared within 100msec is observed. Voltage dropped to "0.97pu during the fault.	1) 220 KV Adani RenewPark_SL_FGRAH_FBT1 (AREPR1)-AHEJ4L PSS 4 HB_FGRAH_FBT1 (AHEJ4L) (AREPR1) Ckt-1
17	GI-2	Rajasthan	30-Jan-24 08:15	30-Jan-24 08:50		0	700	0.000	1.192	46655	58711	i) 400/220kV Chittorgarh(RS) has double main single breaker bus scheme at 220kV side and one and half bus scheme at 400kV side. There are three (03) 315MVA ICTs at Chittorgarh and five 220kV feeders: Sawa-I & II, Pratapggarh, Nimbahera, Chittorgarh and Debari. 220/315kV (Debari) is having two (02) source: one is from 400/220kV Chittorgarh and another from 220kV Amberi. ii) During antecedent condition, 315MVA ICT-1, 220kV feeders to Debari, Pratapggarh were connected to 220kV Bus-1 and 315MVA ICT-2&3, 220kV feeders to Sawa-I&II, Nimbahera were connected to 220kV Bus-2. 400/220kV 315MVA ICT-1, 2 & 3 were carrying approx. 232MW, 228MW & 244MW respectively. iii) As reported, at 08:15:22hrs, 220kV Chittorgarh-Sawa ckt-2 tripped on B-N phase to earth fault. Fault occurred due to snapping of B-ph jumper at tower location no. 74 at distance approx. 26km from Chittorgarh end. iv) As per PMU, B-N phase to earth fault with delayed clearance of 280msec is observed. As per DR of 220kV Chittorgarh-Sawa ckt-2 of Chittorgarh end, B-N fault in 2-3 picked up by distance protection. Further after ~200msec, 2-4 picked up and tripping initiated to B-ph pole. However, simultaneously A/R blocked and all three phase tripped. (Reason of A/R block and 3-ph tripping yet to be shared by Rajasthan) v) At the same time, 400/220kV 315MVA ICT-2 at Chittorgarh and bus coupler at 220kV side also tripped. As informed, ICT-2 tripped with LBB re-trip flag and bus coupler tripped on over current protection. (Exact reason of tripping of ICT-2 and bus coupler along with its DR not received). vi) With the tripping of bus coupler CB, 220kV Bus-1&2 got split. 315MVA along with 220kV feeders to Pratapggarh & Debari at 220kV Bus-1 and 315MVA ICT-3 along with 220kV feeders to Nimbahera, Sawa-1 at 220kV Bus-2. vii) Load of ICT-2 also shifted on ICT-3. ICT-3 also got overloaded due to shifting of complete load of Pratapggarh on ICT-3. As per DR of ICT-3, its loading increased to ~325MVA. viii) At 08:15:33:410hrs (as per DR), 315MVA ICT-3 also tripped on over current protection operation. Loading recorded in DR just before the tripping was ~400MW. ix) After tripping of ICT-3, load of Nimbahera not started drawing power from Pratapggarh which led to further overloading of 315MVA ICT-1 and 220kV Chittorgarh-Pratapggarh feeder. As per DR of ICT-1, loading of ICT-1 increased from 325MVA to 495MVA within span of 250msec. x) At 08:15:33:790hrs, 220kV Chittorgarh-Pratapggarh ckt tripped on over current protection from Pratapggarh end. With the tripping of Pratapggarh feeder, loading of ICT-1 relieved. xi) As per SCADA, total change in demand of approx. 700MW observed at 08:15hrs.	1) 220kV Chittorgarh-Sawa ckt-2 2) 400/220 KV 315 MVA ICT 2 at Chittorgarh(RS) 3) 400/220 KV 315 MVA ICT 3 at Chittorgarh(RS) 4) 220kV Chittorgarh-Pratapggarh ckt
18	GD-1	Rajasthan	30-Jan-24 09:00	30-Jan-24 09:54		0	450	0.000	0.764	45401	58932	i) 400/220kV Chittorgarh(RS) has double main single breaker bus scheme at 220kV side and one and half bus scheme at 400kV side. There are three (03) 315MVA ICTs at Chittorgarh S/s and five (05) 220kV feeders: Sawa-I & II, Pratapggarh, Nimbahera, Chittorgarh and Debari. 220/315kV (Debari) is having two (02) source: one is from 400/220kV Chittorgarh and another from 220kV Amberi. ii) As reported, at 08:50hrs, 220kV Chittorgarh-Pratapggarh ckt was charged and loading of ICT-1 at Chittorgarh again started increasing. ICT-1 was feeding load of Pratapggarh, Nimbahera via Pratapggarh and Debari. iii) At 08:58:45:770hrs, over current protection in B-ph of ICT-1 started. And at 09:00hrs, 220kV Amberi-Debari ckt also tripped. iv) With the tripping of 220kV Amberi-Debari ckt, total load of Debari shifted to Chittorgarh ICT-1. v) At 09:00:21:660, 315MVA ICT-1 at Chittorgarh also tripped on overcurrent protection operation. As per DR, loading recorded was ~365MVA. vi) As per SCADA, change in demand of approx. 450MW observed at 09:00hrs.	1) 220kV Amberi-Debari ckt 2) 400/220 KV 315 MVA ICT 1 at Chittorgarh(RS)
19	GD-1	Delhi	31-Jan-24 03:17	31-Jan-24 04:54		261	350	0.873	0.950	29888	36823	i) During antecedent condition, 400kV interconnectors 41952 and 42352 between 400kV Buses at Bawana(DTL) and 400kV Buses at Bawana CCGT(DTL) were in closed position. 400/220kV 315MVA ICT-1, 4, 5 & 6 were connected to Bawana CCGT(DTL) and 315 MVA ICT-2 & 3 were connected to Bawana(DTL). 216 MW Bawana GPS - UNIT 4 (GT-4) and 253 MW Bawana GPS - UNIT 6 (STG-2) were generating approx. 165MW and 96MW respectively. ii) As reported, at 03:17 hrs, 400 KV Bawana-Munda (DV) Ckt-1 & 2 tripped from Munda end only on R-N phase to ground fault with fault distance of 22.42 km and 24.09 km respectively from Bawana end. iii) At the same time, 400 KV Bawana CCGT(BTL) Bahadurgarh(PG) (PG) Ckt and 400 KV Abdullapur(PG)-Bawana(DV) (PG) Ckt tripped from both the ends on distance protection operation. iv) Rest of the 400kV ckt connected at Bawana CCGT(DTL) and Bawana(DTL) tripped from the remote end only. (Exact reason, nature and location of fault yet to be shared). v) As communicated telephonically by Delhi, current through interconnectors was towards Bawana CCGT(BTL), hence it is suspected that fault was on Bawana CCGT(BTL) bus side and bus bar protection did not operate, hence line tripped from remote end only. (Reason of non-operation of bus bar protection yet to be shared). vi) Due to tripping of all 400kV ckt, both the 400kV buses became dead at Bawana(DTL) and Bawana CCGT(BTL) and blackout occurred at 400/220/66kV Bawana(DTL) & 400kV CCGT Bawana (DTL) S/s. vii) As reported by SLDC Delhi, the load of 220kV Shalimar Bagh, SEIN, DSODC Bawana and Rishini & B. Bawana and Kharjwalda S/s got affected. viii) As per SCADA, no change in demand is observed in Delhi control area, but as reported by SLDC Delhi, load loss of approx. 350MW is observed out of which approx. 250MW was normalized within 30 minutes and remaining load was changeover to alternate sources. ix) As per PMU, R-N phase to ground fault with delayed fault clearing time of 560ms is observed. x) Further as reported, 216 MW Bawana GPS - UNIT 4 (GT-4) and 253 MW Bawana GPS - UNIT 6 (STG-2) tripped during the same time and generation loss of approx. 261MW is observed.	1) 400 KV Bawana-Munda (DV) Ckt-1 2) 400 KV Bawana-Munda (DV) Ckt-2 3) 400 KV Maharanibagh(PG)-Bawana(DV) (DTL) Ckt-1 4) 400 KV Maharanibagh(PG)-Bawana(DV) (DTL) Ckt-2 5) 400 KV Abdullapur(PG)-Bawana(DV) (PG) Ckt 6) 400 KV Deshpur(HWT)-Bawana(DV) (PG) Ckt 7) 400 KV Bawana CCGT(DTL)-Bhawani(PG) (PG) Ckt 8) 400 KV Bawana CCGT(DTL)-Bahadurgarh(PG) (PG) Ckt 9) 216 MW Bawana GPS - UNIT 4 (GT-4) 10) 253 MW Bawana GPS - UNIT 6 (STG-2)

20.Details of Grid Events during the Month of January 2024 in Northern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (H:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t. Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
20	GD-1	Delhi	31-Jan-24 21:22	31-Jan-24 22:37		0	160	0.000	0.323	34570	49549	i) During antecedent condition, 220 KV Ballabhgarh(BB) BTPS(DTL) (BB) Ckt-1 & 2 and 220 KV Tughlakabad(TG) BTPS(DTL) Ckt-1 & 2 were catering the part load of 220KV Okhla and 220KV Sarita Vihar through 220 KV BTPS(DTL)-Okhla Ckt-1 & 2 and 220 KV BTPS(DTL)-Sarita Vihar Ckt-1 & 2. 220 KV BTPS(DTL)-Alwar Ckt and 220 KV BTPS(DTL)-Noida Sec 38 Ckt were on no-load. 220KV bus coupler at BTPS was in closed condition. ii) As reported, at 21:22 Hrs, 220 KV Ballabhgarh(BB) BTPS(DTL) (BB) Ckt-1 tripped on Y-N phase to ground fault with fault distance of 17.01 km and fault current of 4.2 kA from Ballabhgarh end; zone-1 distance protection operated at Ballabhgarh end and zone-4 distance protection operated at BTPS end. On inspection, 220KV Bus-2 PT isolator Y-ph LA jumper was found broken at BTPS S/s. iii) At the same time, all other 220KV ckt connected at BTPS(DTL) tripped on zone-4 distance protection operation at BTPS end (reason of non-operation of bus bar protection yet to be shared). iv) Due to tripping of all 220KV ckt, both the 220KV buses became dead at BTPS(DTL) and blackout occurred at 220KV BTPS(DTL) S/s. v) As reported by SLDC Delhi, the load of Okhla, Badli, Malviya Nagar, Shivaji, Simfort, DC Saket, Select City mall, Sarita Vihar, Meethapur, Jamia, Sarai Julana, Jasola got affected. vi) As per SCADA, change in demand of approx. 220MW is observed in Delhi control area out of which approx. 90MW is restored within 10 minutes. But as reported by SLDC Delhi, load loss of approx. 160MW is observed. vii) As per PMU, Y-N phase to ground fault with delayed fault clearing time of 160ms is observed. viii) Further as reported, at 21:29 hrs, load of Okhla was normalized through 220KV Tughlakabad Okhla Ckt-1 & 2. At 21:45 hrs, load of 220KV Sarita Vihar was attempted to normalize through 220KV Maharanibagh-Sarita Vihar ckt at Sarita Vihar, but line could not hold and a blast occurred in 220KV Bus coupler CB at Sarita Vihar; R-ph pole was found damaged. Later at 23:20 hrs, load of 220KV Sarita Vihar was normalized through 220KV Maharanibagh-Sarita Vihar ckt.	1) 220 KV Ballabhgarh(BB) BTPS(DTL) (BB) Ckt-1 2) 220 KV Ballabhgarh(BB) BTPS(DTL) (BB) Ckt-2 3) 220 KV Tughlakabad(TG) BTPS(DTL) Ckt-1 4) 220 KV Tughlakabad(TG) BTPS(DTL) Ckt-2 5) 220 KV BTPS(DTL)-Okhla Ckt-1 6) 220 KV BTPS(DTL)-Okhla Ckt-2 7) 220 KV BTPS(DTL)-Sarita Vihar Ckt-1 8) 220 KV BTPS(DTL)-Sarita Vihar Ckt-2 9) 220 KV BTPS(DTL)-Alwar Ckt 10) 220 KV BTPS(DTL)-Noida Sec 38 Ckt
21	GD-1	Uttarakhand	31-Jan-24 18:13	31-Jan-24 19:19		0	173	0.000	0.306	44228	56581	i) As reported, at 18:13 Hrs, 220 KV Pithoragarh-Jauljivi (PG) Ckt-1 tripped on R-N phase to ground fault with fault distance of 8.2 km and fault current of 1.5kA from Pithoragarh end. ii) Further as reported, at the same time, 220 KV Pithoragarh-Jauljivi (PG) Ckt-2 also tripped on Y-N phase to ground fault with fault distance of 9.7 km and fault current of 2.27kA from Pithoragarh end. iii) Due to tripping of 220 KV Pithoragarh-Jauljivi (PG) Ckt-1 & 2, supply to 220/132 KV 100 MVA ICT 1 & 2 at Pithoragarh(PG) and all 132KV ckt were lost, resulting in tripping of all the elements connected at both 220KV and 132KV level at Pithoragarh(PG) which led to 220/132KV Pithoragarh(PG) S/s blackout. iv) As per PMU, 3-phase to ground fault with fault clearing time of 80ms is observed. v) As per SCADA, load loss of approx. 173MW is observed in Uttarakhand control area. vi) As reported by DSCA, the load of Pithoragarh (~50MW), Almora (~53MW), Bhawan (~40MW), Baribhat (~20MW) and Bageshwar (~10MW) S/s got affected.	1) 220 KV Pithoragarh-Jauljivi (PG) Ckt-1 2) 220 KV Pithoragarh-Jauljivi (PG) Ckt-2 3) 220/132 kv 100 MVA ICT 1 at Pithoragarh(PG) 4) 220/132 kv 100 MVA ICT 2 at Pithoragarh(PG) 5) 132 KV Pithoragarh(PG)-Pithoragarh(PTCU) (PTCU) Ckt 6) 132 KV Lohaghat(PTCU)-Pithoragarh(PG) (PTCU) Ckt 7) 132 KV Pithoragarh(PTCU)-Almora(PTCU) (PTCU) Ckt 8) 25 MVAR Bus Reactor No 1 at 220 KV Pithoragarh(PG)

20.Details of Grid Events during the Month of January 2024 in Western Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI for GI-2/GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GI-1	WR	23:25 / 03-01-2024	01:26 / 04-01-2024	02:01	-	180	-	0.33%	64045	54071	At 23:25 Hrs/03.01.2024, Y phase conductor of 220 kV Ponda-Bus-2 snapped. At 220 kV Ponda substation, Bus coupler was in open condition for load management. The 220kV Bus-1 was fed by 220kV Mapusa-Ponda 1 and 220kV Amosa-Ponda 1 line. As the busbar protection failed to operate at Ponda 5/5, two lines connected to 220 kV Ponda-Bus-2 tripped from Mapusa & Amosa remote ends. 220 kV Mapusa-Ponda tripped from Mapusa end only on Zone-2 DFR operation. 220 kV Amosa-Ponda tripped from Amosa end only on Earth fault relay operation. Due to 220kV Bus-II failure at Ponda 5/5, the other elements connected to the Bus i.e. 220/110kV, 100MVA ICT-2, 220/110kV, 100MVA ICT-3, 220/33kV, 50MVA ICT and 220kV Ponda-Kaldem (PWR) line tripped. 180 MW load loss occurred due to the event.	Tripping of following elements- 1. 220 kV Amosa-Ponda 2. 220 kV Mapusa-Ponda 3. 220/110 kV Ponda-ICT 2&3 (100 MVA) 4. 220/33 kV Ponda-ICT 1 (50 MVA) 5. 220 kV Ponda-PWR
2	GD-1	WR	18:29 / 04-01-2024	20:53 / 06-01-2024	50:24	57	-	0.08%	-	75329	64691	At 18:29 Hrs/04.01.2024, 220 kV Bhuj-Gadhisa ckt tripped on Bph-E fault alongwith both ICTs. As informed by Gadhisa, 57 MW generation loss occurred during the event due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Bhuj-Gadhisa ckt 2. 220/33 kV Gadhisa-ICT-1&2
3	GD-1	WR	22:36 / 06-01-2024	23:39 / 06-01-2024	01:03	310	-	0.46%	-	66756	52747	At 22:36 Hrs/04.01.2024, due to fire in R-ph wave trap of 220kV Bhuj-Bharanda line at Bhuj end, 220kV Bhuj-Gadhisa line tripped on TEF protection operation at Gadhisa end only and 220 kV Bhuj-Baranda line tripped at Bhuj end only. Generation loss of about 310 MW at Gadhisa (Renew Power) and Baranda (ASPL) wind power plant occurred due to the event.	Tripping of following elements- 1. 220 kV Bhuj-Gadhisa ckt 2. 220 kV Bhuj-Baranda ckt
4	GD-1	WR	06:58 / 09-01-2024	19:17 / 09-01-2024	12:19	240	0	0.32%	-	73939	596917	At 06:58 Hrs/09.01.2024, 220 kV Bhuj-Gadhisa ckt tripped on Bph-E fault. As informed by Gadhisa, 240 MW generation loss occurred during the event due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Bhuj-Gadhisa line
5	GD-1	WR	14:42 / 10-01-2024	01:44 / 11-01-2024	11:02	85.6	0	0.11%	-	76101	65309	At 14:42 Hrs/10.01.2024, 220 kV Bhuj-Vadva line tripped due to Y-B fault. Generation loss of 85.6 MW occurred due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Bhuj-Vadva line
6	GD-1	WR	13:01 / 10-01-2024	16:34 / 10-01-2024	03:33	42.36	0	0.05%	-	78501	66133	At 13:01 Hrs/10.01.2024, 220 kV Rajpur - Sherisha line tripped due to 33 kV Sherisha solar line insulator failure. Generation loss of 42.36 MW occurred due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Rajpur - Sherisha line
7	GD-1	WR	06:52 / 10-01-2024	08:04 / 10-01-2024	01:12	280	-	0.39%	-	72488	58803	At 06:52 Hrs/10.01.2024, 220 kV/33 kV Ostro ICT-1 tripped due to Rph LA jumper opened at 33 kV end due to high wind and 220 kV/33 kV Ostro ICT-2 also tripped due to overload at same time. As per SCADA Generation loss of 280 MW observed.	Tripping of following elements- 1. 220 kV/33 kV ICT-1&2 at Ostro
8	GD-1	WR	17:11 / 11-01-2024	00:20 / 12-01-2024	07:09	25	0	0.03%	-	74283	65055	At 17:11 hrs/11.01.2024, 220 kV Bhuj-Vadva ckt tripped on Y-B fault. As per SCADA, around 25 MW generation loss occurred due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Bhuj-Vadva line
9	GD-1	WR	08:13 / 12-01-2024	09:25 / 12-01-2024	01:12	-	145	-	0.22%	76995	66977	At 08:13 Hrs/12.01.2024, fire occurred in 220kV B-phase CT on 220kV Bhilad-DFCC-1 line at DFCC Daheli End & the line tripped on Zone-2 Y-B Fault and 220kV Bhilad SS become dead due to tripping all connected 220 kV Lines from remote end or Zone-4 operation. 220kV Bhilad-Vapi (PG)-1, 2 & 3 tripped on Zone-4 Y&B Phase from Bhilad end (as informed by Bhilad SS) and Zone-2 B-H fault from Vapi(PG) end (as informed by RTAMC Vadodra) 220 kV Bhilad-Tarapur line tripped from Tarapur end on back up Over current relay 220 kV Bhilad -Atul tripped from opposite end zone-2 protection.	Tripping of following elements- 1. 220kV Bhilad-Vapi(PG)-1 2. 220kV Bhilad-Vapi(PG)-2 3. 220kV Bhilad-Vapi(PG)-3 4. 220kV Bhilad-Vapi 5. 220kV Bhilad-Tarapur 6. 220kV Bhilad-Atul 7. 220kV Bhilad-DFCC-1 & 2 8. 100MVA, 220/66kV ICT-1, 2 & 3 9. 160MVA, 220/66kV ICT-4
10	GD-1	WR	13:24 / 12-01-2024	09:30 / 13-01-2024	20:06	0	-	-	-	73975	67402	At 13:24 Hrs/12.01.2024, 220kV Baranda Bhuj ckt tripped on B-E fault, Bhuj end: 52.8 km, 3.97 kA. 220kV Baranda SS become dead due to tripping of single connected 220 kV ckt to Baranda end.	Tripping of following elements- 1. 220kV Baranda Bhuj ckt
11	GD-1	WR	12:49 / 12-01-2024	13:19 / 12-01-2024	00:30	0	-	-	-	76706	69182	At 12:49 Hrs/12.01.2024, 220 kV/33 kV Ostro ICT-1&2 tripped due to LA failure and conductor snapping in 33 kV feeder. No generation loss due to the event.	Tripping of following elements- 1. 220/33 kV ICT-1&2 at Ostro
12	GD-1	WR	19:19 / 13-01-2024	18:55 / 14-01-2024	23:36	0	-	-	-	73870	62979	At 19:19 Hrs/13.01.2024, 220kV Vadva Bhuj ckt tripped on Yph-Bph fault, Bhuj end: 57.2 km, I _{ph} =4.25 kA, I _{br} =4.21 kA. 220kV Vadva SS become dead due to tripping of single connected 220 kV ckt from Bhuj end	Tripping of following elements- 1. 220 kV Bhuj-Vadva line
13	GD-1	WR	11:30 / 14-01-2024	19:56 / 14-01-2024	08:26	22.51	-	0.03%	-	72277	65637	At 11:30 Hrs/14.01.2024, 220kV Baranda Bhuj ckt tripped on Bph-E fault, Baranda end: 19.7 km, 0.80kA. 220kV Baranda SS become dead due to tripping of single connected 220 kV ckt from Bhuj end to Baranda	Tripping of following elements- 1. 220kV Baranda Bhuj ckt
14	GI-1	WR	16:54 / 15-01-2024	17:15 / 15-01-2024	00:21	-	115	-	0.18%	71639	63592	At 16:54 Hrs/15.01.2024, 220 kV Nashik-Navsari(GI)-1 tripped on Zone-2, R-E fault. As informed by SLDC Maharashtra, kite thread found in location number 02 during line patrolling. Although the fault was cleared, LBB operated and all elements at 220 kV Navsari(GI) except 220 kV Navsari(GI)-Navsari(PG)-1&2 tripped. Load loss of 115 MW occurred due to the event.	Tripping of following elements- 1. 220 kV Nashik-Navsari(GI) 2. 220 kV Sachin-Navsari(GI) 3. 220 kV Pimpri-Navsari(GI) 4. 220 kV Chikali-Navsari(GI) 5. 220 kV Atul-Navsari(GI) 6. 220/66 kV Navsari-ICT-1, 2, 3, 4, 5, 6&7
15	GD-1	WR	23:14 / 15-01-2024	00:16 / 16-01-2024	01:02	-	-	-	-	61547	53934	At 23:14 Hrs/15.01.2024, 220 kV Indore-Pritamnagar tripped on R-E fault. No Generation was present at Pritamnagar (AWEMP2PL) Wind Power Plant.	Tripping of following elements- 1. 220 kV Indore(PG) Pritamnagar(AWEMP2PL)
16	GI-2	WR	18:04 / 22-01-2024	18:24 / 22-01-2024	00:20	215	-	0.30%	-	72276	60758	At 18:04 Hrs/22.01.2024, 400 kV RGPL Bus 1 and all the connected main bays tripped on Bus bar protection operation. As seen from the above PMU plot, there was no physical fault in the system and the tripping seems to be a mal-operation. Main and Tie bays of Unit 3A was under open condition due to unit outage and resulted in tripping of 400 kV RGPL - Nagthane 2. Tie bay of Unit 2A was in open condition (Unit 2A synchronized at 15:32 Hrs through main bay but tie bay was not closed by RGPL) and resulted in tripping of RGPL Unit 2A during BB protection operation. As informed by RGPL, 215 MW generation loss occurred due to the event.	Tripping of following elements- 1. 400 kV RGPL Bus 1 2. 400 kV RGPL - Nagthane 2 3. RGPL Unit 2A (332 MW)
17	GD-1	WR	18:22 / 23-01-2024	16:56 / 24-01-2024	22:34	36.37	-	0.05%	-	68650	65698	At 18:22 Hrs/23.01.2024, 220 kV Bhuj-Baranda tripped on B-E fault. Due to loss of evacuation path, 36.37 MW generation loss occurred at 220 kV Baranda (ASPL) WPP	Tripping of following elements- 1. 220 kV Bhuj-Baranda line 2. 220/33 kV ICT-1&2
18	GD-1	WR	13:23 / 25-01-2024	15:35 / 26-01-2024	26:12	0	-	0.00%	-	75627	66835	At 13:23 Hrs/25.01.2024, 220 kV Bhuj-Baranda tripped on B-E fault. Due to loss of evacuation path, 220 kV Baranda (ASPL) WPP went dark. As informed by ASPL, no generation at 220 kV Baranda during the event, hence no generation loss	Tripping of following elements- 1. 220 kV Bhuj-Baranda line 2. 220/33 kV ICT-1&2
19	GD-1	WR	19:28 / 25-01-2024	15:16 / 26-01-2024	19:48	20	-	0.03%	-	73407	63093	At 19:28 Hrs/25.01.2024, 220 kV Bhuj-Naranagar tripped on B-E fault. Due to loss of evacuation path, 220 kV Naranagar WPP went dark. As informed by Naranagar, 20 MW generation loss occurred during the event.	Tripping of following elements- 1. 220 kV Bhuj-Naranagar 2. 220/33 kV ICT-1&2
20	GD-1	WR	04:48 / 26-01-2024	06:26 / 26-01-2024	01:38	137	-	0.21%	-	64388	53326	At 04:48 Hrs/26.01.2024, 220 kV/33 kV Ostro ICT-2 tripped on overcurrent due to high wind. As 220 kV/33 kV Ostro ICT-1 was already under planned shutdown. Therefore Ostro 5/5 got complete dead due to loss of evacuation path. Generation loss of 137 MW reported.	Tripping of following elements- 1. 220/33 kV ICT-2 AT Ostro
21	GI-2	WR	12:28 / 27-01-2024	14:25 / 27-01-2024	01:57	-	-	-	-	77493	73407	At 12:28 Hrs/27.01.2024, 400kV Bus-1 at Parli-I (MH) tripped due to LBB operation due to slipping of steel ring from the pulley and falling in the vicinity of Bus-1 isolator during stringing arrangements for line gangry of Parli (MG) Parli (MH) Ckt-2. It resulted in tripping of 400kV Parli (MH) New Parli-I, 400kV Parli (MH) - Nanded line, 400kV Parli (MH) - Karjat line, 400kV Parli (MH) - Solapur line and 400/220kV ICT-3. 400 kV Parli (MH) bus-1 revived through 400 kV Parli New Parli - Parli MH line at 14:25 Hrs/27.01.2024. No load loss reported.	Tripping of following elements- 1. 400kV Parli (MH) New Parli-I, 2. 400kV Parli (MH) Nanded line, 3. 400kV Parli (MH) - Karjat line-1, 4. 400kV Parli (MH) - Solapur line 5. 400/220kV ICT-3
22	GD-1	WR	15:11 / 30-01-2024	16:42 / 30-01-2024	01:31	4.6	-	0.01%	-	76707	69308	At 15:11 Hrs/30.01.2024, 220/33 kV Kotda Madh ICTs 1&2 tripped on HV side overcurrent protection operation during R&B phase fault in 33 kV feeder no. 30. Generation loss of 4.6 MW occurred due to the event.	Tripping of following elements- 1. 220/33 kV Kotda Madh ICT-1&2

20.Details of Grid Events during the Month of January 2024 in Southern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (H:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Karnataka	13-01-2024 16:14	13-01-2024 16:36	00:22	0	430	0.00%	0.81%	46883	53116	Complete outage of 220kV/110kV Ranebennur SS, 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and Tripping of 220kV Bus-1 of 400kV/220kV Guttur SS and 220kV Bus-1 of 220kV/66kV Honnali SS of KPCL. During the antecedent conditions, 220kV Ranebennur Shiralkoppa was under outage. 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and 220kV Bus-1 of 220kV/66kV Honnali SS were being radially fed through 220kV Bus-1 of 400kV/220kV Guttur SS. The triggering incident is the spurious tripping of bus coupler at 220kV level led to the bus split and subsequent over loading of ICT-1 at Guttur. This led to the SP operation and tripping of 220kV Guttur-Ranebennur feeder leading to the complete outage of 220kV Ranebennur. However, since 220kV Guttur-Ranebennur was connected to Bus-2 which was connected to ICT-2 did not relieve the ICT-1 loading and tripped on HV side OC operation leading to the Bus-1 outage and the radial stations connected to 220kV Bus-1	1. 220kV Guttur Bus Coupler 2. 400/220kV Guttur ICT-1 3. 220kV Guttur-Ranebennur 4. 220kV Guttur Bus-1
2	GD-1	Andhra Pradesh	15-01-2024 02:40	15-01-2024 06:48	04:08	330	0	1.09%	0.00%	30331	36422	Complete Outage of 400kV RYTPP Generating station of APGENCO: At RYTPP end, Generating Unit-6 tripped due to loss of DC supply to furnace supervisory safeguard system. Subsequently, 400kV Kakili RYTPP line-1&2 tripped at 02:58hrs and 02:40hrs respectively on operation of over voltage protection at RYTPP end. Tripping of both lines resulted in complete outage of 400kV RYTPP Generating station.	1. 400kV Kakili RYTPP Line-1&2
3	GD-1	Tamil Nadu	16-01-2024 21:54	17-01-2024 02:45	04:51	0	0	0.00%	0.00%	37007	37798	Complete outage of 230kV Ettayapuram Solar Plant: As per the reports submitted, the triggering incident was R-N fault in 230kV TTGS Ettayapuram Line-1 and the line tripped. Tripping of the only connected line resulted in complete outage of 230kV Ettayapuram Solar Plant.	1. 230kV TTGS Ettayapuram-1
4	GD-1	Andhra Pradesh	17-01-2024 08:40	18-01-2024 00:57	16:17	520	0	1.06%	0.00%	49082	52209	Complete Outage of 400kV HNPCL Generating station: As per the reports submitted, 8-phase tension insulator connected to 400kV Bus-1 at Hinduja failed and fell on 400kV Bus-2 causing fault in 400kV Bus-1 and Bus-2. Immediately 400kV BBP operated and all the elements connected to the bus tripped. However, 400kV Bus-1 failed to clear the fault and remote ends of the lines connected to Bus-2 cleared the fault in zone-2 protection. Tripping of all 400kV lines resulted in complete outage of 400kV HNPCL Generating station. At the same time, 400kV Simhadri Kalpaka Line-2 tripped on operation of Over Voltage Stage-2 protection at Simhadri end.	1. 400kV HNPCL Kalpaka Line-1&2 2. 400kV Gudlugudem Hinduja Line-1&2 3. HNPCL Unit-1
5	GD-1	Tamil Nadu	18-01-2024 18:08	18-02-2024 18:20	00:12	0	120	0.00%	0.27%	41548	45199	Complete outage of 230kV Tondiarpet and tripping of Bus-1 at Basin Bridge: As per the reports submitted, the triggering incident is the R-phase CT failure of 230kV Tondiarpet-Basin Bridge at Basin Bridge end. At Basin Bridge end BBP operated and Bus-1 got isolated. However, at Tondiarpet the fault was sensed in Z2 and the breaker failed to open and LBB was sensed. The LBB trip was not extended to all bays leading to Z3 operation at remote ends leading to complete outage of 230kV Tondiarpet SS	1. 230kV Tondiarpet-Basin Bridge 2. 230kV Bus-1 at Basin Bridge 3. 230kV Tondiarpet-NCTPS-1 4. 230kV Tondiarpet-NCTPS-2 5. 230kV Tondiarpet-Pulanthope 6. 230kV Tondiarpet SS
6	GD-1	Karnataka	21-01-2024 09:05	21-01-2024 10:13	01:08	0	336	0.00%	0.60%	49321	54156	Complete outage of 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and Tripping of 220kV Bus-1 of 220kV/66kV Honnali SS of KPCL: During the antecedent conditions, 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and 220kV Bus-1 of 220kV/66kV Honnali SS were being radially fed through 400kV/220kV Guttur SS through 220kV Guttur-Davanagere-1,2 and 3 lines of which line-3 was under forced outage. The triggering incident is the tripping of 220kV Guttur-Davanagere-1&2 leading to the complete outage of 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and Tripping of 220kV Bus-1 of 220kV/66kV Honnali SS of KPCL.	1. 220kV Guttur-Davanagere-1 2. 220kV Guttur-Davanagere-2
7	GD-1	Karnataka	21-01-2024 13:36	21-01-2024 14:41	01:15	0	340	0.00%	0.65%	49478	52578	Complete outage of 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and Tripping of 220kV Bus-1 of 220kV/66kV Honnali SS of KPCL: During the antecedent conditions, 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and 220kV Bus-1 of 220kV/66kV Honnali SS were being radially fed through 400kV/220kV Guttur SS through 220kV Guttur-Davanagere-1,2 and 3 lines of which line-2 was under forced outage. The triggering incident is the tripping of 220kV Guttur-Davanagere-1&3 leading to the complete outage of 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS and Tripping of 220kV Bus-1 of 220kV/66kV Honnali SS of KPCL.	1. 220kV Guttur-Davanagere-1 2. 220kV Guttur-Davanagere-3
8	GD-1	Tamil Nadu	26-01-2024 19:52	27-01-2024 09:50	13:58	0	0	0.00%	0.00%	40707	41337	Complete outage of 765kV Ariyalur and 765kV NCPS of TANTRANSCO: 765kV Ariyalur and 765kV NCPS are being radially fed from 765kV Ariyalur-Thiruvallam-2. The triggering incident is the tripping of 765kV Ariyalur-Thiruvallam-2 at Thiruvallam end on over voltage leading to Complete outage of 765kV Ariyalur and 765kV NCPS of TANTRANSCO	1. 765kV Ariyalur-Thiruvallam-2
9	GD-1	Karnataka	26-01-2024 19:13	26-01-2024 19:01	02:48	62	102	0.12%	0.18%	53806	55344	Complete Outage of 220kV/110kV Ghataprabha SS of KPCL: As per the reports submitted, the triggering incident was 220kV BBP maloperation at 220kV/110kV Ghataprabha SS. Immediately, all the lines connected to the buses tripped. This resulted in complete outage of 220kV/110kV Ghataprabha SS.	1. 220kV Ghataprabha Narendra Line-1&2 2. 220kV Ghataprabha Mughalshahod Line-1&2 3. 220kV/110kV Ghataprabha Transformer-1&2
10	GD-1	Tamil Nadu	29-01-2024 19:58	29-01-2024 23:54	03:56	0	0	0.00%	0.00%	43798	45172	Complete outage of 765kV Ariyalur and 765kV NCPS of TANTRANSCO: 765kV Ariyalur and 765kV NCPS are being radially fed from 765kV Ariyalur-Thiruvallam-2. The triggering incident is the tripping of 765kV Ariyalur-Thiruvallam-2 at Thiruvallam end on over voltage leading to Complete outage of 765kV Ariyalur and 765kV NCPS of TANTRANSCO	1. 765kV Ariyalur-Thiruvallam-2
11	GD-1	Karnataka	28-01-2024 10:13	28-01-2024 13:01	02:48	62	102	0.12%	0.18%	53806	55344	Complete Outage of 220kV/110kV Ghataprabha SS of KPCL: As per the reports submitted, the triggering incident was 220kV BBP maloperation at 220kV/110kV Ghataprabha SS. Immediately, all the lines connected to the buses tripped. This resulted in complete outage of 220kV/110kV Ghataprabha SS.	1. 220kV Ghataprabha Narendra Line-1&2 2. 220kV Ghataprabha Mughalshahod Line-1&2 3. 220kV/110kV Ghataprabha Transformer-1&2
12	GD-1	Karnataka	31-01-2024 14:40	31-01-2024 15:11	00:31	93	1000	0.18%	1.70%	52510	58782	Complete outage of 220kV Kadur, Arasikere, Bettedevarakere, Shiralkoppa and tripping of Bus-2 at Honnali, KB Cross and 400kV Bus-1 and Bus-2 at Talaguppa of KPCL: In the antecedent conditions, 220kV Kadur, Arasikere, Bettedevarakere, Shiralkoppa and Bus-2 at Honnali, KB Cross were radially connected to 220kV Shimoga SS. The triggering incident is the R-N fault in 220kV Shimoga-Arhasanahalli line. After which 400/220kV ICTs at Talaguppa and 220kV Hassan Shimoga loading increased and crossed the pick-up current for 400/220kV Talaguppa ICTs of 450A and tripped in around 80s and subsequently 220kV Shimoga-Hassan tripped at Shimoga end - P&B observed and tripped on Z1. With this tripping an island was formed and under frequency envisaged for Bangalore Islanding scheme operated at Shimoga which led to the complete outage of stations radially connected to Shimoga. However, 220kV Shimoga-Varahi-3 did not trip which led to formation of another smaller island which survived and the island was synchronised to the grid.	1. 220kV Shimoga-Arhasanahalli 2. 220kV Shimoga-Hassan 3. 400/220kV Talaguppa ICT-1 4. 400/220kV Talaguppa ICT-3
13	GI-2	Tamil Nadu	02-01-2024 04:49	02-01-2024 07:25	02:36	0	0	0.00%	0.00%	31480	38447	Tripping of 230kV Bus-2 of 230kV Neyveli TS-2: As per the reports submitted, the triggering incident was R-N fault in 230kV Neyveli TS-II -Ilundurpet line. At Neyveli end, B-pole of the circuit breaker failed to open causing LBB to operate and all the lines connected to the 230kV Bus-2 of 230kV Neyveli TS-II tripped.	1. 400kV/230kV NLC ICT-1
14	GI-1	Andhra Pradesh	02-01-2024 07:16	02-01-2024 10:29	03:13	0	0	0.00%	0.00%	39148	51328	Tripping of 400kV Bus-1 and 2 of 400kV/220kV Nellore SS of APTRANSCO: As per the reports submitted, the triggering incident was BBP Central Unit maloperation at 400kV/220kV Nellore SS. Immediately, all the main breakers connected to Bus-1 and Bus-2 tripped. Since the lines which are in the same dia with the ICTs were under outage and Tie breakers were in open condition, this resulted in tripping of 400kV/220kV Nellore ICT-1&2.	1. 400kV/220kV Nellore ICT-1&2
15	GI-1	Tamil Nadu	10-01-2024 15:24	10-01-2024 19:26	04:02	0	0	0.00%	0.00%	43288	51609	Tripping of 230kV Bus at 230kV/110kV Sembattur SS of TANTRANSCO: As per the reports submitted, the triggering incident was Y phase jumper failure between Breaker and 230 kV Bus isolator in 230kV Checkanurani Sembattur line causing a bus fault at 230kV Sembattur. Immediately, BBP operated and all the elements connected to the bus tripped. 110kV Was intact during this event.	1. 230kV Mywad Sembattur 2. 230kV Sembatti Checkanurani 3. 230kV Sembatti Theni 4. 230kV Sembatti Karakudi 5. 230kV/110kV Sembattur Auto Transformer-1
16	GI-1	Karnataka	11-01-2024 00:49	11-01-2024 02:19	01:30	0	42	0.00%	0.11%	34308	38800	Tripping of 220kV Bus-1 of 220kV/66kV Chintamani SS of KPCL: As per the reports submitted, the triggering incident was 220kV Bus-1 LBB maloperation causing all the lines connected to 220kV Bus-1 to trip.	1. 220kV Kolar Chintamani Line-1 2. 220kV Chintamani Srinivasapura Line-2 3. 220kV/66kV Chintamani Auto Transformer-1
17	GI-1	Tamil Nadu	13-01-2024 09:25	13-01-2024 13:45	04:20	0	0	0.00%	0.00%	50040	54467	Tripping of 230kV Bus at 230kV/110kV Sembattur SS of TANTRANSCO: As per the reports submitted, the triggering incident was Y phase jumper failure between Breaker and 230 kV Bus isolator in 230kV Mywad Sembattur line causing a bus fault at 230kV Sembattur. Immediately, BBP operated and all the elements connected to the bus tripped. 110kV Was intact during this event.	1. 230kV Mywad Sembattur 2. 230kV Sembatti Checkanurani 3. 230kV Sembatti Theni 4. 230kV Sembatti Karakudi 5. 230kV/110kV Sembattur Auto Transformer-1

20.Details of Grid Events during the Month of January 2024 in Eastern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Biharsharif	14.01.202 44:22	14.01.2024 05:06	00:44	0	128	0.00%	0.80%	23916	15853	At 04:04 Hrs on 14.01.2024, 220 KV Biharsharif-Mokama-1 tripped due to R_N fault. While attempting to charge the line at 04:22 Hrs, all emanating lines tripped from remote end and total power interrupted at 220/132 KV Biharsharif S/s. Around 128 MW load loss reported at Biharsharif.	220KV-BIHARSARIFF-MOKAMA-1 220KV-BIHARSARIFF-MOKAMA-2 400KV/220KV 315 MVA ICT 1 AT BIHARSARIFF(PG) 400KV/220KV 315 MVA ICT 2 AT BIHARSARIFF(PG) 400KV/220KV 315 MVA ICT 3 AT BIHARSARIFF(PG) 400KV/220KV 500 MVA ICT 4 AT BIHARSARIFF(PG) 220KV-BIHARSARIFF-TTPS-1 220KV-BIHARSARIFF-Khizersarai-1 220KV-BIHARSARIFF-Khizersarai-2 220KV-BIHARSARIFF-Fatuha-1 220KV-BIHARSARIFF-Fatuha-2 220/132KV 160MVA ICT1 at Biharsharif(BH) 220/132KV 160MVA ICT2 at Biharsharif(BH) 220/132KV 160MVA ICT3 at Biharsharif(BH)
2	GD-1	Kahalgaon	30.01.2024 06:48	30.01.2024 09:03	02:15	473	110	1.69%	0.53%	27872	20477	At 06:47 Hrs, while synchronizing 210 MW U#2 at Kahalgaon, 400 KV B ph CB of its tie bay burst. During this fault, differential protection of 2*400/132 KV 200 MVA ICTs at Kahalgaon operated. But one of the ICTs failed to open and LBB operated leading to tripping of 400 KV Bus#2 (Both ICTs are connected to Bus#2). At the same time, 400 KV Bus#4 also tripped as bus sectionalizer didn't open. 500 MW U#6 tripped due to loss of auxiliary supply as its entire auxiliary load was fed through station transformer at 132 KV (ST supply lost due to tripping of 400/132 KV ICTs) due to unavailability of UAT at that time. Later at 07:12 Hrs/07:13 Hrs, 500 MW U#7 and U#5 tripped due to low vacuum pressure. Several auxiliaries were fed through station transformers, supply of which was lost due to tripping of 400/132 KV ICTs. All three units of Kahalgaon St. 2 tripped.	400KV MAIN BUS - 2 AT KAHALGAON 400KV MAIN BUS - 4 AT KAHALGAON 400KV-DURGAPUR-KAHALGAON-1 400KV-DURGAPUR-KAHALGAON-2 400KV-KHSTPP-BANKA (PG)-2 400KV-KHSTPP-BANKA (PG)-1 400KV-LAKHSARAI-KHSTPP-2 400KV-KHSTPP-LAKHSARAI-1 400KV-KHSTPP-BARH-1 132KV-KHSTPP-SABOUR-1 132KV-KHSTPP-KAHALGAON(BSEB)-1 132KV-KHSTPP-LALMATIA-1 400KV/132KV 200 MVA ICT 1 AT KHSTPP 400KV/132KV 200 MVA ICT 2 AT KHSTPP 132 KV Bus at KHSTPP KHSTPP Units (500MW)

20.Details of Grid Events during the Month of January 2024 in North-East Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GI 1	AGTCCPP	01-01-2024 02:27	02-01-2024 04:00	01:33:00	15	0	1.04%	0.00%	1440	1445	AGTCCPP Unit-1 tripped at 02:27 Hrs on 01.01.2024 & AGTCCPP Unit-2,5 tripped at 03:04 Hrs on 01.01.2024 due to High Air Inlet Differential Pressure. Revision done from Block no. 17 on 01.01.2024	AGTCCPP Unit-1,2,5
2	GI 2	Kopili	02-01-2024 16:59	02-01-2024 17:41	00:42:00	115	0	3.77%	0.00%	3053	2376	Kopili Unit-2,3,4 tripped at 16:59 Hrs on 02.01.2024 due to Bus Bar operation at Kopili Bus-I. Revision done at 19:09 Hrs on 02.01.2024	220 KV Misa-Kopili-II, 220/132 KV ICT-I at Kopili, Kopili Unit-2,3,4 & Bus Coupler at Kopili
3	GI 2	Kopili	02-01-2024 17:50	02-01-2024 18:26	00:36:00	110	0	3.26%	0.00%	3379	2608	Kopili Unit-2,3,4 tripped at 17:50 Hrs on 02.01.2024 due to Bus Bar operation at Kopili Bus-I. Revision done at 19:09 Hrs on 02.01.2024	220 KV Misa-Kopili-II, 220/132 KV ICT-I at Kopili, Kopili Unit-2,3,4 & Bus Coupler at Kopili
4	GI 2	Kopili	05-01-2024 17:05	05-01-2024 20:21	03:16:00	95	0	3.05%	0.00%	3117	2529	Kopili Unit-2,3 tripped at 17:05 Hrs on 05.01.2024 due to Bus Bar operation at Kopili Bus-I. Revision done at 19:52 Hrs on 05.01.2024	220 KV Misa-Kopili-II, 220/132 KV ICT-I at Kopili, Kopili Unit-2,3 & Bus Coupler at Kopili
5	GI 2	Kopili	08-01-2024 18:03	08-01-2024 18:37	00:34:00	100	0	3.23%	0.00%	3097	2641	Kopili Unit-2,3 tripped at 18:03 Hrs on 08.01.2024 due to Bus Bar operation at Kopili Bus-I. Revision done from Block no. 83 on 08.01.2024	220 KV Misa-Kopili-II, 220/132 KV ICT-I at Kopili, Kopili Unit-2,3 & Bus Coupler at Kopili
6	GI 2	AGBPP	14-01-2024 12:41	14-01-2024 14:00	01:19:00	20	0	0.97%	0.00%	2061	1925	AGBPP Unit-6 tripped at 12:41 Hrs on 14.01.2024 due to rotor earth fault. Revision done from Block no. 57 on 14.01.2024	AGBPP Unit-6
7	GI 2	AGBPP	19-01-2024 06:20	19-01-2024 08:00	01:40:00	20	0	1.22%	0.00%	1636	1886	AGBPP Unit-5 tripped at 06:20 Hrs on 19.01.2024 due to restricted earth fault. Revision done from Block no. 33 on 19.01.2024	AGBPP Unit-5
8	GI 2	AGBPP	22-01-2024 10:30	22-01-2024 12:00	01:30:00	3	0	0.16%	0.00%	1826	2037	AGBPP Unit-5 tripped at 10:30 Hrs on 22.01.2024 due to restricted earth fault. Revision done from Block no. 49 on 22.01.2024	AGBPP Unit-5
9	GI 2	AGBPP	27-01-2024 19:26	27-01-2024 21:00	01:34:00	12	0	0.41%	0.00%	2941	2589	AGBPP Unit-8 tripped at 19:26 Hrs on 27.01.2024 due to Low Vacuum pressure. Revision done from Block no. 85 on 27.01.2024	AGBPP Unit-8

21. NEW ELEMENTS COMMISSIONED* IN JANUARY 2024							
COMMISSIONING OF NEW GENERATING UNITS							
REGION	S.NO.	ENTITY	UNIT NAME / LOCATION	ADD. CAPACITY (MW)	UNIT NO.	INS. CAPACITY (MW)	DATE®
NR	1	ONEVOL TENERGY PRIVATE LIMITED	Onevolt Energy Private Limited(Solar)/Rajasthan	100	-	100	03-01-2024
	2	GRIAN ENERGY PRIVATE LIMITED	Grian Energy Private Limited(Solar)/Rajasthan	100	-	100	04-01-2024
	3	AMPLUS AGES PRIVATE LIMITED	Amplus Ages Private Limited(Solar)/Rajasthan	100	-	100	05-01-2024
	4	RENEW SURYA PRATAP PRIVATE LIMITED	Renew Surya Pratap Private Limited(Solar)/Rajasthan	175	-	197	06-01-2024
	5	RENEW SURYA VIHAAN PRIVATE LIMITED	Renew Surya Vihaan Private Limited(Solar)/Rajasthan	98	-	98	06-01-2024
	6	ALTRA XERGI POWER PVT LTD	Altra Xergi Power Pvt Ltd(Solar)/Rajasthan	120.4	-	380	28-01-2024
SR	7	AYANA SIX RENEWABLES	Ayana Six Renewables(Wind)/Gujarat	57.6	-	208.8	03-01-2024
	8	AYANA SIX RENEWABLES	Ayana Six Renewables(Wind)/Gujarat	18	-	226.8	03-01-2024
	9	RSRPL	RSRPL(Wind)	13.86	-	235.62	05-01-2024
	10	AYANA SIX RENEWABLES	Ayana Six renewables(Wind)/Gujarat	75.6	-	302.4	20-01-2024
	11	JSW	JSW Savalperi (Wind)/Tamilnadu	8.1	-	59.4	12-01-2024
ER	12	IBEUL	IBEUL(Thermal)/Odissa	350	II	700	13-01-2024
®RE (Solar & Wind generation added capacity FTC is in phases. Date mentioned is the date by which complete (Added) capacity							
COMMISSIONING OF NEW ICTs							
REGION	S.NO.	ENTITY	SUB-STATION	VOLTAGE (kV)	ICT NO.	CAPACITY (MVA)	DATE
NR	1	RRVPNL	Chittorgarh	400/220/33	III	315	06-01-2024
	2	HPPTCL	AD hydro	220/33	I	31.5	08-01-2024
	3	PSTCL	Dhanansu	400/220/33	I	500	31-01-2024
WR	4	KVTPL	Vikholi	400/220	I	500	31-01-2024
	5	KVTPL	Vikholi	400/220	II	500	31-01-2024
SR	6	POWERGRID	Trichy	400/230	II	500	31-01-2024
	7	TANRANSO	Ariyalur	765/400	II	1500	29-01-2024
	8	KNTL	Koppal	400/220	V	500	23-01-2024
	9	KNTL	Koppal	400/220	IV	500	23-01-2024
	10	JSW RENEW ENERGY	Vilathikulam	230/33	I	210	17-01-2024
	11	RSOPL	Koppal	220/33	II	160	11-01-2024
	12	RSOPL	Koppal	220/33	I	160	11-01-2024
	13	TANTRANSCO	Ariyalur	765/400	II	1500	10-01-2024
	14	POWERGRID	Nizamabad	765/400	III	1500	09-01-2024
	15	TSTRANSCO	Suryapet	400/220	III	500	03-01-2024
NER	16	AEGCL	Kahilipara	132/33	V	50	09-01-2024
COMMISSIONING OF NEW TRANSMISSION LINES							
REGION	S.NO.	ENTITY	LINE NAME	CKT NO.	VOLTAGE (kV)	DATE	
SR	1	TANTRANSCO	765kV-Ariyalur-NCPS	II	765	18-01-2024	
	2	TANTRANSCO	765KV-Ariyalur-NCPS-1	I	765	06-01-2024	
	3	TANTRANSCO	765KV-NCPS-NCTPS STG3	II	765	13-01-2024	
	4	TANTRANSCO	765KV-NCPS-NCTPS STG3	I	765	13-01-2024	
NER	5	STAR CEMENT NORTHEAST LTD	220kV Sonapur - SCNEL	-	220	19-01-2024	
LILO OF TRANSMISSION LINES							
REGION	S.NO.	ENTITY	LINE NAME (LILOed LINE)	LILO AT	VOLTAGE (kV)	DATE	
NR	1	POWERGRID,PTCL	400kV Kurukshetra-Dhanansu-I (400 KV Kurukshetra-Jalandhar)	Dhanansu	400	31-01-2024	
	2	POWERGRID,PTCL	400kV Dhanansu-Jalandhar-I (400 KV Kurukshetra-Jalandhar)	Dhanansu	400	31-01-2024	
SR	3	APTRANSCO	400kV-Krishnapatnam_SEIL_P2-I (Krishnapatnam - Nellore_AP-I)	SEIL_P2_AP section	400	23-01-2024	
	4	APTRANSCO	400kV-Nellore_SEIL_P2-I (Krishnapatnam - Nellore_AP-I)	SEIL_P2_AP section	400	23-01-2024	
ER	5	IBEUL/OGTPL	400kV IBEUL-OPGC-I (400kV OPGC-Jharsuguda -II)	IBEUL(Ind-Barath)	400	10-01-2024	
	6	IBEUL/OGTPL	400kV IBEUL-OPGC-I (400kV IBEUL-Jharsuguda-I)	IBEUL(Ind-Barath)	400	09-01-2024	
COMMISSIONING OF BUS/LINE REACTORS							
REGION	S.NO.	ENTITY	SUB-STATION / LINE (@SUB-STATION)	VOLTAGE (kV)	CAPACITY (MVAR)	DATE	
WR	1	KBTL	Bus Reactor-I@Khavda PSI SectionI	400	125	05-01-2024	
	2	TANTRANSCO	Ariyalur-NCPS-II@Ariyalur	765	240	20-01-2024	
SR	3	TANTRANSCO	Ariyalur-NCPS_II@NCPS	765	240	18-01-2024	
	4	TANTRANSCO	Ariyalur-NCPS-I @ Ariyalur	765	240	10-01-2024	
	5	TANTRANSCO	Ariyalur-NCPS_I@NCPS	765	240	06-01-2024	
	6	TANTRANSCO	Thiruvalam-Ariyalur_I @ Ariyalur	765	240	06-01-2024	
	7	TSTRANSCO	Bus Reactor-I @Kamalapuram	400	125	18-01-2024	
	8	TANTRANSCO	Bus Reactor-I @NCTPS STG III	765	240	11-01-2024	
ER	9	WBSETCL	Bus Reactor-II@Kharagpur	400	125	25-01-2024	
COMMISSIONING OF HVDC/FACTS DEVICE/SYSTEM							
REGION	S.NO.	NAME	SUB-STATION / CONSTITUENT	RATINGS	VOLTAGE (kV)	DATE	
*Includes first time charging (FTC) as well							

22.Voltage Profile

विद्युत विभव रूपरेखा माह - जनवरी 2024

VOLTAGE PROFILE - JANUARY 2024

Date of reporting: 05-Feb-24

क्र.स. Sl. No.	क्षेत्र REGION	उपकेंद्र SUBSTATION	VOLTAGE < V(lower)* (V=380,728 kV)	V(lower) < VOLTAGE < V(upper)*	VOLTAGE > V(upper)* (V=420,800 kV)	Voltage Deviation Index (%age of time voltage is outside range)	Number of hours voltage was outside IEGC band during month	उच्चतम (कि.वो.) Maximum(kV)	निम्नतम(कि.वो.) Minimum(kV)	औसत (कि.वो.) Average(kV)
1	पूर्वी क्षेत्र ER	अंगुल ANGUL	0%	100%	0%	0%	0	799	748	776
2		दालिपली DARLIPALI	0%	100%	0%	0%	0	791	764	776
3		गया GAYA	0%	100%	0%	0%	0	787	740	766
4		जेरत JEERAT	0%	100%	0%	0%	0	785	732	759
5		झारसूगड़ा JHARSUGUDA	0%	100%	0%	0%	1	803	766	784
6		मिदनापुर MEDINIPUR	0%	100%	0%	0%	0	787	744	766
7		रांची RANCHI	0%	100%	0%	0%	0	794	759	778
8		सासाराम SASARAM	0%	100%	0%	0%	0	779	734	759
1	उत्तरी क्षेत्र NR	आगरा AGRA	0%	100%	0%	0%	0	799	739	776
2		आगरा (फतेहाबाद) AGRA (FATEHABAD)	0%	100%	0%	0%	0	796	736	771
3		अजमेर AJMER	0%	100%	0%	0%	0	804	743	777
4		अलीगढ़ ALIGARH	0%	100%	0%	0%	0	802	737	770
5		अनपरा सी ANPARA-C	0%	100%	0%	0%	0	777	747	762
6		अनपरा डी ANPARA-D	0%	100%	0%	0%	0	777	749	762
7		अंता ANTA	0%	100%	0%	0%	0	792	722	773
8		बालिया BALLIA	0%	100%	0%	0%	0	787	734	764
9		बारा BARA	0%	100%	0%	0%	0	782	751	767
10		भाडला BHADLA	0%	100%	0%	0%	0	797	727	772
11		भाडला-2 BHADLA-2	0%	100%	0%	0%	0	800	747	776
12		भिवानी BHIWANI	0%	100%	0%	0%	1	802	742	778
13		बिकानेर BIKANER	0%	100%	0%	0%	0	804	739	774
14		बरेली BAREILLY	0%	100%	0%	0%	0	801	732	775
15		चित्तौड़गढ़ CHITTORGARH	0%	100%	0%	0%	0	802	751	779
16		फतेहगढ़-2 FATEHGARH-2	0%	100%	0%	0%	0	796	750	776
17		फतेहपुर FATEHPUR	0%	100%	0%	0%	0	795	740	768
18		ग्रैंड नोड्डा GREATER NOIDA	0%	100%	0%	0%	0	804	742	776
19		हापड़ HAPUR	0%	100%	0%	0%	1	804	735	774
20		जवाहरपुर JAWAHARPUR	0%	100%	0%	0%	1	802	741	772
21		झाटकुली JHATKARA	0%	100%	0%	0%	0	795	739	773
22		कानपुर जैआइस KANPUR GIS	0%	100%	0%	0%	0	792	739	770
23		खेतनी KHETNI	0%	100%	0%	0%	0	806	744	776
24		कोटेश्वर KOTESHWAR	0%	100%	0%	0%	0	797	739	772
25		ललितपुर LALITPUR	0%	100%	0%	0%	0	780	737	759
26		लखनऊ LUCKNOW	0%	100%	0%	0%	0	794	738	771
27		मैनपुरी MAINPURI	0%	100%	0%	0%	0	801	739	771
28		मेरठ MEERUT	0%	100%	0%	0%	1	802	742	776
29		मेरठ(UP) MEERUT (UP)	0%	100%	0%	0%	0	802	735	773
30		मोगा MOGA	0%	100%	0%	0%	1	805	725	768
31		ओबरा सी OBRA C	0%	100%	0%	0%	0	776	741	756
32		उरई ORAI	0%	100%	0%	0%	2	804	742	781
33		फागी PHAGI	0%	100%	0%	0%	0	802	744	775
34		रामपुर RAMPUR	0%	99%	0%	0%	4	804	717	764
35		उन्नाव UNNAO	0%	100%	0%	0%	0	789	727	761
36		वाराणसी VARANASI	0%	100%	0%	0%	0	790	746	772
1	पश्चिमी क्षेत्र WR	अकोला AKOLA	0%	100%	0%	0%	0	791	751	772
2		औरंगाबाद AURANGABAD	0%	100%	0%	0%	0	797	741	776
3		भोपाल (बीडीटीसीएल) BHOPAL (BDTCL)	0%	100%	0%	0%	0	792	739	770
4		भुज BHUJ	0%	100%	0%	0%	0	800	750	781
5		भुज-2 BHUJ-II	0%	100%	0%	0%	0	790	750	774
6		बिलासपुर BILASPUR	0%	100%	0%	0%	0	783	747	768
7		बिना BINA	0%	100%	0%	0%	2	800	740	778
8		बनसकंठा BANASKANTHA	0%	100%	0%	0%	1	801	756	785
9		चांपा CHAMPA	0%	97%	3%	3%	20	805	764	787
10		धुले (बीडीटीसीएल) DHULE (BDTCL)	0%	100%	0%	0%	0	800	748	774
11		धर्मजगढ़ DHARAMJAGARH	0%	100%	0%	0%	0	791	751	772
12		रायपुर पूलिंग RAIPUR POOLING	0%	100%	0%	0%	0	793	756	780
13		एकतनी EKTUNI	0%	100%	0%	0%	0	792	741	770
14		गदरवाड़ा GADARWARA	0%	100%	0%	0%	0	802	756	783
15		गवालियर GWALIOR	0%	100%	0%	0%	3	803	737	779
16		इंदौर INDORE	0%	100%	0%	0%	0	794	741	771
17		जबलपुर JABALPUR	0%	98%	2%	2%	14	806	750	784
18		खंडवा KHANDWA	0%	100%	0%	0%	0	795	745	773
19		कोरडी KORADI	0%	100%	0%	0%	0	782	751	768
20		लकाडिया LAKADIYA	0%	100%	0%	0%	2	802	757	783
21		रायगढ़ पूलिंग RAIGARH POOLING	0%	100%	0%	0%	0	799	763	782
22		पडुछे PADGHE	0%	100%	0%	0%	1	806	738	776
23		परली PARLI	0%	99%	1%	1%	4	805	752	781
24		पुणे PUNE	0%	100%	0%	0%	3	808	740	779
25		राजनन्दागव राजनान्दागव RAJNANDGAON	0%	100%	0%	0%	3	803	757	785
26		सासन SASAN	0%	100%	0%	0%	0	789	755	775
27		सतना SATNA	0%	100%	0%	0%	2	801	751	782
28		सिवनी SEONI	0%	100%	0%	0%	0	793	740	773
29		सोपल SIPAT	0%	100%	0%	0%	0	782	751	769
30		सोलापुर SOLAPUR	0%	99%	1%	1%	9	805	752	784
31		तिरुठा TIRORA	0%	100%	0%	0%	0	775	753	765
32		तामनार TAMNAR	0%	100%	0%	0%	0	796	765	781
33		वडोदरा VADODARA	0%	100%	0%	0%	1	803	742	776
34		विंध्याचल पूलिंग VINDHYACHAL PS	0%	100%	0%	0%	0	793	759	779
35		वधा WARDHA	0%	100%	0%	0%	1	802	743	781
36		वरौरा WARORA	0%	100%	0%	0%	0	800	746	782
1	दक्षिणी क्षेत्र SR	अरियलूर ARYALUR	1%	99%	1%	1%	9	809	716	766
2		कडपा CUDDAPAH	0%	94%	6%	6%	43	811	742	784
3		चिल्कलुरिपेट CHILAKALURIPETA	0%	100%	0%	0%	3	805	735	775
4		कनूल KURNOOL	0%	100%	0%	0%	1	802	738	776
5		महेश्वर MAHESWARAM	0%	89%	11%	11%	83	815	742	785
6		निजामाबाद NIZAMABAD	0%	80%	20%	20%	148	817	741	788
7		नेल्लूर पूलिंग NELLORE PS	0%	100%	0%	0%	0	793	741	774
8		नॉर्थ चेन्नै पूलिंग NORTH CHENNAI PS	0%	85%	15%	15%	114	821	731	783
9		रायचूर RAICHUR	0%	100%	0%	0%	2	802	744	779
10		श्रीकाकुलम SRIKAKULAM	0%	98%	2%	2%	17	807	743	783
11		तिरुवेलम THIRUVALEM	0%	100%	0%	0%	1	803	732	774
12		वेंमगिरि VEMAGIRI	0%	98%	2%	2%	19	809	741	779
13		वारंगल WARANGAL	0%	98%	2%	2%	17	808	741	783
1	पूर्वोत्तर क्षेत्र NER	बालिपारा BALIPARA (400 kV)	0%	89%	7%	7%	53	429	395	410
2		बिस्वनाथ चरिआल BISWANATH CHARIALI (400 kV)	0%	100%	0%	0%	0	418	398	409
3		बोंगागाँव BONGAIGAON (400 kV)	0%	100%	0%	0%	0	420	400	410
4		बोंगागाँव टोपिएस BONGAIGAON TPS (400 kV)	0%	100%	0%	0%	0	417	396	406
5		इम्फाल IMPHAL (400 kV)	0%	98%	2%	2%	12	422	403	412
6		बर्निहाट BYRNHAT (400 kV)	0%	100%	0%	0%	0	416	390	404
7		कामेंग KAMENG (400 kV)	0%	100%	0%	0%	0	402	393	398
8		अज़ारा AZARA (400 kV)	0%	100%	0%	0%	0	415	392	403
9		मिसा MISA (400 kV)	0%	100%	0%	0%	0	419	395	407
10		नू कोहमा NEW KOHIMA (400 kV)	0%	100%	0%	0%	0	416	401	407
11		पालाताना PALATANA (400 kV)	0%	100%	0%	0%	0	414	393	402
12		प्रबुक्चनबारी PK BARI (400 kV)	0%	96%	4%	4%	29	424	392	410
13		रंगानादी RANGANADI (400 kV)	0%	100%	0%	0%	0	419	399	409
15		सुबानसिरी SUBANSIRI (400 kV)	0%	100%	0%	0%	0	412	393	401
16		सुज्यामनिनगर SURJYAMANINAGAR (400 kV)	0%	100%	0%	0%	0	418	394	406
17		थोबाल THOUBAL (400 kV)	0%	100%	0%	0%	0	414	390	402

All listed stations are 765 kV stations unless otherwise mentioned

*Percentage is calculated w.r.t. Time of one month.

23.ALL TIME HIGHEST

31-01-2024

	Maximum Demand Met during the day (MW)	Demand Met during Evening Peak hrs(MW)	Energy Met (MU)	Hydro Gen. (MU)	Wind Gen. (MU)	Solar Gen. (MU)
NR	81220 04-09-2023	76274 04-09-2023	1796 04-09-2023	443 01-08-2023	86 07-08-2023	147 10-05-2023
WR	74799 29-01-2024	65518 13-10-2023	1618 02-09-2023	167 18-12-2014	273 07-08-2023	70.69 12-05-2023
SR	64314 15-03-2023	52366 18-04-2023	1329 19-04-2023	208 31-08-2018	298 21-07-2023	142.7 21-03-2023
ER	29641 28-07-2023	28579 18-04-2023	639 08-06-2023	157 14-09-2022	-	7 23-03-2023
NER	3712 04-09-2023	3510 12-08-2022	72 15-09-2023	41 29-08-2023	-	2 22-06-2022
All India	239978 01-09-2023	215592 28-08-2023	5224 02-09-2023	877 30-08-2022	611 02-08-2023	351 10-05-2023
Regions	States	Max. Demand Met during the day (MW)		Energy Consumption (MU)		
			As on date		As on date	
NR	Punjab	15237	23-06-2023	343.5	24-06-2023	
	Haryana	12692	18-08-2023	273.1	18-08-2023	
	Rajasthan	17949	20-01-2024	371.6	04-09-2023	
	Delhi	7770	29-06-2022	153.5	28-06-2022	
	UP	28553	25-07-2023	580.0	03-09-2023	
	Uttarakhand	2594	14-06-2022	55.8	17-06-2023	
	HP	2235	20-01-2024	40.7	22-08-2023	
	J&K(UT) and Ladakh(UT)	3636	29-06-2023	66.8	26-01-2024	
	Chandigarh	426	08-07-2021	8.4	08-07-2021	
Railways NR ISTS	-	-	-	-		
WR	Chhattisgarh	5832	02-09-2023	133.5	28-08-2023	
	Gujarat	24544	01-09-2023	511.6	02-09-2023	
	MP	17819	29-12-2023	335.9	10-11-2023	
	Maharashtra	28846	14-04-2022	636.1	21-06-2023	
	Goa	776	07-06-2023	17.1	08-06-2023	
	DD & DNH	1327	17-08-2023	30.9	10-08-2023	
	AMNSIL	1083	10-01-2024	21.0	31-05-2022	
	Balco	-	-	-	-	
SR	Andhra Pradesh*	13039	19-10-2023	263.8	16-06-2023	
	Telangana*	15497	30-03-2023	303.4	03-03-2023	
	Karnataka	16950	25-08-2023	309.5	20-04-2023	
	Kerala	5024	18-04-2023	103.3	19-04-2023	
	Tamil Nadu	19387	20-04-2023	415.4	20-04-2023	
	Pondy	512	17-05-2023	11.6	18-05-2023	
ER	Bihar	7578	27-07-2023	161.3	28-07-2023	
	DVC	3751	18-04-2023	81.2	22-04-2022	
	Jharkhand	1923	16-06-2023	43.3	17-06-2023	
	Odisha	7192	19-04-2023	145.5	25-09-2022	
	West Bengal	11868	18-04-2023	245.7	18-04-2023	
	Sikkim	137	03-02-2022	2.5	28-01-2020	
	Railways ER ISTS	-	-	-	-	
NER	Arunachal Pradesh	181	31-01-2024	3.2	04-09-2023	
	Assam	2498	16-09-2023	50.0	15-09-2023	
	Manipur	258	31-01-2022	3.7	10-02-2022	
	Meghalaya	415	09-02-2022	7.8	31-01-2022	
	Mizoram	162	27-01-2024	2.5	27-01-2024	
	Nagaland	179	28-09-2023	3.29	05-09-2023	
	Tripura	370	07-06-2023	7.0	06-06-2023	
*SR	Andhra Pradesh (Undivided)	13162	23-03-2014	284.8	22-03-2014	

24.System Reliability Indices Report for the month of January 2024

Percentage (%) of times ATC was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	74	18.50	2.49
2	ER-NR	1	0.25	0.03
3	Import of NR	10	2.50	0.34
4	NEW-SR	0	0.00	0.00
5	NER Import	0	0.00	0.00

Percentage(%) of times (N-1) Criteria was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	36	9.00	1.21
2	ER-NR	0	0.00	0.00
3	Import of NR	3	0.75	0.10
4	NEW-SR	0	0.00	0.00
4	NER Import	0	0.00	0.00

Remarks: Flows crossing Total Transfer Capability (TTC) on interregional corridors has been worked out as a proxy for (N-1) violation.



POWER SYSTEM OPERATION CORPORATION LIMITED

पावर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड

NATIONAL LOAD DESPATCH CENTRE – राष्ट्रीय भार प्रेषण केन्द्र

First floor, B-9, Qutab Institutional Area,
Katwaria Sarai, New Delhi-110016.